

1.1 Ideas about the cause of disease and illness

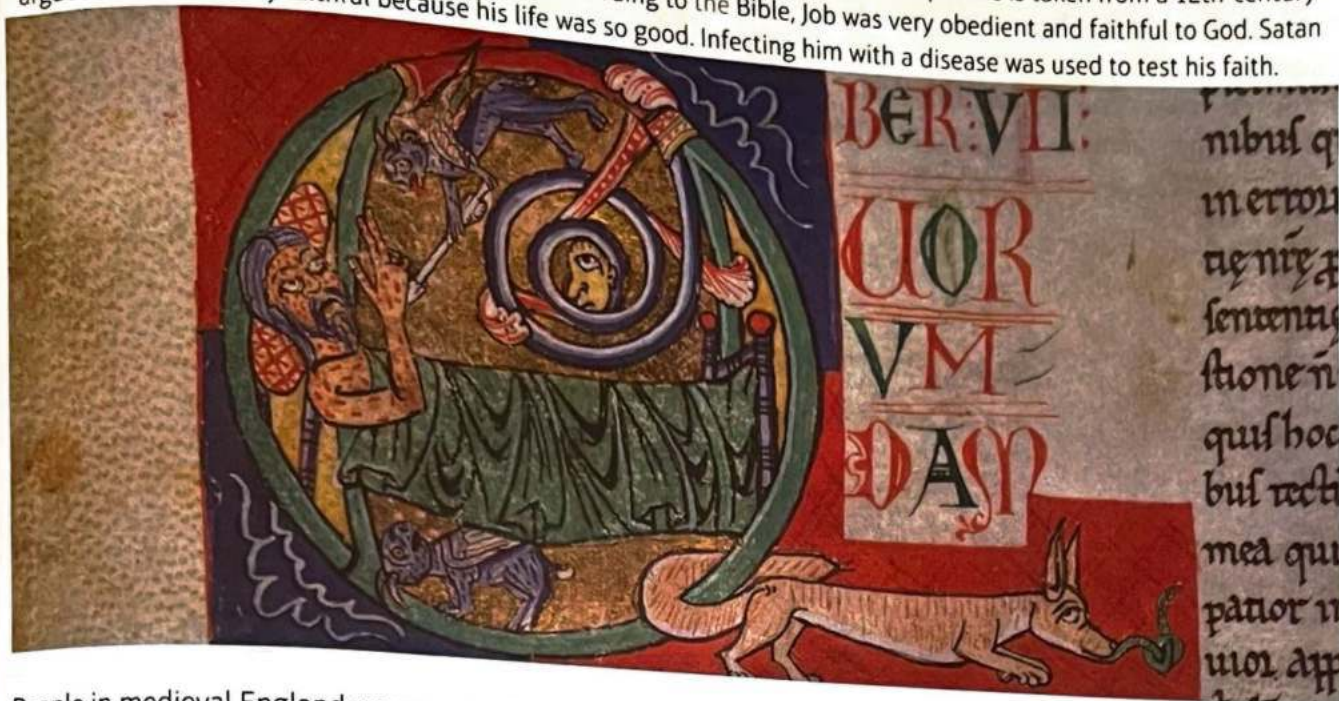
Learning outcomes

- Understand different ideas about the cause of disease before 1500, including the Theory of the Four Humours.
- Know the different influences on ideas about the cause of disease before 1500.

Supernatural and religious explanations of the causes of disease

Source A

The devil, or Satan, was often held responsible for sending diseases to people. This picture is taken from a 12th-century manuscript. In the image, Satan is infecting Job. According to the Bible, Job was very obedient and faithful to God. Satan argued that he was only faithful because his life was so good. Infecting him with a disease was used to test his faith.



People in medieval England were very religious. The vast majority of people in England followed the teachings of the Catholic Church. They attended services regularly and were expected to give a sum of money to the Church each month. This was known as a **tithe**. The Church also owned large amounts of land in England, where it built churches, monasteries and convents. These became important centres of the community: as well as praying, monks and nuns of the Church provided basic medical care, looking after people who were not able to care for themselves. The Church used the tithes given by ordinary people to pay for the care of the community.

Illness was not uncommon. Malnutrition*, particularly in times of famine*, made people more likely to fall ill. A lack of scientific knowledge at this time meant that the causes of disease and illness were a mystery. The Church used religion to answer the questions people had about illness and disease.

Ordinary people received most of their teaching from the Church, as they didn't receive any formal education. The majority of people at this time could not read or write. Instead, they learned from the stories they heard, or the paintings they saw on the wall of their church. One thing they learned was that sin was very dangerous. The Church taught that those who committed a sin could be punished by God. They also taught that the devil could send disease to test someone's faith, as seen in Source A.

Key terms

Malnutrition*

An illness caused by lack of food.

Famine*

Food shortage, usually due to bad harvests.

The Church often explained famine by saying that God had sent it as a punishment for sin. Therefore, it was logical also to blame people's sins for their illnesses. This meant that, when people recovered, the Church was able to declare that a miracle had happened, thanks to the patient's prayers. Therefore, blaming sickness on God acted as 'proof of the divine': it provided evidence of God's existence. This explains why the Church supported the idea that God sent disease as a punishment.

Although disease was mainly seen as a result of sin, the Church also taught that disease was sent by God to cleanse one's soul of sin. If you became ill, God could be sending the illness to purify your soul, or to test your faith. Since they had learned that God controlled every aspect of the world, this was very believable to people at this time.

Leprosy

The Bible tells many stories of how God sent disease as a punishment – leprosy in particular was included in the Bible as an illustration of a punishment for sin. Leprosy usually began as a painful skin disease, followed by paralysis* and eventually death. Fingers and toes would fall off, body hair would drop out and ulcers would develop both inside and outside the body.

Key term

Paralysis*

Being unable to move either all or part of your body as a result of illness, poison or injury.

There was no cure for leprosy, so lepers were banished from their communities. They usually had to move to leper houses or to isolated island communities. If they were allowed to stay in their home towns, they had to wear a cloak and ring a bell to announce their presence, and they were banned from going down narrow alleys, where it was impossible to avoid them.

Source B

A painting of a leper from around 1400. Lepers were made to wear a cloak to cover their diseased bodies and ring a bell to warn people when they were nearby. The bell would also have acted as a way to ask for alms, or charitable donations. The words say, 'Some good, my gentle master, for God's sake'.



This was because it was believed their breath was contagious. Although this was not true (leprosy was spread only by very close contact with the infected), it does show that medieval people had some correct ideas about how some diseases were transmitted.

Although there was no formal care for lepers, a few leper houses did help people suffering from leprosy. Leper houses were more commonly known as leper colonies.

Astrology

Along with the role of God, the alignment of the planets and stars was also considered very important when diagnosing* illness. A physician* would consult star charts, looking at when the patient was born and when they fell ill, to help identify what was wrong.

Traditionally, the Church frowned upon the idea of using astrology in diagnosing illness, as it seemed only one step away from predicting the future, or fortune telling. However, after the Black Death arrived, astrology became more popular and the Church became more acceptant of it. Many people believed the Black Death was caused by a bad alignment of the planets.

Astrology was a **supernatural** explanation for disease. During the period c1250–c1500, the impact of the stars and planets on health was considered important because of the influence of Hippocrates (see page 17), who had been a leading physician from Ancient Greece.

The Theory of the Four Humours

Not all explanations for disease were supernatural or religious. A very popular idea, first put forwards by the Ancient Greeks, was the **Theory of the Four Humours**.

The theory stated that, as the universe was made up of the four basic elements – fire, water, earth and air – the body must also be made up of four humours, which were all created by digesting different foods. The four humours were:

- **blood**
- **phlegm** – the watery substance coughed up or sneezed out of the nose, or expelled in tears
- **black bile** – not one particular substance in the body, but probably referred to clotted blood, visible in excrement or vomit
- **choler**, or yellow bile – this appeared in pus or vomit.

There was a belief that all the humours must be balanced and equal. If the mix became unbalanced, you became ill. Being careful to maintain a good balance of the humours was really important to preserving good health. However, people believed a combination of age, family traits and circumstances, such as the season in which someone was born, usually combined to make one or two of the humours stronger than the others.

Key terms

Diagnosing*

Deciding what is wrong with a patient by considering different symptoms. In medieval England, physicians could also consider star charts to diagnose an illness.

Physician*

Someone who practices medicine. A medieval physician did not have to have the same level of training as a modern physician.

Activities

- 1 Draw a rough outline of a church building. Inside it, list at least three reasons why many people in the Middle Ages believed that the main cause of illness and disease was punishment from God.
- 2 Write a leaflet in the style of a medieval manuscript, offering advice on how to avoid disease and illness.
- 3 If people believed that God sent disease, what sort of treatments do you think they tried? Make a list – you can see how many you correctly predicted in the next section.

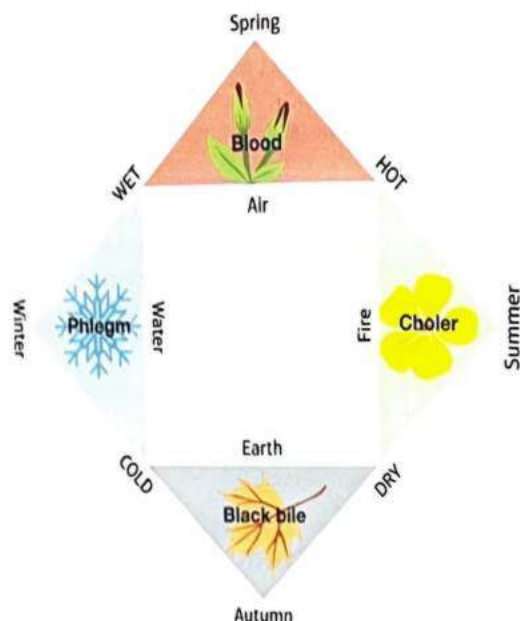


Figure 1.1 The Four Humours.

Humour	Season	Element	Qualities	Ancient name
Blood	Spring	Air	Hot and wet	Sanguine
Cholera	Summer	Fire	Hot and dry	Choleric
Black bile	Autumn	Earth	Cold and dry	Melancholic
Phlegm	Winter	Water	Cold and wet	Phlegmatic

Activity ?

Create a paper fortune teller to help you remember the different parts of the Theory of the Four Humours.

- On the outside, write the names of the four humours.
- On the first layer inside, write the characteristics of each humour – hot, cold, wet, dry.
- On the next layer, write anything else that is related to the humour – characteristics, seasons and star signs.

According to the theory, each humour was linked to certain characteristics that physicians would look for when carrying out their diagnosis. For example, a person suffering from a fever had a temperature, causing the skin to go hot and red because, physicians believed, they had too much blood. This was a hot and wet element. Meanwhile, a person suffering from a cold had too much phlegm, which was cold and wet. They would shiver and the excess phlegm would run out of their nose.

The humours were linked with the seasons. For example, in winter, which is cold and wet, it was thought that the body produces too much phlegm, causing coughs and colds as the patient tries to get rid of it. The star signs for each season were associated with its humour, too: Capricorn, Aquarius and Pisces were linked with phlegm. Astrology was considered an important part of the Theory of the Four Humours, as the humours were connected with star signs and seasons, and each one had its own ruling planet.

The humours were also linked with certain personality traits. For example, a quick tempered, argumentative person was said to have choleric characteristics, while an optimistic, calm person had cheerful tendencies (sanguine). What we would today recognise as depression was blamed on an excess of black bile (melancholic) in medieval times.

The origins of the theory

The Theory of the Four Humours was created by an Ancient Greek physician named Hippocrates in the 5th century BCE. The word 'humour' comes from the Greek word for fluid – humon. Hippocrates was very careful to observe all the **symptoms** of his patients and record them. The Theory of the Four Humours fitted with what he saw.

Galen, a physician in Ancient Rome during the 2nd century CE, liked the ideas of Hippocrates and developed them further. He had been a physician in a gladiator school and later became the personal physician of the Roman Emperor. This meant that he had lots of time to experiment, ponder philosophy and write. By the time he died, he left behind a very large body of work – more than 350 books.

Extend your knowledge**Galen's character**

Galen had a reputation for being quite arrogant and a 'know-it-all'. He wrote about this in his works, using himself as an example of how one humour could be more dominant than another. He claimed that he had a naturally choleric temperament, which made him short-tempered. He thought he had inherited this from his mother, who was so bad tempered that she used to bite her servants!

Galen developed the Theory of the Four Humours to include the idea of balancing the humours by using the **Theory of Opposites**. For example, he suggested that too much phlegm, which was linked to water and the cold, could be cured by eating hot peppers; a fever, or an excess of blood, could be treated with cucumber, which would cool the patient down when eaten. Galen also theorised that the circulatory system circulated blood generated in the liver, and the blood was then distributed around the body.

Why was the theory so popular?

The Theory of the Four Humours was very detailed and could be used to explain away almost any kind of illness – physical or mental. It was important that the theory covered almost every type of illness that occurred, because there was no other scientific explanation for the cause of disease. Often, physicians twisted what they saw to fit with the logic of the theory.

Activities ?

- Create an advice poster to explain how the Four Humours led to illness. Make sure you give examples of different illnesses caused by different imbalances.
- Hippocrates and Galen were both physicians who had a huge impact on medieval medicine. Write a short paragraph to summarise their ideas and explain how Galen built on the work of Hippocrates.

If you are suffering from too much...

phlegm



blood



...then treatment is something...

hot and dry



cold and wet



Figure 1.2 Examples of the Theory of Opposites.

Classical thinking in the Middle Ages

Hippocrates and Galen were both very popular figures in medieval medicine. Although written in the time of Ancient Greece and Rome, Latin translations of their texts only started to appear in Europe from the 11th century – almost 800 years after they were written. These translations were copied and recopied by monks, who passed them on to new medical universities. The first European medical school had been established in Salerno in the 9th century, and taught students based on these texts, rather than practical experience.

Extend your knowledge

The *Articella*

By the mid 13th century, medical students in European universities relied on the *Articella* to tell them everything they needed to know about medicine. This medical textbook included some directly translated works from Hippocrates and Galen. It also included the work of a 9th-century Persian doctor, Hunayn ibn-Is'haq, who had studied the work of both Hippocrates and Galen.

Other classical works were also popular among physicians and medical students, including those of the Greek thinker, Aristotle, and the Persian philosopher and physician, Avicenna. Physicians were expected to have a good background in the liberal arts, such as philosophy, before studying medicine.

Galen's influence

However, classical texts like those of Galen continued to be very influential in the Middle Ages for three reasons: the influence of the Church, the importance of book learning and the lack of alternative theories.

The influence of the Church

Galen wrote that the body was clearly designed for a purpose and that the different parts of the body were meant to work together in balance, as first proposed by Hippocrates. Galen also believed in the idea of the soul. This theory fitted in very well with the ideas of the Church, who believed that God created man in his image, and so they promoted Galen's

teachings and, by extension, those of Hippocrates. Since books were produced in monasteries, and libraries were maintained by the Church, their choice of texts were the ones that were widely read, preached and believed. In the early Middle Ages, the Church controlled medical learning in universities, too.

The importance of book learning

Many people could not read in the Middle Ages. This meant that being widely read was a sign of intelligence. A good physician was considered to be one who had read many books, rather than one who had treated a lot of patients. Having read the works of Hippocrates and Galen was proof that a physician was worth the money he was being paid. The authority of these classical texts was so strong that people believed them even when there was actual, physical evidence that suggested they were wrong.

The lack of alternatives

There was a lack of scientific evidence to support any other kind of theories of the causes of disease. Dissections were mostly illegal, because the Church taught that the body needed to be buried whole in order for the soul to go to heaven. Very occasionally, physicians were able to dissect executed criminals, or criminals who had been sentenced to death by vivisection*. When this happened, the physician would sit far away from the body, reading from the works of Galen, while the actual cutting and examining was done by a barber surgeon*. This meant that Galen's ideas were preserved: anything in the body that didn't agree with Galen's writings could be explained away, since the body was that of a criminal and therefore imperfect, and the physician himself never did any of the examinations.

Key terms

Vivisection*

Criminals sentenced to death by vivisection had their bodies cut open (dissected) and examined by physicians and medical students.

Barber surgeon*

Barbers worked with sharp knives, so as well as giving people haircuts, they also carried out medical procedures such as blood-letting (see page 23). Over time, they took on smaller surgeries.

Exam-style question, Section B

Explain why there was continuity in ideas about the cause of disease during the period c1250–c1500.

You may use the following information in your answer:

- the Church
- Galen.

You must also use information of your own. **12 marks**

Source C

A picture of a medieval dissection. The physician is sitting high up, away from the body. He is reading aloud from the works of Galen. The body is being dissected by somebody else. This painting appeared in the first illustrated anatomy book to be printed – the *Fasciculus Medicinae*, written by Joannes de Ketham, an Austrian professor of medicine, in 1491.



Exam tip

There are **six marks** available here for your knowledge and **six marks** available for how well you explain your answer.

- Make sure you use your **own knowledge** as well as what is suggested by the bullet points. If you don't, you are limiting yourself to a maximum of eight marks.
- Think about **structure** and **coherence**. Before you start writing, think carefully about the order of your points and make links between them.

Other ideas about the cause of disease

Miasma

A **miasma** was bad air that was believed to be filled with harmful fumes. Hippocrates and Galen both wrote about miasmata (which is the plural of miasma) and suggested that swamps, corpses and other rotting matter could transmit disease.

Extend your knowledge

The Romans

Over a thousand years before the 13th century, the Romans had built their settlements away from swamps and smelly places in an effort to avoid miasmatic diseases, such as malaria. Indeed, the word malaria comes from the Italian words 'mala' – which means bad, and 'aria' – which means air.

Smells and vapours like miasmata were also, unsurprisingly, associated with God. A clean and sweet-smelling home was a sign of spiritual cleanliness, and incense was burned in churches to purify the air. Homes that smelled badly suggested sinfulness and corruption and, if a person was unwashed, other people would avoid them, in case they breathed in the bad miasma and contracted a disease. This was also why people avoided lepers. Although many believed leprosy was a punishment from God, they also believed the disease was contagious.

More on miasma

It is worth noting that medieval physicians didn't use the word 'miasma' to describe disease-spreading air, as far as we can tell from the evidence we have. Instead, they referred to it as 'corruption of the air', 'pestilential air' or 'putrefaction of the air'. 'Miasma' was not used regularly until later.

Urine charts

Although medieval physicians didn't blame people's urine for making them ill, they did carefully examine the urine in order to make their diagnosis. It was thought to be one of the best ways to check on the balance of the humours inside the body. Samples of a patient's urine could be sent to a physician, where it would be examined and compared with a urine chart.

The physician would carefully check the colour, thickness, smell and even taste of the urine before making his diagnosis. This was seen as a very important part of medieval medicine. Norwich Cathedral Priory, for example, employed a full-time physician to examine urine.

Influences on ideas about the cause of disease

The Middle Ages was a time when there was **continuity** in ideas about the cause of disease. There were only a few small changes. For example, the use of astrology became more widespread, which meant it was adopted by many people, but did not expand upon Galen's original theories. On the whole, ideas remained the same.

Individuals and the Church

The Church was very important in maintaining the status quo at this time. This means that they did not like change, and wanted to keep things the way they were. The Church controlled medical learning. It chose which books were copied and distributed. The Church liked the Theory of the Four Humours because it fitted with their teachings, so it promoted this theory. The Church strongly discouraged anybody from criticising the theory.

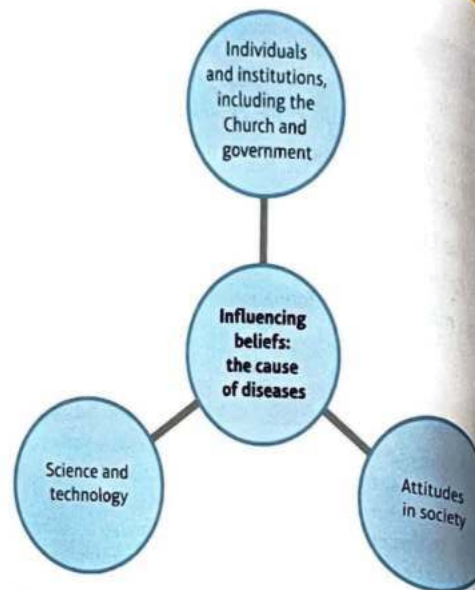


Figure 1.3 Three main factors to understanding change and continuity in early diagnosis of disease.

Hippocrates and Galen were important individuals in the Middle Ages, even though they lived many years before this time. Their books had been preserved by Arabic scholars and Latin translations were available in Europe by the Middle Ages. Galen in particular was popular with the Church, which meant that his work was widely promoted.

Science and technology

During the Middle Ages, a lack of scientific understanding meant that new knowledge was limited. Physicians and medical students tried to make new discoveries fit into the old theories, rather than experimenting to explain the discoveries.

One important piece of technology invented near the end of the Middle Ages was the **printing press**. It was invented in around 1440 by Johannes Gutenberg in what is now Germany. Although this was not directly related to advancing medieval medicine, it eventually led to much faster and easier sharing of medical texts. However, it did not have a huge impact during the medieval period.

Attitudes in society

Medieval people had a strong belief in God and did not want to risk going to hell by being critical of the Church. Physicians who did not follow the old ideas found it hard to get work, because everybody knew a 'good' physician would follow the Theory of the Four Humours. A famous 13th-century military surgeon, Henri de Mondeville, was among those who questioned the old ways of practising medicine. Mondeville is even quoted as stating that classical figures like Hippocrates and Galen were like an old dog that needed to be put down, but whose owner could not face replacing it with a younger, healthier dog. However, even he continued to practise medicine in the same way as everybody else – he probably wouldn't have found employment otherwise.

Many people believed that, since medicine had always been done this way, there was no need to change it.

Activity

This activity will help you to decide the importance of each key idea about the cause of illness in the years c1250–c1500.

- On slips of paper, write down the key ideas about the causes of illness at this time: God; imbalanced humours; miasma; alignment of the planets/stars.
- Now put these into a hierarchy. Which do you think was the most important? Which was the least important? Do we still believe any of them today? Were some of these ideas more important to particular groups of people, or at different times?
- Look back over the information in this section if you need to jog your memory. Discuss your ideas with another person or in a small group.

Summary

- Many people believed that disease was a punishment or test from God because of sin.
- The Theory of the Four Humours was very popular. It stated that an imbalance in the humours could cause illness.
- Other ideas about the causes of disease and illness included miasma (bad air) and a misalignment in the planets or stars.

Checkpoint

Strengthen

- Create a spider diagram or a bullet point list to show the different ideas people had between 1250 and 1500 about what caused illness and disease.
- List the four humours and their properties.
- Describe, in detail, the role of Hippocrates and Galen in medieval medicine.

Challenge

- Explain both **why** and **how** the Church had an impact on medieval medicine.

If you are not confident about this question, form a group with other students, discuss the answer and then record your conclusion. Your teacher can give you some hints.

1.2 Approaches to treatment and prevention

Learning outcomes

- Understand different approaches to treatment and prevention before 1500.
- Understand how people cared for the sick, including treatments in hospitals and in the home before 1500.

Religious and supernatural treatments

As the Church taught that disease was sent by God as a punishment for sin, it followed that the cure should also involve the supernatural. As well as looking for medical treatment for disease, it was important to undergo a course of spiritual healing. Religious treatments included:

- healing prayers and incantations (spells)
- paying for a special mass* to be said
- fasting (going without food)

Pilgrimages* to the tombs of people noted for their healing powers also became extremely popular. Once the pilgrimage was complete, there were a few suggested actions those with diseases could take (see Figure 1.4).

Key terms

Mass*

Roman Catholic service where bread and wine is given. Catholics believe that this involves a miracle: the bread and wine is turned into the body and blood of Christ.

Pilgrimage*

A journey to an important religious monument, shrine or place.

Touching holy relics, such as a piece of the 'true' cross on which Jesus was crucified, or the bones of a saint.

Presenting an offering at a shrine – usually an image of the body part to be healed, made from anything from wax to precious metals and jewels, depending on how wealthy you were.

Lighting a candle proportionately as tall as you (or as long as the body part you wanted to heal).

Praying for God to help heal your ailment.

As well as making the locals a lot of money, as the pilgrims brought with them money to spend in the towns of holy sites, the ability to heal the sick was also important to the local churchmen. In order to be declared a saint, the dead person had to be associated with a certain number of miracles; so, the more people who were 'healed' after touching the relics, the better.

Extend your knowledge

The king's healing power

It was widely believed that the king had the power to heal certain illnesses. During the coronation of a monarch, his hands would be rubbed with special holy oil and this, it was said, gave him supernatural healing powers. The king's touch was considered particularly effective for scrofula, a form of tuberculosis. Kings took this very seriously, as it was a good way to demonstrate their divine right (the idea that God had chosen them to be king). Edward I, for example, aimed to touch up to 2,000 people a year during his reign (1272–1307).

If prayers and offerings did not work, there were other supernatural remedies available, although the Church did not approve of them. Chanting incantations and using charms or amulets to heal symptoms and ward off diseases were fairly common throughout this period.

Sometimes the sick were discouraged from seeking cures. After all, if God had sent the disease to purge the soul, it was important for the disease to run its course. Taking medicine to cure the disease might keep you alive, but it would mean that your soul would still be stained with sin. That meant risking not being admitted into heaven when you died.

Astrology

Physicians consulted star charts when diagnosing illness. These were also important when prescribing treatment. Treatments varied according to the horoscope of the patient. The alignment of the planets was then checked at every stage of the treatment prescribed: herb gathering, bleeding, purging, operations and even cutting hair and nails all had to be done at the right time.

1.2 Approaches to treatment and prevention

Humoural treatments

Today, when we fall ill, doctors assess the symptoms, make a diagnosis and treat the infection. For example, if a patient catches a chest infection, the treatment prescribed will be to attack the germ, rather than to stop the patient from coughing. This is because modern medicine recognises that treating the cause of the illness will eventually treat the symptom.

Medieval physicians did not work in the same way. Each symptom was broken down and treated separately, as they believed each symptom represented an imbalance in the humours. Therefore, conflicting remedies might be provided.

Source A

Advice from John of Gaddesden's medical book, the *Rosa Anglica*. John, a very well-respected English physician, wrote this very popular medical text in the 14th century. Here, he explains how to cure lethargy [extreme tiredness].

It is necessary for lethargies that people talk loudly in their presence. Tie their extremities lightly and rub their palms and soles hard; and let their feet be put in salt water up to the middle of their shins, and pull their hair and nose, and squeeze the toes and fingers tightly, and cause pigs to squeal in their ears; give them a sharp clyster [an enema] at the beginning... and open the vein of the head, or nose, or forehead, and draw blood from the nose with the bristles of a boar. Put a feather, or a straw, in his nose to compel him to sneeze, and do not ever desist from hindering him from sleeping; and let human hair or other evil-smelling thing be burnt under his nose...

Blood-letting

Phlebotomy, or blood-letting/bleeding, was the most common treatment for an imbalance in the humours. The idea behind it was that bad humours could be removed from the body by removing some of the blood.

Phlebotomy was so common that most physicians didn't even bother to carry out the procedure themselves – and monks were forbidden from bleeding their patients.

Figure 1.4 Pilgrims suffering from disease visit a holy site.

Instead, it was usually done by barber surgeons and wise women. Demand was so high that even some people with no medical background offered the service.

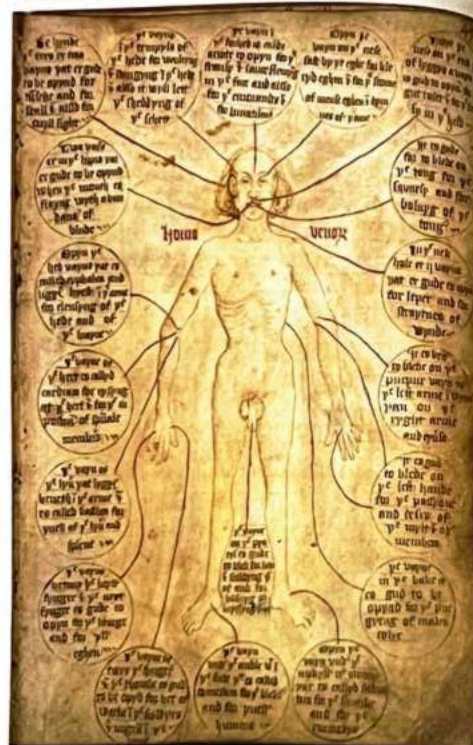
Bleeding was carried out in several different ways.

Type	Method	Uses
Cutting a vein	This involved cutting open a vein with a lancet or other sharp instrument. Blood was usually let from a vein near the elbow, because it was easy to access.	The most straightforward method of bleeding. Phlebotomy charts like the vein man (see Source B) were used to show points in the body where bleeding was recommended for specific illnesses.
Leeches	Freshwater leeches were collected, washed and kept hungry for a day before being placed on the skin. Bleeding might continue for up to 10 hours after the leech was full.	Used for people whose age or condition made traditional bleeding too dangerous.
Cupping	The skin was pierced with a knife or a pin, or even scratched with fingernails, until it was bleeding. A heated cup was placed over the cuts to create a vacuum. This drew blood out of the skin.	Used for women, children and the very old. People believed different areas treated different illnesses. For example, people believed that cupping on the back of the neck was good for eye trouble, bad breath and facial acne.

Sometimes patients were bled for too long and died as a result. Evidence suggests that this was quite common and it was probably seen as a necessary hazard. In 1278, court records from an inquest in London show that William le Paumer had collapsed and died due to a blood-letting procedure that had taken place the previous day. The court did not hold anybody responsible for the death, and didn't even name the person who had carried out the bleeding.

Source B

A vein man, or phlebotomy chart. This picture was printed in a manual belonging to the York Barbers in the late 15th century. It shows points on the body where blood-letting should happen, matching different ailments with different places on the body. For example, if a patient was suffering from depression, the recommendation was to bleed them from a vein in the back.



Purging

Because it was believed that the humours were created from the foods eaten, a common treatment was purging the digestive system to remove any leftover food. This was done by giving the patient either something to make them vomit (an emetic), or a laxative or enema to clear out anything left over in the body.

Emetics usually consisted of strong and bitter herbs like scammony, aniseed and parsley. Sometimes they contained poisons like black hellebore, so it was best to vomit them up quickly.

Laxatives were very common. Some well-known examples included mallow leaves stewed in ale, and linseeds fried in hot fat. Linseeds are still used today as a digestive aid.

Sometimes people needed a bit more help to purge, and the physician would administer a **clyster** or enema. For example, John of Arderne, a famous English surgeon, mixed water with honey, oil, wheat bran, soap and herbs such as mallow and camomile. He would squirt it into the patient's anus using a greased pipe fixed to a pig's bladder, while the patient rubbed his stomach. This would clear out any stubborn blockages.

Remedies

Sick people in the period c1250–c1500 were also treated with remedies – usually herbal infusions to drink, sniff or bathe in.

Some of these are still in use today. For example, aloe vera was prescribed to improve digestion. Other ingredients featured regularly included mint, camomile and rose oils, tamarind, almonds, saffron, butter, absinth, turpentine and corals. Sometimes the ingredients were expensive and difficult to find.

A common remedy mixed and sold at this time was **theriaca**. This was a spice-based mixture that could contain up to 70 ingredients, including quite common things like ginger, cardamom, pepper and saffron, but also some unusual ingredients like viper flesh and opium. Galen had written a book on theriacs, looking particularly at their use in treating snake bites and poisons. Over time, they became widely popular and were used for many different illnesses.

Different foods were prescribed to encourage the balance of the humours – remember that the humours were thought to be created from the digestion of food. A dish called **blanc mangier**, made from chicken and almonds, was regularly recommended for medieval invalids because the ingredients were warm and moist.

Bathing

Warm baths were regularly prescribed to help the body draw in heat to help dissolve blockages in the humours. This gave the body the opportunity to steam out impurities and ease aching joints. Herbal remedies could also be given this way.

Various plants and herbs were added to the bath water to help. For example, people suffering from bladder stones were advised to stir in mallow and violets.

Some of the remedies were less pleasant and were based purely on superstition: those suffering from paralysis were advised to boil a fox in water and then bathe in it! This was because it was thought that the quick and nimble properties of the fox would be transferred to the patient through the bath water.

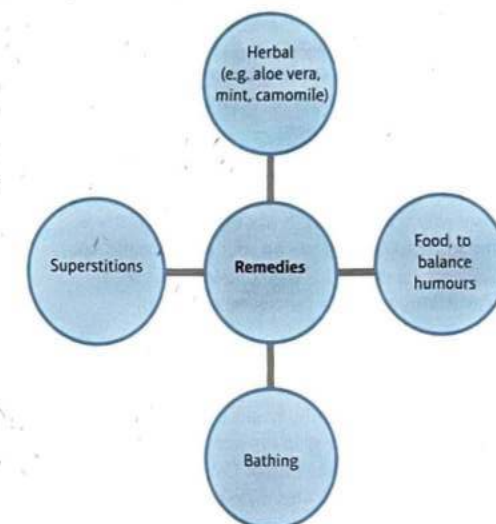


Figure 1.5 Common remedies, c1250–c1500.

Extend your knowledge

Popular books

There were many books around at this time that gathered together knowledge of the healing powers of herbs and plants. These were generally known as *Materia Medica*. The most popular book had been written by a Roman called Dioscorides in the 1st century CE. He had catalogued around 600 plants, roots, trees, minerals, essences and balms, and their properties.

Another popular herbal manual was the *Circa Instans*. Published in the 12th century, it included details of the use of minerals and metals. It also gave recommendations on how to use the ingredients in spells and incantations.

Preventing disease

Although a physician could expect to be paid a lot more money for providing treatments for disease, there was a strong focus on following various regimes to prevent getting sick in the first place. This was seen as a far safer plan of action, since cures and treatments were hit-and-miss in their effectiveness.

The Church

Most people believed that the best, and most important, way of preventing disease was to lead a life free from sin. Regular prayers, confessions and offering tithes to the Church worked together to ensure that any minor sins were quickly forgiven.

Hygiene

Once your spiritual health was taken care of, it was important to concentrate on your bodily health, too. Guidance on doing this was contained in a set of instructions known as the *regimen sanitatis*.

Regimen Sanitatis

The *regimen sanitatis* was a loose set of instructions provided by physicians to help a patient maintain good health. It first appeared in the work of Hippocrates, where it was later picked up by Galen and Arabic scholars like Avicenna. This meant that the advice was widespread and very common by 1250. A lot of the advice is familiar to us today.

Ideally, a physician would provide a *regimen sanitatis* written especially for their patient, taking into account their predominant humours and lifestyle. However, in practice, this service was only used by the very rich, because it took a long time to write such a detailed set of instructions for every patient.

Bathing was an important treatment for disease. It was also used as a preventative measure: bad smells indicated a miasma. However, only the wealthy could afford a private bath of hot water. Public baths, or *stewes*, were available for a fee. Poorer people swam in rivers, where possible, to keep themselves fresh. Although only the wealthy bathed their whole bodies regularly, everybody – no matter how poor – washed their hands before and usually after every meal. They believed that cleanliness was next to godliness, so it was important to stay clean.

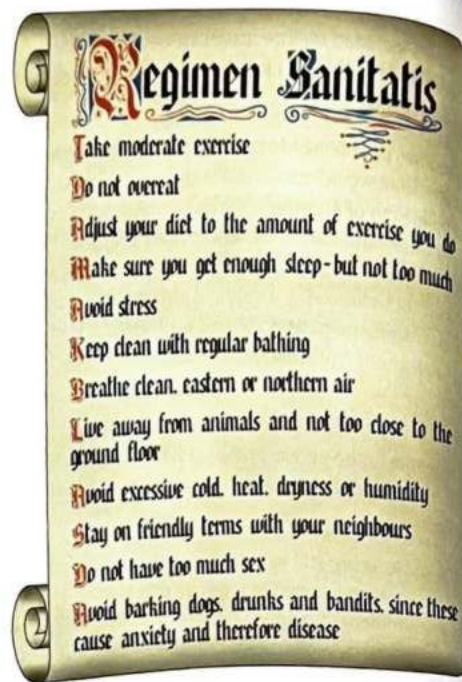


Figure 1.6 Advice given in the *regimen sanitatis*.

As well as keeping themselves clean, people also tried to make sure their homes smelled sweet and fresh, too. Floors were swept regularly and rushes were laid down to soak up any mess. Sometimes sweet smelling herbs, like lavender, were spread with the rushes to keep the air free of miasmata.

Diet

Since the humours were thought to be produced by digestion, what and when you ate were both considered very important in preventing an imbalance. Eating too much was strongly discouraged.

According to the chroniclers of the time, several medieval kings died as a result of eating too much, or having too rich a diet. Edward I died of dysentery*, for example.

Key term

Dysentery*

Very severe diarrhoea.

Fear of digestive problems leading to death was so great that many people purged themselves, either by vomiting or using laxatives, as a way of preventing disease as well as treating it. Hippocrates recommended using an emetic once a fortnight in the winter, and to use enemas in the summer.

Purifying the air

Medieval people attempted to keep the air free from miasmata by purifying it. They did this by spreading sweet herbs, such as lavender. Sometimes this might be carried as a bunch of flowers (posy), or placed inside a decorative piece of jewellery called a pomander (a large locket, which would be worn around the waist).

Local authorities, usually under the direction of magistrates or noblemen, also tried to tackle miasmata outside of the home, putting into place measures to keep towns clean. For example, they tried to make sure there were no rotting animals left lying around and pulled down or cleaned particularly smelly public toilets.

Medieval 'medics'

Most people in the Middle Ages would have been treated at home by a female family member. Women did most of the treatment at home, caring for the sick and mixing remedies themselves. Women also acted as midwives: evidence from medieval sources suggests that only women attended births.

Asking for medical advice cost a lot of money. Since the treatments weren't guaranteed, most people were not willing to spend this money even if they had it. However, there were other treatment options for people willing to pay.

Physicians

During the Middle Ages, new universities were set up across Europe, including Oxford, Cambridge, Paris, Bologna, Montpellier and Padua. Medicine became more professional. A medical degree took between seven and ten years to complete, depending on the level and university.

Medieval doctors were known as physicians – the word 'doctor' did not become common until the 17th century.

The main role of a physician was to diagnose illness and recommend a course of treatment. They rarely got involved in treating the patients themselves – this was left up to less educated midwives, apothecaries or barber surgeons.

If you were lucky enough to be able to afford a physician, he might attend in person to examine you, but this wasn't really considered necessary. The consultation would follow three stages.

- 1 The physician would look at a sample of the patient's urine, faeces and blood, all of which would be collected and sent to him.
- 2 He would also consult the astrological charts under which the patient was born and at the time they fell sick.
- 3 Based on this, and the humoral tendencies of the patient (whether they were naturally sanguine, choleric, phlegmatic or melancholic), the physician would create a course of treatment.

It was then up to less trained and lower paid professionals to carry out the treatment. For most of this period, this was due to the fact that many physicians were clergymen, who were forbidden from carrying out procedures such as bleeding. From 1215 onwards, any operations likely to involve cutting the patient were also forbidden for clergymen.

During this time, new universities and centres for medical learning were set up in Europe without religious sponsorship. Foreign physicians who had no connection to the Church were able to both diagnose and treat their patients.

Physicians were very expensive, because there weren't many of them. This was mainly due to the training taking a long time. Royalty and the very wealthy often employed a physician full time. Others paid for them when they needed them.

Apothecaries

Apothecaries mainly mixed herbal remedies. They had a good knowledge of the healing power of herbs and plants thanks to studying herbal manuals such as *Materia Medica*. They usually had a good amount of knowledge from their own experience, or passed down from family members.

Apothecaries were not considered as skilled or knowledgeable as physicians. Physicians prescribed the medication and apothecaries were just there to mix the remedy. However, since doctors were expensive and apothecaries were comparatively cheap, lots of people would see an apothecary as an alternative to a doctor. This meant that doctors saw them as a threat to their livelihood.

Source C

A picture of an apothecary's shop taken from an Italian medical text published in the 14th century. The shelves are filled with large, ornate jars, which held theriac.



Apothecaries did not just create remedies, they also prescribed poison. This went against an idea fundamental to physicians – that they should do no harm. This rule went all the way back to Hippocrates and still exists today: all doctors must swear a Hippocratic Oath before practising medicine. Apothecaries were not bound by this rule, so they could not be trusted to do the best for their patient. Nor did they need to attend university to set up their business.

Finally, many apothecaries also dabbled in alchemy and the supernatural, providing amulets and charms for patients who wanted something extra to cure a disease. This was frowned upon by the Church and, since many physicians were also priests, this meant that the gap between physicians and apothecaries became even wider.

Surgeons

Barber surgeons were probably the least qualified medical professionals in England. Since good barbers had sharp knives and a steady hand, they regularly performed small surgeries as well, such as pulling teeth and bleeding patients.

Extend your knowledge

Barber surgeons

Until 1307, barber surgeons advertised their services by putting a bowl of blood in their shop windows; after that, they displayed the sign of a bandaged, bloodied arm, which later became the red and white striped barber pole still used today.

Some surgeons were highly trained: in Europe, some physicians were encouraged to study surgery alongside medicine, so many learned their skills at university. In fact, the quality of surgery was arguably higher than the quality of medical advice, because it was usually based on experience rather than knowledge learned from books. In medieval England, a skilled surgeon could set a broken limb, remove an arrow or even successfully remove cataracts from the eyes.

Activities

- 1 Draw a cartoon stick figure to represent each different medical professional. Add labels to explain what sort of treatments each person carried out.
- 2 Explain how each of the different professionals was trained to do their job.
- 3 Why do you think there was so much friction between the different types of medics in medieval England?

Caring for the sick: hospitals and the home

Hospitals

The number of hospitals in England was on the rise during the Middle Ages. By 1500, there were an estimated 1,100 hospitals, ranging in size from just a few beds to hundreds. Bury St Edmund's, for example, had at least six hospitals to cater for lepers, the infirm and the old. The city had a shrine famous for its healing powers and therefore attracted a lot of sick people. However, many hospitals did not actually treat the sick. Instead, they offered hospitality to travellers and pilgrims, which is how hospitals got their name.

About 30% of the hospitals in England were owned and run by the Church in the Middle Ages. These were run by the monks and nuns who lived in nearby monasteries.

The rest were funded by an **endowment**, where a wealthy person had left money in their will for the setting up of a hospital. Since charity was a foundation of religion and the Church taught that charitable donations could help to heal disease, it is not surprising that there were so many. The Church was in charge of running many of these hospitals, too.

Medieval hospitals that did treat the sick were not the same as the hospitals we have today.

Medieval hospitals were good places to rest and recover. The space would have been kept very clean and the bed linens and clothing of the patients changed regularly. It was a large part of the nuns' duties to do the washing and make sure everywhere was kept clean. This meant that, for people not suffering from terminal disease, hospitals were probably quite successful.

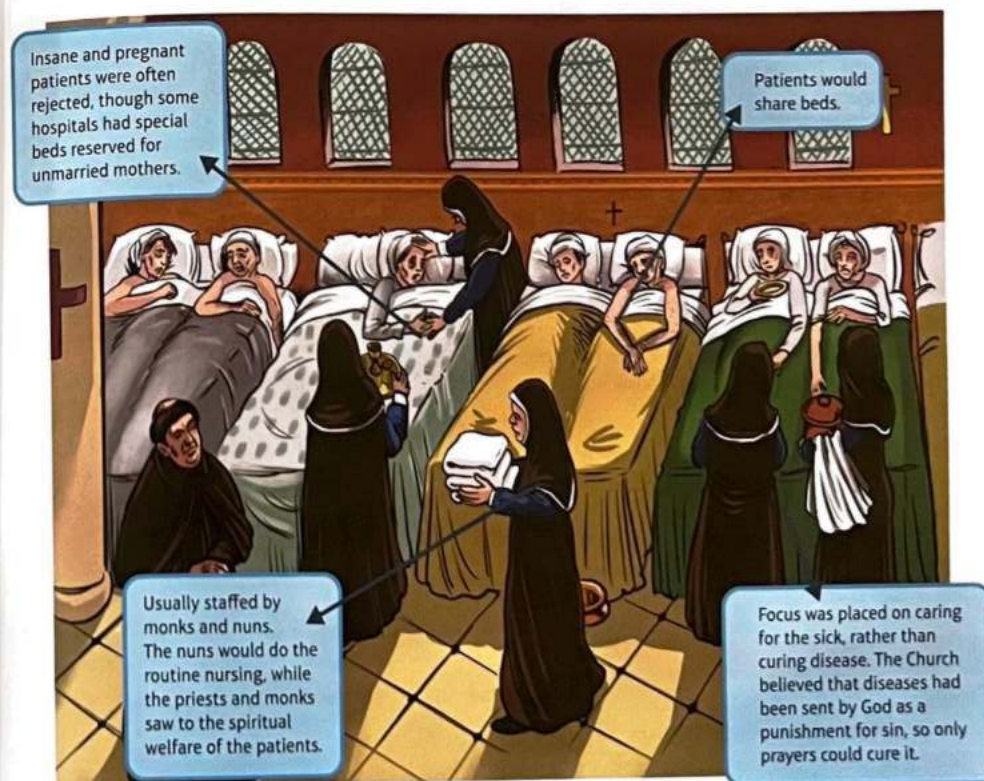


Figure 1.7 Common features of medieval hospitals.

Source D

A picture of a medieval hospital, from 1482. Some of the patients are sharing beds, which was normal at this time. The only patient allowed their own bed was a dying woman. Henry VII's famous hospital, the Savoy, opened in 1512. It was unique in offering all patients their own beds.



Naturally, this is what the Church wanted: a recovery was further proof of the existence of God and the importance of prayer.

Many European hospitals employed physicians and surgeons, but there is no evidence to suggest that English hospitals did the same. Since religious men were forbidden from cutting into the body, treatment was very limited.

Infectious or terminal patients were often rejected from hospitals, as prayer and penance* could do nothing for these people. However, patients who had a chance of recovery were able to see the altar and even participate in Church services from their beds, to help with the healing of their souls.

Key term

Penance*

A punishment inflicted on yourself to show that are sorry for your sins.

The home

Although many hospitals were established in medieval England, the vast majority of sick people were cared for at home. It was expected that women would care for their relatives and dependents when needed. This care would have involved making the patient comfortable, preparing restorative foods and mixing herbal remedies.

Women would also be responsible for the garden, in which they were expected to grow various plants known for their healing properties, such as marigolds and clover.

Some historical sources hint that women in the home were well-respected for their healing skills. Letters written in 1464 between Lady Margaret Paston and her husband Sir John, when he was sick in London and she was at home worrying about him, show that neither of them trusted doctors and both would have been happier if he had been at home receiving treatment from her (see Source E).

Source E

An extract from a letter sent from Margaret Paston to her husband, John Paston, in 1464. The Pastons were wealthy landowners living in Norfolk. At the time that this letter was written, John Paston was staying in London.

For God's sake beware of any medicine that you get from any physicians in London. I shall never trust them because of what happened to your father and my uncle, whose souls God forgive.

Women likely had many more healing skills than just mixing herbal remedies and keeping the patient clean, warm and well-fed. There is some suggestion that they carried out minor surgeries and bleedings – however, records are very patchy. This might be because it was taken for granted that women cared for the sick, so nobody bothered to record it when it happened.

Exam-style question, Section B

'Hospital treatment in England in the period from 1250 to 1500 was very rare'.

How far do you agree? Explain your answer.

You may use the following information in your answer:

- charity hospitals
- care in the home.

You must also use information of your own. 20 marks

Exam tip

This question gives an extra four marks for good spelling, grammar and punctuation, and the use of specialist terms. Take extra care over things like capital letters for key words.

Summary

- Religious treatments included prayer, fasting and pilgrimages.
- Supernatural treatments included saying spells or carrying amulets, although these were discouraged by the Church.
- There were a large number of treatments aimed at rebalancing the humours. This was normally done by eating a particular food, taking herbal remedies or by purging the body to remove bad humours, either by making the patient vomit or go to the toilet.
- Because there were no guaranteed treatments, medieval people were advised to avoid getting ill by living a healthy lifestyle and keeping clean.
- Physicians, apothecaries and barber surgeons all provided different treatments.
- Hospitals followed religious teachings. Patients were cared for and prayers were said, but they rarely received any medical treatment. However, most sick people were cared for in the home by a female family member.

Checkpoint

Strengthen

- S1 What were the three different types of blood-letting?
- S2 List the different ways people tried to prevent disease in medieval England.
- S3 List the different sources of help sick people had in medieval England.

Challenge

- C1 You need to show links between the treatments and remedies used in the Middle Ages, and the ideas people had about what caused disease. With this in mind, create a revision resource, matching what you learned about the causes of disease with what you have learned about treatments. This could take the form of a table, a poster or a 'pairs' game to play when you are revising.
- C2 Add to your revision resource about treatments, linking the different sources of treatment to different beliefs about disease.
- C3 Identify which different sources of treatment were available to different groups of people (for example: the rich, the poor, pregnant women, lepers).

If you do not feel confident answering any of these questions, discuss them with a partner or in a group.

1.3 Dealing with the Black Death, 1348–49

Learning outcomes

- Understand what the Black Death was and how it affected people in England during the years 1348–49.
- Understand the disputed causes, treatments and preventative measures used during the time of the Black Death.

In 1348, a new disease reached the shores of England. It had spread from the Far East along trade routes, arriving in Sicily in 1347, quickly spreading across the whole of Europe. The **Black Death**, as it eventually came to be known, was a new plague that was unfamiliar to the ordinary people of England, as well as English physicians. Within months, it had spread the length and breadth of England, killing thousands of people. It was absolutely devastating: it didn't matter if you were rich, poor, a city dweller or a country farmer – the plague did not discriminate. Those who caught it could expect to die within a matter of days.

Source A

This engraving from the 14th century shows somebody suffering from the Black Death.



The disease still occurs every so often in modern times but it is easily treated with antibiotics and patients usually make a full recovery, as long as it is caught in time. In the Middle Ages, treatments like this did not exist. People were completely unprepared, and they did not know how prevent and treat the 'scourge'.

Key term

Scourge*

A person or thing that causes great suffering.

Source B

In this extract from a report on the Black Death written in 1347, Italian chronicler Marchione di Coppo Stefani describes how helpless people felt in the face of the epidemic (outbreak).

Neither physicians nor medicines were effective. Whether because these illnesses were previously unknown or because physicians had not previously studied them, there seemed to be no cure. There was such fear that nobody seemed to know what to do.

The Black Death

The Black Death was an outbreak of the bubonic plague. The bacteria were carried in the digestive system of fleas who arrived in England on rats carried by merchant ships. It was probably spread by flea bites, although some recent evidence suggests that it was also spread in the air. The main symptom was buboes, which was swelling in the armpit or groin, filled with pus.

Once caught, it was unlikely that you would survive the disease. It usually killed its victims in three to five days. At its height in London, 200 people were being buried every day. Contemporary accounts estimate that a third of the population of England died. Where the plague spread, it was common for more than half of a population of a town or city to die.

1.3 Dealing with the Black Death, 1348–49



Figure 1.8 Symptoms of the Black Death.

The Black Death is the name given by Victorian historians to the particular outbreak of the plague in 1348. After this, the plague returned every 10–20 years, although it killed fewer people with each subsequent outbreak.

People applied the same knowledge they had about the causes of disease and illness to the plague. New treatments appeared and advice on how to avoid catching the disease spread quickly.

Causes of the Black Death

Religious and supernatural

Many believed that the Black Death was the result of God deserting mankind – that is, the Black Death was a punishment for the sin in the world.

In 1345, there was an unusual positioning of the planets Mars, Jupiter and Saturn, which astrologers interpreted as a sign that something wonderful or terrible was about to happen.

Natural causes

The main natural cause attributed to the Black Death was impure air. Breathing in this miasma caused

corruption to the body's humours. People believed that this impure air may have originated from poisonous fumes released by an earthquake or a volcano.

Common beliefs

For the everyday people living in the cities and countryside, the spread of the Black Death was terrifying and they would have heard many conflicting ideas about what caused it. In Europe, many people blamed the Jewish population, but the Jews had been expelled from England in the 13th century, so this was not the case in England.

Treatments of the Black Death

Supernatural

The main recommendation to treat the Black Death was to confess your sins, and ask God for forgiveness through prayer. However, there was an air of inevitability about the disease: once caught, people believed it was clearly God's will; if it was his will that the patient should die, there was no cure that would work.

Natural

To begin with, physicians tried bleeding and purging – the same things they would usually do to correct a humoral imbalance. Unfortunately, that didn't work and, in fact, seemed to make people die more quickly.

As well as this, physicians recommended strong-smelling herbs like aloe and myrrh, which were believed to have cleansing properties. They often prescribed theriac, as it was believed to work for lots of ailments. Lighting a fire and boiling vinegar could also drive off the bad air. Physicians or surgeons sometimes lanced the buboes – occasionally, people whose buboes burst survived.

Common beliefs

Everyday people were willing to try anything to survive the Black Death. They held strong Christian beliefs, and so would have gone to confession and prayed, as well as seeking traditional cures like bleeding. However, it quickly became clear that neither priests nor physicians were capable of curing the disease.

Activity ?

In groups of four, use the information above to create a television news report about the possible causes of the Black Death and what advice sufferers were given. One of you should play the part of a journalist, while the others take on the roles of priest, physician and an everyday person. Remember to give advice on both how to treat the disease and how to prevent it.

Apothecaries sold remedies and herbs were mixed in the home, based on old recipes, but they had uncertain and unpredictable results. Nobody came up with a cure that definitely worked in all cases.

The lack of medical knowledge about what **caused** the disease meant it was impossible to know how to **cure** it.

Preventing the Black Death

Supernatural means

The main advice given by priests was for people to:

- pray to God and fast
- go on a pilgrimage and make offerings to God
- show God how sorry you are by self-flagellation (whipping yourself). Large groups of flagellants wandered the streets of London, chanting and whipping themselves.

Natural means

Escaping the plague was the best advice for prevention. Guy de Chauliac, who was the physician to the Pope, advised people to: 'Go quickly, go far, and return slowly'. It was essential to escape the foul air to stay healthy.

If this kind of movement was impossible, people believed it was essential to carry a posy of flowers or fragrant herbs and hold it to your nose. This helped to avoid breathing in the miasma. Unlike the usual advice on preventing disease, people were advised to avoid bathing. It was believed that water would open the skin's pores to the corrupted air.

Common beliefs

One physician in Italy recommended doing joyful things, listening to cheerful music and avoiding anything sad as a protection against the disease. This is a clue as to just how desperate most physicians felt – they were willing to suggest anything to prevent their patients from catching the disease.

Much like the physicians, people did not know how to prevent the disease. However, they did stop visiting family members who had caught the plague – the common belief of the need to avoid those with the disease was so great that even their houses were avoided.

Government action

Local authorities attempted to take action to prevent the plague from spreading. New quarantine* laws were put in place to try to stop people from moving around too much. People new to an area had to stay away from everybody else for 40 days, to ensure they were not carrying the disease. Authorities also started to quarantine houses where the plague had broken out. They considered banning preaching and religious processions, to stop large crowds of people gathering.

However, since the local government did not have a great deal of power at this time, they could not fully enforce these laws: rich people, for example, moved around quite freely and the Church continued to run as normal.

Because of the belief in bad air causing disease, the local authorities also stopped cleaning the streets. They believed that the foul stench of the rubbish and rotting bodies would drive off the miasma causing the plague.

Interpretation 1

Writer Sean Martin shares his views, and those of other historians, on the impact of the Black Death, in his book, *A Short History of Disease* (2015).

One immediate effect of the pandemic [an infectious disease spread across a large region] was the invention of quarantine. Historian GM Trevelyan argued that the Black Death was at least as important as the industrial revolution, while David Herlihy argued that the Black Death was 'the great watershed', [an important turning point] without which there would have been no Renaissance, and with no Renaissance, no industrial revolution.

THINKING HISTORICALLY Cause and Consequence

The language of causation

Study these words and phrases. They are useful in describing the role of causes and how they are related to each other and to events.

motivated	precondition	prevented	determined the timing	deepened the crisis	led to
exacerbated	allowed	triggered	impeded	catalyst	developed
underlying	created the potential	influenced	enabled	accelerated	sparked

- 1 Create a table with the following column headings: 'word', 'meaning', 'timing'. Write out each word or phrase in a separate row of the 'word' column.
- 2 Discuss the meaning of each word or phrase with a partner. For each word or phrase, write a short definition in the 'meaning' column.
- 3 Is each word or phrase more likely to describe a short-term, medium-term or long-term cause? In the 'timing' column, write 'short', 'medium' or 'long'.
- 4 Look at the following incomplete sentences. Write three versions of each sentence, using different words and phrases from the table to complete them. You can add extra words to make the sentences work. For each sentence, decide which version is the best.
 - a The power of the Church ... the development of medical understanding.
 - b Following a regimen sanitatis ... the prevention of disease.
 - c The lack of medical understanding ... the spread of the Black Death.
- 5 You can also describe the importance of causes. Place the following words in order of importance. You should put words that suggest a cause is very important at the top, and words that suggest a cause is less important at the bottom of your list.

necessary	contributed to	added to	marginal	fundamental	influenced	supported	negligible
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Key term

Quarantine*

Separating the sick from the healthy to stop the spread of a disease. Those who are sick are not allowed to leave the quarantined area.

2.1 Ideas about the cause of disease and illness

Learning outcomes

- Understand the different ideas about the causes of disease that changed and stayed the same, c1500–c1700.

Ideas about disease and illness: change and continuity



Figure 2.1 Change and continuity surrounding the causes of disease and illness, c1500–c1700.

People who fell ill during the period 1500 to 1700 were likely to believe the same things about the cause of their illness as their medieval ancestors. Very little really changed in the practice of medicine during this period.

However, all across Europe, enormous shifts were taking place in other areas of daily life. Beautiful art was being created in new styles and with new techniques; beliefs were changing, with new forms of Christianity and a more secular* society developing; and understanding of the surrounding world was increasing with scientific discoveries.

Key word

Secular*

Not religious or in any way connected with spiritual beliefs.

Because of these changes, medical knowledge grew with the changing attitudes of ordinary people. The general population of Europe wanted better answers to the questions about what caused disease. Epidemics of the plague and other killer diseases, such as smallpox, the Great Pox (syphilis) and sweating sickness, could not be easily explained by the Theory of the Four Humours. They affected everybody in the same way and were not cured by traditional humoral treatments, like blood-letting and purging.

There was still a widespread belief in miasmata as a cause of humoral imbalance and disease. A miasma could be the product of rotten vegetables, decaying bodies of humans or animals, excrement or any swampy, smelly, dirty place. However, even this did not provide a satisfactory explanation for the spread of diseases when people took such care to avoid miasma.

Some people came up with new ideas about the causes of disease and illness. They included new ideas based on alchemy* and new discoveries about the body. Some of these new ideas are listed in the table below, together with the name of the scientist or doctor who discovered them.

Key word**Alchemy***

This was an early form of chemistry. Alchemists tried to turn one material into another: mostly, they were trying to discover a way of making gold.

New ideas and discoveries in the period c1500–c1700

New ideas about disease and illness	Influential individual
In the 16th century, the Theory of the Four Humours was rejected by some radical physicians. Disease was seen as something separate from the body, which needed to be attacked. New chemical treatments started to appear, influenced by the increasing popularity of alchemy.	Paracelsus, a Swiss scientist and medical professor.
In 1546, a new text called <i>On Contagion</i> theorised that disease was caused by seeds spread in the air.	Girolamo Fracastoro, an Italian physician.
In 1628, a new theory was published in Britain, which suggested that blood circulated around the body instead of being made in the liver, as taught by Galen.	William Harvey, an English scientist
A better understanding of the digestive system developed. This meant that people gradually stopped believing disease was caused by eating the wrong things. Urine was no longer seen as an accurate way of diagnosing illness.	Jan Baptiste van Helmont, a Flemish physician.
New microscopes were being developed, which allowed for much clearer magnification. A new book, <i>Micrographia</i> , published in 1665, showed many detailed images, including a close-up drawing of a flea, copied from a magnified image.	Robert Hooke, an English scientist and head of experiments at the Royal Society.
In 1676, the medical textbook <i>Observationes Medicae</i> was published. This theorised that illness was caused by external factors, rather than the four humours.	Thomas Sydenham, an English physician. (see page 44)
By 1683, more powerful microscopes had been developed to allow for the observation of tiny 'animalcules' or little animals in plaque scraped from between the teeth. The images were not very clear, but they were visible. This was the first recorded observation of bacteria.	Antony van Leeuwenhoek, a Dutch scientist (see page 46).

So, due to all these new discoveries and ideas, by c1700:

- the Theory of the Four Humours had been discredited – however, it was still being followed by the general population of Britain
- other ideas about causes of disease had been discovered (for example, 'animalcules').

Even though some of these ideas were very close to what we now know as the truth, they had very little impact at the time. This was due to several reasons.

A better understanding of human anatomy (the makeup of the body) was developing all the time. However, there was no point studying correct drawings of the internal organs when it was impossible to diagnose or treat internal problems on a living patient.

Also, the lack of quality medical instruments, such as microscopes, prevented any rapid change in people's beliefs about the causes of disease. The new theories might have been very convincing, but without scientific proof they were just that – theories.

Because the general public believed in the Theory of the Four Humours, most physicians stuck to the old methods. They were in the business of healing the sick, not coming up with better methods of doing it. Even those who did look for new ideas still needed to work, and patients did not want to pay physicians to experiment on them.

Changing ideas

The key point to note here is that, while the **practice** of medicine did not change much at this time, **ideas** were starting to change. Scientists like Galileo and Copernicus were challenging the authority of the Church in other areas of scientific understanding. This encouraged medical scientists to start looking beyond the works of Galen and Hippocrates. By the end of the 17th century, doctors and scientists had lots of new ideas about the causes of illness and disease – it just wasn't applied to everyday medical practice.

Below is a chart that shows how ideas about the causes of disease and illness changed during the Medical Renaissance.

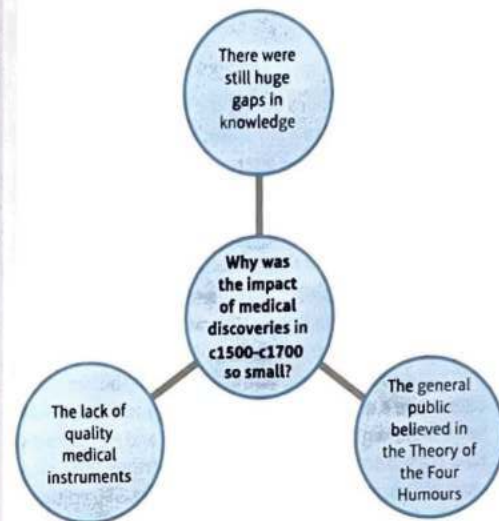


Figure 2.2 Why was the impact of scientific discoveries up to 1700 so small?

Ideas that changed a lot	Ideas that changed a little	Ideas that stayed the same
The Theory of the Four Humours. Very few physicians believed this by the end of the 17th century, though it was still used when diagnosing disease, because patients understood it. The human body. There was a much better understanding of anatomy.	The use of medical books. Physicians carried out more observations of their patients. However, they still relied on texts for looking up symptoms. The influence of the Church. Most people now recognised that God did not send disease. However, in times of epidemics, such as during the Great Plague, religious causes were still considered.	Miasmata. The idea that disease was spread by bad smells and evil fumes was constant throughout this period – and even became more widespread during epidemics.
Diagnosis using urine. Physicians now understood that urine was not directly related to a person's health.	Supernatural. Although astrology was much less popular from 1500, in times of epidemics people still wore charms to ward off the disease.	

Activity

Draw two columns and label the first one 'Old ideas', and the second one 'New ideas'. Using the information in this chapter, and things you have already learned in Chapter 1, make brief notes on ideas about the causes of illness and disease in the years c1500–c1700 that were the same as c1250–c1500, and the ideas that were different.

Exam-style question, Section B

Explain **one** way in which ideas about the cause of disease and illness were similar in the 14th and 17th centuries.

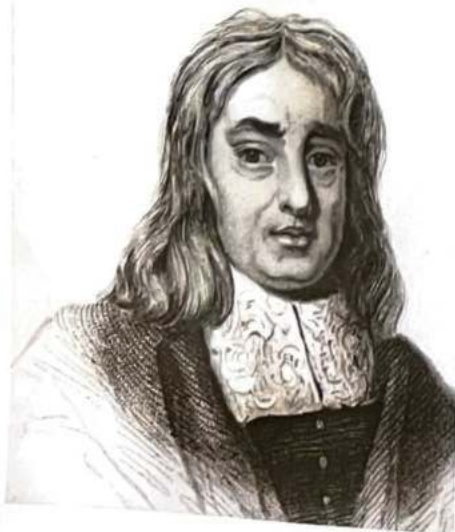
4 marks

Exam tip

This answer should not be very long, but it does need to have specific information for each time period. Try to make a general point that covers both time periods – for example, the idea of miasmata – and then give a specific example for each time period.

Source A

A drawing of Thomas Sydenham, by an unknown artist, after his death.



A scientific approach to diagnosis

One of the key changes during the Renaissance was the rise of **humanism**. Humanism was characterised by a love of learning, a new interest in classical scholars and the belief that human beings could make up their own minds when it came to discovering the truth of the world around them.

Humanism also represented a break with some of the old medieval traditions. Humanists rejected the Christian view that God was responsible for everything that happened, but they hadn't yet figured out an alternative explanation. People returned to the original texts of ancient scholars such as Galen and Hippocrates. New translations of the works of Hippocrates and Galen started to appear. During the 16th century, 590 editions of Galen's writings were published.

During the 17th century, more experimentation began to take place in the field of medicine. This, in part, was because the Church had less authority in everyday life. Proof that Galen had been wrong about the human anatomy was becoming more widespread, thanks to the work of Vesalius (see pages 52–55). New ideas were starting to gain more support, although it would be a long time before this had an impact on everyday medical treatment.

Thomas Sydenham

Thomas Sydenham was nicknamed 'the English Hippocrates'. He was a well-respected doctor in London in the 1660s and 1670s. The Theory of the Four Humours was still being used at this time, but it was starting to lose popularity. Sydenham's work was very important in moving medicine in Britain away from the classical ideas of Galen and Hippocrates, and into the new era.

Sydenham refused to rely on medical books when diagnosing a patient's illness. Instead, he made a point of closely observing the symptoms and treating the disease causing them. This was a change from the medieval method of treating each of the symptoms separately, instead of seeing them all as side effects of one cause.

Source B

Thomas Sydenham published his theories about disease and his observations of various epidemics in a book called *Observationes Medicae* in 1676. In this extract from the introduction, he explains that doctors should devote as much time to identifying different types of disease as botanists spend identifying different types of plant.

In the first place, it is necessary that all diseases be reduced to definite and certain species, and that, with the same care which we see exhibited by botanists [specialists in the scientific study of plants]; since it happens, at present, that many diseases... are... different in their natures, and require a different medical treatment. We should have known the cures of many diseases before this time if physicians, whilst with all due good-will they communicated their experiments and observations, had not mistaken one species for another.

One of Sydenham's most controversial ideas was that diseases were like plants and animals, in that they could be organised into different groups. According to the Theory of the Four Humours, a patient's disease was personal to them: it was caused by any number of individual factors, including the weather, diet and the patient's particular balance of humours, which differed from person to person. This meant that treatments also varied from person to person.

Sydenham did not believe this. He encouraged his students to observe their patients, note down their symptoms in detailed descriptions and then look for remedies to tackle the disease. He theorised that the nature of the patient had little to do with the disease.

This was a very modern idea and laid the foundations for a more scientific approach to medicine from the 18th century onwards. Sydenham was not able to isolate and identify the various microorganisms that caused the diseases that he was observing. However, he was able to identify that measles and scarlet fever were separate diseases.

Extend your knowledge

Thomas Sydenham's new treatments

Thomas Sydenham developed a number of new treatments to tackle the diseases he identified. Instead of using the popular 'sweating' method for treating smallpox, he prescribed airy bedrooms, light blankets and cold drinks. He also popularised the use of cinchona bark from Peru to treat malaria. Cinchona contains quinine which is still used to treat malaria today.

Activities

- 1 List the effects humanism had on medicine during the Medical Renaissance.
- 2 Draw a stick figure to represent Thomas Sydenham. Add thought bubbles to detail his ideas about medicine.
- 3 Investigating the causes of disease and isolating different types was very important in the fight against disease. Imagine you are a doctor in the 17th century. Write a proposal or letter addressing other doctors, explaining how you think Sydenham's ideas could be used to improve medical treatments.

Improved communications

One of the changes across Europe during the Medical Renaissance was the improvement in literacy: more people than ever before were able to read and write. This meant new ideas could spread further and more quickly.

The influence of the printing press

In around 1440, Johannes Gutenberg, a German goldsmith, created the world's first printing press*. It didn't take long for the popularity of his new invention to grow: by 1500, there were hundreds of presses in Europe.

Key term

Printing press*

A machine for printing text or pictures. It had movable letters so that many copies of the same text could be printed.

The new printing press enabled information to be spread accurately and quickly. Text no longer had to be copied by hand, meaning there were fewer inconsistencies in the same version of a text. It also meant that scientists could publish their work and share it across Europe much faster than when the work had to be copied by hand.

The printing press also took book copying out of the hands of the Church. This meant that a much wider variety of subjects were written about, whereas, before this time, most books were about religious topics. The Church was no longer able to prevent ideas they disapproved of being published. For example, physicians could now publish works criticising Galen.

The work of the Royal Society

The desire to explain the world in secular terms led to a big increase in the number of experiments being carried out. Scientists wanted to talk to each other about their new discoveries and share new ideas. This led to the founding of the **Royal Society**.

The Royal Society met for the first time at Gresham College in London, in 1660. Its aim was to promote and carry out experiments to further the understanding of science. They also heavily promoted the sharing of scientific knowledge and encouraged argument over new theories and ideas. In 1662, the society received its royal charter* from Charles II, who had a keen interest in science.

Key term

Royal charter*

A document from the monarch, granting a right or power to a particular person or group. A Royal Charter shows that the monarch is supportive of a particular project.

The support of the king gave the Royal Society credibility: if the king approved of and supported them, clearly they were doing something right. It also raised their profile. More people sent their work in to be published, or were willing to donate money supporting the scientific work of the Royal Society.

Extend your knowledge

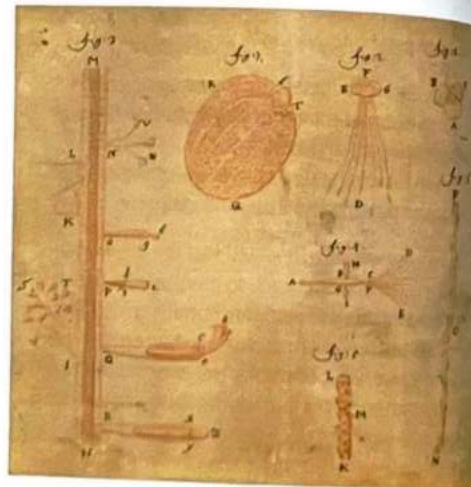
The Royal Society

The motto of the Royal Society was 'Nullius in verba'. This means 'Take nobody's word for it'. It shows that at the heart of the Royal Society was a desire to find new scientific discoveries through experimenting and evidence.

In 1665, the Society began publishing their scientific journal, called *Philosophical Transactions*. It was the world's first scientific journal, and it continues to be published today, celebrating its 350th anniversary in 2015.

Source C

This picture of 'animalcules' was drawn to go with a letter Antony van Leeuwenhoek had published in *Philosophical Transactions* in 1702. The letter was titled 'Concerning green weeds growing in water, and some animalcula growing about them'.



When it was first published, the journal consisted of letters, book reviews and summaries of experiments and observations carried out by European scientists. *Philosophical Transactions* provided a vitally important platform from which scientists could share their work, and therefore contributed a great deal to the spread of medical ideas. For example, they published several letters from Leeuwenhoek, in which he described his observations of 'animalcules'.

The Royal Society offered funding for translations of European scientific texts. It encouraged its members to write their reports in English instead of Latin, and in straightforward language, to make it accessible for everyone. It also requested that members provide a

copy of any of their submissions, which would be put into a reference library and made available for anybody to study.

The Royal Society made it possible for physicians and scientists to access and study each others' research. It was therefore very important in the development of new medical ideas.

When Leeuwenhoek's work (see Source C) was received by the Royal Society, they gave it to their microscope enthusiast, Robert Hooke. Hooke used his own microscope to confirm what Leeuwenhoek had seen. Because it was published in *Philosophical Transactions*, news of the new discovery spread quickly and widely.

THINKING HISTORICALLY Change and continuity (2a)

Distinguishing between change and continuity

Change is happening all the time. Every day, things change slightly in lots of different ways. For example, ten more people in the country might be unemployed today compared to yesterday, or mobile phone batteries might have become 1% more powerful. Most of these changes are small, and not important enough to be called 'change' by historians.

Study the following events that demonstrate changes in understanding the causes of disease.

- 1526** Paracelsus theorised that disease was caused by problems with chemicals inside the body
- 1546** Fracastoro wrote a text called *On Contagion*, suggesting that disease was caused by seeds that spread in the air
- 1628** Harvey proved that blood circulates around the body
- 1648** Van Helmont claimed that digestion happened because of stomach acid, rather than anything to do with the Four Humours
- 1665** Hooke developed a more powerful microscope and published a book of images from his observations
- 1676** Thomas Sydenham published *Observationes Medicae*, theorising that diseases were separate from the patient, rather than being caused by something the patient did
- 1683** Leeuwenhoek developed a microscope and observed 'animalcules', or 'little animals'

- 1 Using only three events in the list, write a paragraph explaining how understanding about the causes of disease changed during this period. Think carefully about which events you include.

A period of 'continuity' is a period of time when very little (if anything) changes.

- 2 Can you identify any periods of continuity in the timeline when looking at ideas about the causes of disease?
- 3 If you identified a period of continuity, write down:
 - a what it was that stayed the same over that period
 - b what, if anything, changed.
- 4 Write your own definitions of 'change' and 'continuity'.

Exam-style question, Section B

Explain why there were changes in the way ideas about the causes of disease and illness were communicated in the period c1500–c1700.

You may use the following in your answer:

- the printing press
- the Royal Society

You **must** also use information of your own. **12 marks**

Summary

- There was very little change in medical practice. Methods of diagnosing illness remained similar to medieval times.
- Lots of new ideas about the causes of disease and illness started to appear, but they were slow to have an impact on patients.
- By the end of the 17th century, most doctors no longer believed that humoral imbalance caused disease, although they still referred to the Theory of the Four Humours when diagnosing disease.
- There was a new fashion for careful observation and believing what you saw over what you read. Thomas Sydenham was important in promoting this method in England.
- The invention of the printing press made it easier for physicians to share their work.
- From the middle of the 17th century onwards, the Royal Society promoted scientific experimentation, funded research and provided a hub for sharing ideas and discoveries.

Checkpoint

Strengthen

- What new ideas about the causes of disease and illness started to appear at this time?
- What was humanism and how did it affect ideas about the causes of illness and disease?
- Who was Thomas Sydenham and what new ideas did he have?
- How did the Royal Society encourage the development of medical ideas?

Challenge

- How important was new technology in enabling the development of new ideas about the causes of disease and illness?
 - How far did individuals and institutions in the 17th century help lead to new discoveries?
- If you are not confident about any of these questions, form a group with other students, discuss the answers and then record your conclusions. Your teacher can give you some hints.

Exam tip

Make sure that you fully explain your points by using the **Point, Example, Explain, or PEE** method. State your argument, then add some examples from your own knowledge to back it up. Finish off by explaining how that evidence relates to the question.

2.2 Approaches to prevention and treatment

Learning outcomes

- Understand the different approaches to prevention and treatment that changed and stayed the same between c1500–c1700.

Treatment: change and continuity

Since belief in humoral imbalances persisted through to the end of the 17th century and beyond, the old treatments, which were aimed at rebalancing humours, continued as well. Bleeding, purging and sweating were all popular ways of removing too much of a particular humour.

A new popular theory in this period was the idea of **transference** – which meant that an illness or a disease could be transferred to something else. For example, people believed that if you rubbed an object on an ailment (such as a boil), the disease would transfer from you to the object. There was also a popular theory that you could get rid of warts by rubbing them with an onion – through transference, the warts would 'transfer' to the vegetable!

Extend your knowledge

Transference

Some patients with a fever used to sleep with a sheep in the bedroom, hoping that their fever would transfer to the animal.

Herbal remedies continued to be popular, although their use changed slightly. In c1500–c1700, often remedies were chosen because of their colour or shape. For example, yellow herbs, such as radish and saffron, were used to treat jaundice (which turns the skin yellow). Smallpox, which had a red rash as one of the symptoms, was treated with the 'red cure' – drinking red wine, eating red foods and wearing red clothes.

Since this was the age of exploration, new herbal remedies started to appear from other countries. New plants started to appear from the New World*.

Source A

In 1652, Nicholas Culpeper published his book *The English Physitian: Or An Astrologo-Physical Discourse of the Vulgar Herbs of This Nation*. Culpeper brought together all the information from the medieval *Materia Medica* and wrote it in English to make it available for everybody.

Vervain is hot and dry, opening obstructions, cleansing and healing. It helps the yellow jaundice, the dropsy and the gout; it kills and expels worms in the belly, and causes a good colour in the face and body, strengthens as well as corrects the diseases of the stomach, liver and spleen; helps the cough, wheezings and shortness of breath, and all the defects of the veins and bladder, expelling the gravel and stone.

Some physicians believed that within each country were herbal remedies which would cure the diseases that came from that country: the appearance of new remedies opened up a huge number of new possibilities for treatments and cures.

New remedies that started to appear included sarsaparilla from the New World, used to treat the Great Pox, and ipecacuanha from Brazil, later known just as ipecac, which was effective as a cure for dysentery*. Thomas Sydenham popularised the use of cinchona bark, from Peru, in treating malaria. This was an effective remedy as long as patients continued to take it for some time after it seemed as though the disease had gone. Physicians also tested other new arrivals like tea, coffee, nutmeg, cinnamon and even tobacco to see if they had any impact on diseases.

Key words

New World*

North and South America. Europeans were only aware of their existence from 1492.

Dysentery*

A stomach bug that causes severe diarrhoea.

Chemical cures

The growth of alchemy, which laid the foundations for the modern science of chemistry, had an impact on medical treatments. People began to look for chemical cures for diseases instead of relying on herbs and blood-letting. This new science was known as **iatrochemistry**, or **medical chemistry**, and it was extremely popular in the 17th century.

Inspired by Paracelsus, the scientist who experimented with chemical treatments, medical chemists experimented with metals as cures for common ailments. The *Pharmacopoeia Londinensis*, published by the College of Physicians in 1618 as a manual of remedies, included a chapter on salts, metals and minerals. Among its 2,140 remedies were 122 different chemical preparations, including mercury and antimony. In small doses, antimony promotes sweating, which cools the body down. This fitted in with the idea of purging the body of disease. Patients would leave wine in an antimony cup overnight and drink the contents in the morning. In larger doses, antimony was used to encourage vomiting – another type of purge. Although it is poisonous in its pure form, a compound of it, known as antimony potassium tartrate, was said to have cured Louis XIV of France of typhoid fever in 1657, and became wildly popular afterwards.

Source B

This woodcut, published in a French medical textbook by Hannibal Barlet in 1657, shows a patient vomiting after being given antimony as a purge.



Extend your knowledge

Syphilis

Syphilis, also known as the Great Pox, probably arrived in Europe with sailors who had travelled to the New World with Christopher Columbus. Starting in Naples, it spread across Europe very quickly. It was a much more violent strain of the disease than seen today and attacked the body in three phases.

Firstly, sores and spots would appear on a person's genitals and around their mouth. Secondly – often many months later – the patient would suffer from tiredness, headaches, a fever and a steady ache from their bones. Finally, one third of sufferers would develop tumours and abscesses, which would destroy the body from the inside. Somebody in this phase could be blinded, paralysed and possibly go insane.

Doctors were powerless to treat syphilis. A large number of different remedies were tried, including guaiac wood from the West Indies, mercury, and sweating. However, nothing worked.

Prevention: change and continuity

Preventing disease was still considered to be the best way to avoid dying from it: since treatments had not moved on from medieval times, there was still no certainty that a person would recover. The only sure way to avoid dying from a disease, therefore, was not catching one at all.

People believed you could avoid disease by practising moderation in all things, as well as avoiding draughts, exhaustion, rich and fatty foods, too much strong alcohol and being too lazy. Condition at birth was also important – being born small or weak might be used to explain death from an illness in adulthood. This idea of a person's 'constitution' was related to the medieval idea of a person's humours and personality being influenced by the season in which they were born.

Cleanliness was also still important – both the home and the body needed to be kept clean and free from bad smells. However, bathing had become a lot less fashionable in England since the arrival of syphilis.

Syphilis had spread so quickly among people who regularly visited the stewes, or bathhouses, in London that Henry VIII had been forced to close them down in the early 16th century. The spread of syphilis at these places was probably, in part, due to the fact that many bathhouses were also brothels. However, the link between syphilis and bathhouses was not easily forgotten, and it was thought that bathing led to people catching the disease. People in the 16th and 17th centuries were far more likely to keep themselves clean by rubbing themselves down with linen and changing their clothes regularly than by going to public baths.

People continued to try to avoid catching diseases by practising **regimen sanitatis**. However, by the end of the 17th century, avoidance methods were as much about changing your surroundings (moving away from an area with a disease) as they were about looking after yourself. The idea that certain weather conditions, or the surrounding atmosphere, spread disease was becoming more popular. New instruments like **barometers** and **thermometers** were used to measure and record weather conditions over a long period of time, to see if there was a link between the weather and outbreaks of disease.

More steps were also taken to remove miasma from the air. Homeowners in English towns were fined for not cleaning the street outside their house. Projects were set up to drain swamps and bogs. Removing sewage and picking up rubbish from the streets was a punishment given to minor criminals.

Preventing disease: things that were the same (continuity)

People still believed that there were many factors that could prevent disease, including superstitions and prayer...

Cleanliness was still very important...

People continued to practise **regimen sanitatis**...

Miasma was still believed in...

Activities

- 1 There are lots of unusual new words in this chapter, like iatrochemistry. Create a glossary for at least five of them. Write the words and their definition. You could also do this on flash cards.
- 2 Make a list of all the new treatments described in this chapter that appeared between 1500 and 1700. How many of them are newer versions of old treatments and how many of them are completely new?
- 3 What influences encouraged people to try new treatments? Work through the text above, identifying different things that changed medical treatment and matching them to your list from the first activity. For example, you might label sarsaparilla and ipecacuanha 'Influenced by exploration of the New World'.

Exam-style question, Section B

Explain **one** way in which ideas about the treatment of disease were different in the 17th century from ideas in the 13th century. **4 marks**

Exam tip

The difference between half marks and full marks on this question is how precise your knowledge is. Make sure you give a fact that relates to each time period.

Preventing disease: things that were different (change)

...but people also started to believe that other things could help avoid disease, such as practising moderation and your condition at birth.

...but bathing had become a lot less fashionable in England since the arrival of syphilis. People now kept clean by changing their clothes more often.

...but, by the end of the 17th century, people also began to think that disease was also related to other factors (for example, the weather).

...but more steps were now taken to remove miasma from the air (for example, removing sewage and picking up rubbish from the streets).

Medical care: change and continuity

The same range of professionals offered treatment from c1500–c1700 as in the period c1250–c1500: trained physicians, apothecaries and surgeons. However, there were some changes.

Apothecaries and surgeons

Apothecaries continued to mix remedies and surgeons continued to carry out simple operations during the Medical Renaissance. In the period c1250–c1500, apothecaries were organised into **guild systems**. This meant that men would carry out an apprenticeship, and then spend several years practising as a journeyman* under the supervision of a master, before becoming a master surgeon or apothecary himself.

Key term

Journeyman*

An experienced member of a profession who was not yet experienced enough to have his own business. Journeymen usually worked for a master until they had enough expertise to start their own business.

Education for both types of medical professional increased considerably between 1500 and 1700. With wars being fought with new technology, new wounds on the battlefield meant that more surgery was necessary, while the introduction of iatrochemistry introduced new ingredients into the stores of apothecaries. Both surgeons and apothecaries had to possess licences to be able to practice their trade. Surgeons and apothecaries continued to provide services to those unable to afford physicians.

Physicians

Physicians continued to be trained at universities in the period c1500–c1700. Training courses changed very little during this period: there were some new ideas emerging, but, as with diagnosis and treatment, they were slow to take effect.

Although new subjects were introduced into the medical curriculum, like iatrochemistry and anatomy, most learning was still from books and not from practical

experience. Lectures were dictated in Latin. However, new ideas about human anatomy and iatrochemistry started to be shared, doctors were inspired to challenge the old teachings and investigate for themselves. This was particularly the case in the 17th century, when the Hippocratic focus on observation became more prominent.

However, there was still very little practical, hands-on training. Dissection, which had once been banned by the Church, was legalised due to the decline in the power of the Church, but it was still very difficult to get a supply of fresh corpses to dissect. Very few universities had an anatomy theatre, because most of them didn't think it was necessary to train a physician in anatomy – after all, he would only need it if he was supervising a local surgeon.

Luckily, trainee doctors had much better access to medical textbooks and there were a wider variety of these books than ever before. The newly-invented printing press made books easier to find and a lot cheaper. Protestantism rejected highly-decorated churches, so many artists found themselves with hours to spare and in need of work. This meant that they were available to create detailed drawings for these new medical textbooks. For medical students who couldn't afford a whole book, individual copies of pictures were available. These were known as **fugitive sheets**.

Andreas Vesalius

The most famous anatomist of this period was **Andreas Vesalius**. He studied medicine in Paris in 1533. Paris was a centre for the new humanist ideas about medicine. From there he went to Padua, which had a very famous university, where he became a lecturer in surgery. Vesalius had a deep interest in the human body and was keen to share his discoveries with the rest of the world.

His first publication in 1537, *Six Anatomical Tables*, showed the different parts of the human body, labelled in Latin, Greek, Hebrew and Arabic. Three of the six drawings showed a human skeleton, which Vesalius himself had assembled. He used the publication in his lectures: they were very popular with his students and

Source C

A fugitive sheet printed in 1566 as an addition to an anatomical book by Valverde, a Spanish doctor.

Fugitive sheets were prints of anatomical drawings that were made for medical students. They usually featured layers of paper that could be lifted to show the body during various stages of dissection.



Six years later, in 1543, Vesalius published the book for which he is most famous: *De Humani Corporis Fabrica*, or *On the Fabric of the Human Body*. He had been able to carry out a large number of dissections, thanks to a local magistrate who allowed him to use the bodies of executed criminals.

Vesalius noted that Galen had made some errors in his original theory on the human body. He put this down to the fact that Galen dissected animals instead of people. In all, Vesalius found around 300 mistakes in Galen's original work on anatomy. These included:

- the human lower jaw was in one part, not two
- the vena cava (the main vein leading out of the heart) did not lead to the liver
- men did not have one fewer pair of ribs than women
- the human liver did not have five separate lobes
- the human breastbone was in three parts, not seven.

As well as correcting these mistakes, Vesalius encouraged other doctors to base their work on dissection rather than believing old books. He wrote that it was vital that anatomy professors carry out

Source D

This illustration is from Vesalius's book, *On the Fabric of the Human Body*. It is titled 'Fourth Muscle Dissection'. There are 14 muscle dissection pictures in total: eight drawn from the front and six drawn from the back. In each one, Vesalius has stripped back another layer of muscle from the human body to reveal what is underneath. The final picture in each sequence shows the body supported by walls or blocks, as the substance of the limbs is cut away.



dissections for themselves, and claimed that this was really important if further advances in medical knowledge were to be made. Because of this, Vesalius laid the foundation for others to investigate the anatomy of the human body in more detail.

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Vesalius's book was richly illustrated with incredibly detailed drawings of the human body in various stages of dissection. By including so many pictures, Vesalius hoped to present the ideal version of the human body, to which other dissected corpses could be compared.

It was not difficult to find artists willing to do this. Renaissance artists were keen to study anatomy so they could paint the human form more accurately and secure more work.

Made the study of anatomy not only acceptable, but fashionable. Anatomy became a central part of the study of medicine. Doctors and medical professors now carried out dissections themselves, rather than a surgeon.

His work was heavily copied, and even plagiarised. Versions of the drawings from the *Fabrica* appeared in other medical texts and as fugitive sheets.

He caused a huge controversy. A lot of traditional physicians were angry that he had criticised Galen. They said that the differences Vesalius had found were down to the fact that there had been changes in the human body since the time of Galen.



Vesalius

Inspired other anatomists, some of whom went on to correct his own mistakes in *Fabrica*.

He was a trail blazer. After he died, Fabricius discovered valves in human veins. Fabricius shared this work with his students at Padua – one of whom was William Harvey, who went on to discover the circulation of the blood.

Figure 2.3 The impact of Vesalius.

Activities

- Which individuals impacted changes in medical care between 1500 and 1700?
- Draw a table to show changes and continuities in medical training during the 'Medical Renaissance'. How much do you think medical training had changed by 1700 – a little bit, some or a lot? Explain why you think this.
- Vesalius is a key individual in the development of anatomy. Can you explain why he had an impact?

Exam-style question, Section B

'Individuals had the biggest impact on medical training in the 16th and 17th centuries'.

How far do you agree? Explain your answer.

You may use the following information in your answer:

- Vesalius
- the printing press.

You **must** also use some information of your own.

16 marks

Extend your knowledge

Andreas Vesalius

As a humanist, Vesalius did not reject the ideas of Galen outright. There is evidence to suggest that he felt he was just completing a body of work begun by the ancient scholars. The title page to his book *Fabrica* shows Vesalius dissecting a woman while Galen, Hippocrates and Aristotle look on (see Source E). Vesalius hoped that, by correcting the errors in Galen's work on anatomy, more people would begin to use the ancient texts.

Exam tip

This question asks you to consider the impact of one factor on medical training – individuals. Your counter-argument should focus on the impact of other factors. The bullet point is giving you a hint here by mentioning the printing press, a technological impact.

Source E

This is the first page of *Fabrica*. In it, Vesalius is dissecting a body whilst surrounded by a large crowd of people, including Galen. When his book was published, Vesalius dedicated it to the Holy Roman Emperor, Charles V, and presented him with a hand-coloured copy. By doing this he hoped that his book would be more respected by the wider medical community. Charles V must have liked what he saw, because he later offered Vesalius the job of personal physician!



2.2 Approaches to prevention and treatment

prayer, this had begun to change. Patient records suggest that many people went to hospital with wounds and curable diseases such as fevers and skin conditions. They didn't spend very long in the hospital before being discharged: this suggests that they got better.

A patient in a 16th-century hospital could expect:

- a good diet – the restorative effects of food were still important, as many people didn't have access to a lot of food that was good for them
- a visit from a physician – hospitals had contracts with doctors, who would visit the patients sometimes as often as twice a day, to observe the symptoms and prescribe treatments
- medication – many hospitals had their own pharmacies and an apothecary to mix the medicines.

However, the dissolution of the monasteries* in England in 1536 dramatically changed the availability of hospital care in England.

Key term

The dissolution of the monasteries*

Henry VIII split from the Catholic Church in 1533 and created the Church of England. In 1536, he disbanded certain religious institutions, such as monasteries and convents, and confiscated their land.

Since the vast majority of hospitals were connected to the Church, very few were able to stay open after the dissolution of the monasteries. As hospitals were attached to abbeys, monasteries and convents, it was the nuns and monks who administered the medical care. With the convents and monasteries gone, the hospitals also went. Saint Bartholomew's hospital in London, which was founded in 1123, only survived because Henry VIII re-founded it himself in 1546.

Some smaller hospitals opened to fill the gap left by the dissolution of the monasteries, funded by charities, but there was a big change in the amount of medical treatment provided by hospitals.

Many hospitals reopened without their religious sponsors. However, it took a long time for the amount of hospitals to return to what it had been before the dissolution of the monasteries.

Caring for the sick: change and continuity

Hospitals

Some changes had begun to take place in English hospitals by the early 16th century. Whereas before, travellers, pilgrims, the elderly, and a few sick people would have attended hospitals for food, shelter and

Pest houses

One change in hospital care in this period was the appearance of hospitals that specialised in one particular disease. Versions of these existed in the Middle Ages, when there were lazaret houses for people suffering from leprosy. There was a growing understanding that disease could be transmitted from person to person (even though people didn't understand how or why this happened). This meant that new types of hospital began to appear that catered only for people suffering from plague or pox. These were known as **pest houses**, **plague houses** or **poxhouses**.

These new types of hospital provided a much-needed service. Traditional hospitals would not admit patients who were contagious, but people suffering from serious, contagious diseases had to go somewhere or risk infecting their families.

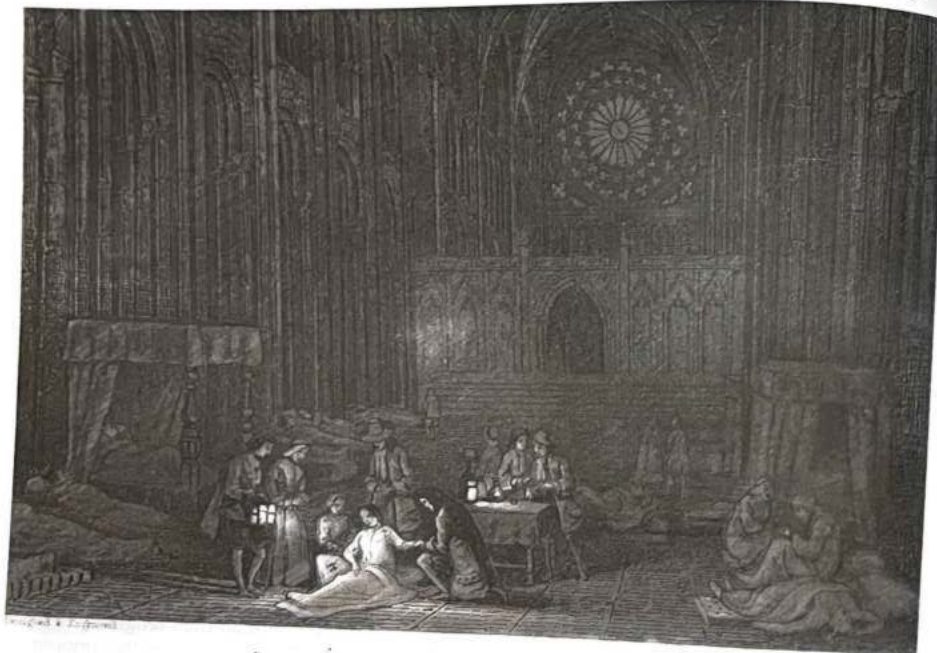


Figure 2.4 This 19th-century English engraving of St Paul's Cathedral shows it being used as a pest house during the Great Plague of London. St Paul's became a pest house during many outbreaks of disease in the city of London.

Community care

In spite of changes to hospitals, most sick people continued to be cared for at home. Local communities were very close-knit, which meant that there were a lot of people around to give advice and even mix remedies.

Women continued to play an important role in the care of the sick. This included rich and well-born ladies like Lady Grace Mildmay (1552–1620), who kept detailed notes of the healing and treatment she carried out. It also included poor women working in big cities to support their families. We don't know a great deal about these women, but we know that a lot of them were prosecuted by the London College of Physicians for practising medicine without a licence. They usually sold simple herbal remedies to purge the body or cure a particular ailment. Records suggest that they were very popular, probably because they were cheaper than going to a licensed physician or apothecary.

2.2 Approaches to prevention and treatment

Interpretation 1

In this extract from an article in the *Bulletin of the History of Medicine* (2008), Deborah E. Harkness describes the attitudes of male medical professionals towards female healers in the 16th century. She suggests that we can see they were popular because male physicians at the time were very critical of them.

The view of women's medical work in London that emerges from urban records... stands in stark contrast to the view conveyed... at the same time by male medical practitioners – namely, the records of proceedings at the College of Physicians, the reports of the Barber-Surgeons' Company, and various printed works. In 1566, for example, physician John Securis surveyed the medical practitioners available to eager Elizabethan consumers – and he did not like what he saw... "presumptuous" women offended his sense of proper medical order. Securis was not alone in his opinion that women and medical work were a problematic combination. A year earlier, John Hall, an eminent Maidstone surgeon who moved to London in the 1560s, described in print how the "true minister" of surgery and the hard-working physician had to compete in the streets with the "smiths, cutlers, carters, cobblers, coopers, curriers of leather, and a great rabble of women" for the attention of potential patients.

Summary

- Methods of treatment mainly stayed the same between 1500 and 1700. Bleeding, purging and other humoral treatments were still popular. Herbal remedies were very common.
- New herbal remedies appeared. Exploration to places previously unknown, like the New World, meant that new plants were available.
- There was a new focus on chemical cures. This was known as iatrochemistry. This reflected a new interest in minerals and chemicals in society.
- People still believed in the importance of cleanliness and tried to avoid miasmata.
- Apothecaries and surgeons continued to treat the sick and received more formal training.
- By 1700, physicians studied anatomy and botany alongside their traditional medical courses.
- Vesalius, an anatomy professor, published an anatomy textbook called *On the Fabric of the Human Body*. It corrected many of Galen's mistakes and encouraged physicians to carry out their own dissections.
- Hospitals had more focus on medical treatment by the end of this period, but there were fewer of them in England because of the dissolution of the monasteries. However, the vast majority of sick people continued to be treated at home by women.

Checkpoint

Strengthen

- 1 Describe the changes in the way that physicians were trained between c1250–c1500 and c1500–c1700.
- 2 List five of the mistakes Vesalius found in the works of Galen.
- 3 Describe the different places that sick people could seek medical treatment between 1500 and 1700.

Challenge

- 1 Attitudes in society were a big factor in both driving and limiting the amount of change in treatment during this period. Select information to show three ways that people's beliefs and suspicions encouraged medical development, and three ways that they held them back.
- 2 Explain the differences between hospital and community care in the period c1250–c1500 and in the period c1500–c1700.

How confident do you feel about your answers to these questions? Share your answers with a partner and see if you can improve them together.

William Harvey was born in 1578. He studied medicine at Cambridge, and then at the famous medical school in Padua. In 1615, he became a lecturer of anatomy at the College of Physicians, and by 1618 he was one of the royal doctors for James I.

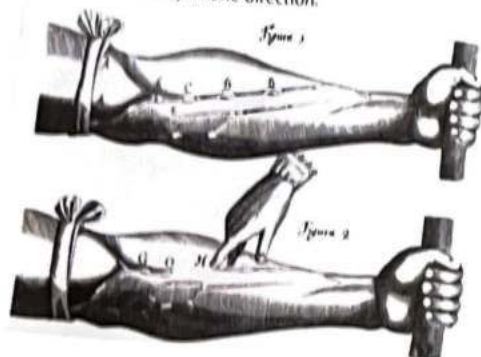
Harvey had a keen interest in dissection and observing the human body, in order to improve his knowledge of human anatomy. Carrying out public dissections was part of his job. He taught his students that it was important to observe the body and believe what they saw, rather than believing what had been written in classical texts. This idea was also followed by Thomas Sydenham, who would work on this theory later that century.

Harvey's research

Harvey was particularly interested in blood. During his studies at Padua, his professors had taught him Vesalius's theory that the veins of the body contained valves, which was proof that the blood in those veins flowed towards the heart. Using dissected bodies, Harvey saw the evidence to prove this theory. When he tried to pump liquids through the veins the other way, it did not work. This proved that the blood only flowed towards the heart, contradicting what Galen had taught about the blood (see page 17).

Source A

This source shows an illustration from Harvey's book, *An Anatomical Account of the Motion of the Heart and Blood in Animals*. In it, he is showing an experiment that proved blood flowed only in one direction.



Harvey looked in more detail at the old Galenic theory. It stated that blood was made in the liver as a product of what a person ate, then flowed through the body to provide energy and was burned up. Harvey worked out that if Galen's theory was correct, the liver would have to make 1,800 litres of blood a day for a person to survive.

Harvey was influenced by new inventions that had appeared during the Renaissance. Fire engines now used mechanical pumps to provide water to put out fires: perhaps the human body worked in the same way.

Discovering the circulation of the blood

Harvey's research involved dissecting human corpses and cutting open cold-blooded animals, which had a much slower heartbeat, to observe the movement of their blood while they were still alive. Through his research, he famously proved that arteries and veins were linked together into one system. This was done by tying a tight cord around somebody's arm and cutting off the blood flow in the artery leading into the arm. Because the artery in the arm is deeper than the veins, loosening the cord a little bit allowed blood to flow into the arm but stopped it from flowing out and the veins swelled with blood. Harvey's theory was that blood must pass from arteries to veins through tiny passages that were invisible to the naked eye. Today we know about these blood vessels – they are called capillaries.

Galen had suggested that blood flowed from one side of the heart to the other through invisible pores in the walls of the ventricles. He also said that veins carried both blood and *pneuma*, which was picked up in the lungs, while arteries carried just blood. Harvey's theory criticised both of these ideas. He showed that the veins carried only blood. He proved that the heart acted as a pump just as the new mechanical fire pumps did.

Key word

Pneuma*

Means 'breath of life'. Galen thought it was both the air that you breathe and your life force, or soul.

Factors enabling Harvey's discoveries

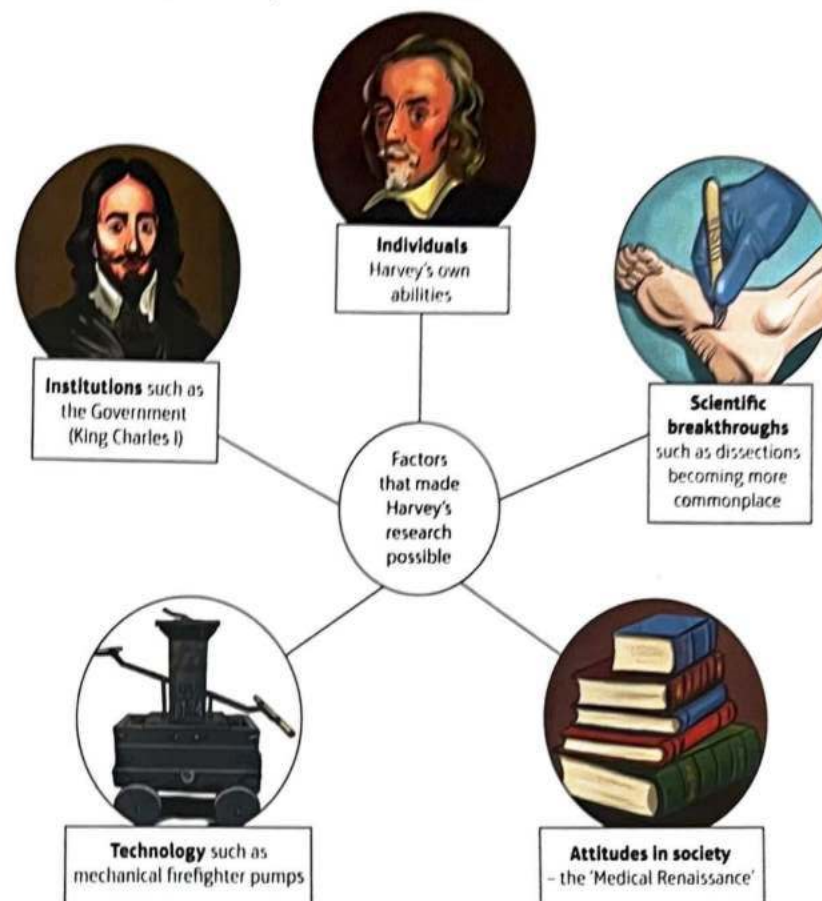


Figure 2.5 Five factors enabling Harvey's discoveries.

Individuals and institutions

Individuals such as Vesalius had previously proved parts of the work of Galen wrong, which made it easier for other scientists and physicians to do the same.

Harvey was employed by Charles I as his personal physician. This gave him credibility, in the same way the royal charter gave credibility to the Royal Society (see page 46). More people heard of Harvey's theory about the circulation of blood.

The decline of the power of the Church also enabled Harvey to be critical of Galen's teachings.

Science and technology

Newly popular technologies, like the pump used when fighting fires, inspired Harvey to look again at how the heart worked.

Attitudes in society

There was more interest in science and in solving some of the puzzles of the human body. People had begun to search for rational explanations for things.

The impact of Harvey

Harvey's arguments relied on careful observations of human and animal anatomy. Many people consider his book, *An Anatomical Account of the Motion of the Heart and Blood in Animals*, to be the beginning of modern physiology. However, Harvey did not consider himself to be a 'modern' scientist. He didn't believe many of Galen's theories, but he did follow the teachings of another classical thinker: Aristotle. Like Aristotle, he believed that the body was designed by a higher power, and thought that the soul was responsible for how the body worked.

The most immediate impact that Harvey's theory had was to encourage other scientists to experiment on actual bodies. For example, Harvey had proved that the liver did not digest food to create blood. If the liver didn't make blood, what did it do? Harvey had proved that blood circulated, instead of being absorbed to provide nourishment for the body – how, then, was the body nourished?

However, understanding the circulation of the blood had little practical use in medical treatment. This meant that the impact of Harvey's discoveries on treatment during the 17th century was quite limited. He may have

Summary

- William Harvey discovered that blood circulated around the body, instead of being made in the liver and absorbed into the body, as previously theorised by Galen.
- Harvey also proved that the heart acted as a pump, propelling blood around the body.
- Harvey's work had little impact at first, because it couldn't be used to improve practical medical treatments.
- Harvey inspired other scientists to carry out further experiments, building on his discoveries about blood and circulation.

Checkpoint

Strengthen

- S1 Name three discoveries that Harvey made.
- S2 Describe the methods Harvey used to investigate his theories.
- S3 What was Harvey's biggest impact in the short term?

Challenge

- C1 Think back to the factors that affect change and continuity in medicine. How many of them can you link to the history of Harvey?
 - C2 Who had the biggest impact on ideas about the human body – Vesalius or Harvey? Select information from this chapter to support your argument.
- To help structure your answer to C2, you might find it useful to create a 'For' and 'Against' list for both individuals.

paved the way for a modern understanding of anatomy and how the human body functions, but a lot of people at the time ignored him. Some even openly criticised him. Nobody liked to be told that they had been doing their job incorrectly. They also reasoned that nobody recovered from disease by simply knowing that new blood flowed to the heart. To many, it had no practical application. English medical textbooks continued to use Galen's account until 1651; Harvey's ideas only began to appear in universities from 1673.

Activities

- 1 Draw an outline of a person. Label it with the different discoveries that Harvey made about the human body and blood.
- 2 Like Vesalius, Harvey is a key individual who had an impact on medicine. Create a fact file about him to use in revision. You should include: details of his background, his experiments and his impact on medicine.
- 3 Write a short paragraph to explain why Harvey's work did not have much impact when it was first published.

2.4 Dealing with the Great Plague in London, 1665

Learning outcomes

- Understand what the Great Plague was and how it affected people in England during the year 1665.
- Understand the disputed causes, treatments and preventative measures used on those suffering, or at risk, from the Great Plague, and how these differed from the Black Death.

In 1665, a serious outbreak of the plague swept across the whole of England. Lasting from June until November, the rate of infection of the Great Plague peaked in September, when 7,000 deaths from the disease were recorded in one week. Across the whole outbreak, 100,000 Londoners died – one in five people. It was the last serious outbreak of the disease in England, but people were as helpless to resist it as they had been over 300 years earlier. As with previous epidemics, the disease was spread by fleas carried on rats.

People believed this miasma had been created by the stinking rubbish and dunghills that were a feature of 17th-century city life. The vapour was present in the soil, where it would stay as long as the weather stayed cold. When the weather turned warmer, however, the vapour would pour out of the earth as a plague-carrying miasma. This fitted the pattern of the infection: plague outbreaks generally appeared when the weather began to turn warmer.

Other people

By 1665, many people believed the correct idea that disease could be spread from person to person, although, as there was no proof that this was the case it was not the most popular theory. However, plague victims were still quarantined. Even people who believed a miasma caused the disease believed that, once people had caught it, they could pass it on to others.

Ideas about the causes of the Great Plague

Although fewer people believed in the Theory of the Four Humours by 1665, it had not been replaced with any proven alternative. Still, nobody knew for certain what caused disease in general. Therefore, most people blamed the same things for the Great Plague in 1665 as they had during the Black Death in 1348.

Astrology

There had been an unusual alignment between Saturn and Jupiter in October 1664, and between Mars and Saturn on 12 November. These were seen as unlucky combinations that suggested there was trouble ahead. Worse still, a comet had been spotted, too.

Punishment from God

Many people believed the Great Plague was a result of mankind's wickedness and that God had sent it to clean up his kingdom.

Miasma

In contrast to the Black Death outbreak in 1348, by far the most popular theory about the spread of the Great Plague in 1665 was that it was caused by a miasma.

Extend your knowledge

Eyam, Derbyshire

The village of Eyam in Derbyshire was infected by the Great Plague in 1665 from fleas on cloth that had been shipped from London. This meant that the village had to be quarantined to stop the disease from spreading.

Food was brought over by residents of nearby villages and left at the edge of the town. The villagers left money in a trough of vinegar to make sure that they did not spread the Great Plague via the money.

Over the course of 14 months, at least 273 people died from the Great Plague, out of a population of around 350. However, the quarantine worked: the plague did not spread outside of Eyam.

Approaches to treatment of the Great Plague

We don't know a great deal about treatments that were given to plague victims in 1665. This is partly because so many of them were shut up with family members in quarantine, so there are not many records of the methods they attempted to keep people alive.

We do know that some of the new ideas about treatment of diseases had an impact on plague treatments in 1665. As was the fashion at the time, physicians advised that patients be wrapped in thick woollen cloths and laid by a fire so that they could sweat the disease out. Transference was also a popular idea – methods such as strapping a live chicken to a bubo, or lancing it with a feather plucked from a live chicken, were meant to draw out the poison and help the patient to recover.

Recipes for herbal remedies continued to be extremely popular. These took the form of medicines, poultices or rubs.

Quack doctors* took advantage of the general panic. They mixed remedies and advertised them as fabulous cures, hoping to make some easy money.

Key term

Quack doctor*

Somebody who did not have any medical qualifications, but who sold their services as a doctor or apothecary.

Source A

This recipe from Culpeper's *Complete Herbal* is for a 17th-century version of medieval theriac. It was known as London treacle and was a popular treatment for, and preventative against, the plague.

Take ... the seeds of citrons, sorrel, peony, basil, of each one ounce; scoridum, coralliana, of each six drams; the roots of angelica, tormentil, peony, the leaves of dittany, bay-berries, juniper berries, of each half an ounce; the flowers of rosemary, marigolds, clove, gilliflowers, the tops of St John's wort, nutmegs, saffron, of each three drams; the roots of gentian, zedoary, ginger, mace, myrrh, the leaves of scabious, devil's-bit, carduus, of each two drams; cloves, opium, of each a dram; Malaga wine as much is sufficient ... mix them according to art.

People still did not understand the cause of the Plague and therefore could not treat it effectively. The best advice was the same as it had always been: make sure you don't catch it in the first place.

Approaches to preventing the Great Plague

Advice from physicians

The College of Physicians recommended a variety of preventative measures that could be taken to avoid catching the Great Plague.

- Prayer and repentance
- Quarantine anybody who had the plague.
- Carrying a pomander was a way to drive away miasma. A pomander was a ball containing perfumed substances.
- Various diets were suggested, from eating almost nothing (fasting) to eating a diet heavy with garlic and sage fried in butter.
- Plague doctors wore special costumes to avoid catching the plague from their patients. They had hooked, birdlike masks, with sweet-smelling herbs to ward off the miasma. Birds were meant to attract the disease, so it was thought that the disease might be attracted to the bird shape and leave the patient. More practically, the physician's cloak would be treated with wax to make sure that none of the patient's blood from the patient soaked into it.

Advice from other healers

Most people turned to local healers for help in warding off the plague. Recipes for plague water, a 'treatment' for the plague, were popular among apothecaries. Some relied on native herbs that would have been used in England for centuries, such as mint and rosemary, while others contained new, exotic ingredients such as nutmeg and sugar. Smoking tobacco was encouraged to ward off the miasma.

Some people thought that, because buboes were symptoms of both syphilis and the Great Plague, catching syphilis would prevent a person from catching the Great Plague. They went out of their way to become infected with syphilis – though this, of course, did not prevent them catching the Great Plague.

Government action

Local governments also tried to prevent the plague from spreading. Unlike the first large outbreak of the plague, this time they did more, and so did the king. Charles II decreed that people should fast regularly and made a list of actions to try to stop the spread of the plague. These were carried out by local government officials of each city, including the mayor. Public meetings, fairs and even large funerals were banned. Theatres were closed. Streets and alleyways were swept and cleaned. Fires were set to burn on street corners, often in barrels of tar or strewn with sweet-smelling herbs, to drive away the evil miasma. Cats, dogs and pigeons were killed if they were seen on the street. Around 40,000 dogs and 200,000 cats were slaughtered because people thought they were helping to spread the disease.

The mayor also appointed searchers and wardens to monitor the spread of the disease. Searchers would go from house to house, checking to see if there were any plague victims in each one. If a household was infected, the inhabitants were either taken to the pest house or quarantined inside the house for 28 days. The house was painted with a red cross together with the words, 'Lord have mercy on us'. The parish officials were in charge of bringing them food and other necessities. Every day, carts would travel through the city to collect the bodies of the dead.



Figure 2.6 A 17th-century street showing different methods used to prevent the Great Plague.

2.4 Dealing with the Great Plague in London, 1665

Many people believed the best way to avoid the Great Plague, was the same as it had been in 1348: run away. People in 1665 still did not know what caused the plague, but they realised that this ignorance could kill them. Since there was no known cure for the disease, people focused on prevention, since methods of prevention had been successful in the past. Rather than attempt wild treatments, they put their energy into stopping the disease from spreading, or into escaping it completely.

Exam-style question, Section B

Explain **one** way in which ideas about preventing the plague were different in the 14th and 17th centuries.

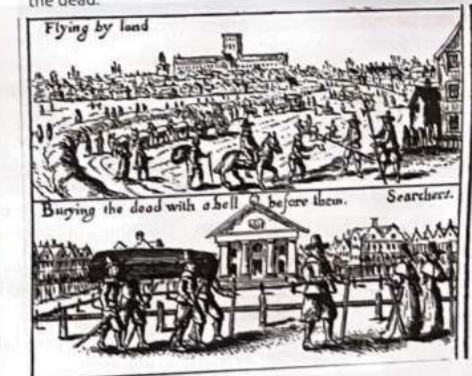
4 marks

Exam tip

The Black Death of 1348 and the Great Plague of 1665 are both case studies for this unit. That means that the examiner will want to see that you recognise the similarities and differences between these two specific outbreaks.

Source B

In this engraving from 1665, you can see how people reacted to the Great Plague: by fleeing London and burying the dead.



3.1 Ideas about the cause of disease and illness

Learning outcomes

- Understand how ideas about the cause of disease changed and stayed the same, 1700–1900.

The 18th century was a very exciting time in the world of science. In 1700, the influence of the Church was not as great as it once was. Many people no longer believed that God was responsible for all worldly events. Instead, they focused on developing scientific explanations. Intellectual movements such as the Enlightenment* made it fashionable to seek answers to questions about the world – including disease and illness.

Key term

The Enlightenment*

A movement in Europe during the 18th century that promoted the idea that people could think for themselves and that traditional authorities, like the nobility and the Church, should not be able to control everyday life.

This fashion for rational explanations touched every part of life – politics, philosophy, history, economics and, of course, science. In fact, the 'Age of Enlightenment' was happening at the same time as the Scientific Revolution, during which developments across all branches of science completely changed the way people lived and the ways they understood the world around them.

Some historians suggest that there were two halves to the Scientific Revolution.

- In the first half, starting with the Renaissance in the 16th century, old theories were discredited.
- In the second half, new ideas began to replace the old. This half of the Scientific Revolution began in c1700.

Society itself was also changing. Cities began to grow as people moved there in search of jobs. The new cities were not well planned and quickly became dirty and disease-ridden. Diseases like tuberculosis, typhus and smallpox were a big threat to this new working population. Therefore, understanding the causes of disease and illness became even more important.

Continuity and change

There were not a lot of new ideas about the causes of disease in the 18th century. Ideas such as the Theory of the Four Humours had been discarded. People still believed in miasma, although this theory was also becoming less popular.

Scientists in the early 18th century developed the theory of **spontaneous generation** as an alternative to theories like the Four Humours. Improvements in the quality of the glass lenses used in microscopes meant that scientists could see microbes* present on decaying matter*. Most people believed that these microbes were the **product** of decay, rather than the **cause** of it.

Key terms

Microbes*

A microbe is any living organism that is too small to see without a microscope. Microbes include bacteria.

Decaying matter*

Material, such as vegetables or animals, that has died and is rotting.

In the 18th century, this was just a theory, and scientists were unable to prove that spontaneous generation was correct. It took until the 19th century for scientists to make a solid link between these microbes and disease.

Medical breakthrough: Germ Theory

Louis Pasteur and the development of Germ Theory

In 1860, the French Academy of Science challenged scientists to come up with evidence to either prove or disprove the theory of spontaneous generation.

By the middle of the 19th century, microscopes had improved even more – it was now possible to magnify substances to a much higher level and keep the image clear enough to see. Because of this, Louis Pasteur, a French scientist, was able to observe unwanted microbes in wine and vinegar, which turned both liquids ‘bad’.

Pasteur published the results of his experiments in 1861. He called his discovery **Germ Theory**.

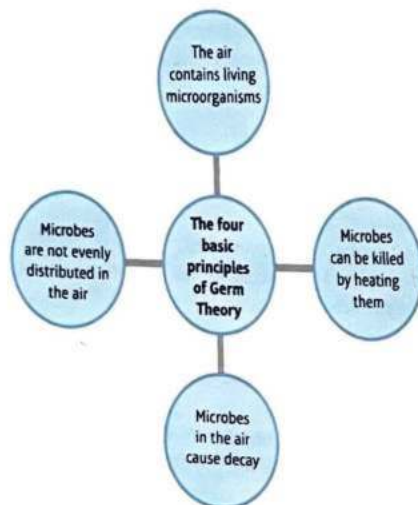


Figure 3.1 The four basic principles of Germ Theory.

Extend your knowledge

Germ Theory

Pasteur called the microorganisms he observed ‘germs’ because he observed the microorganisms growing, or germinating.

Pasteur proved that the idea of spontaneous generation was wrong because decay did not happen to sterilised matter that was left undisturbed. Instead, something in the air was causing the decay.

Pasteur also theorised that, as germs were causing decay, they might also be causing disease in the human body. He observed one particular type of microorganism killing off France’s silkworm population, which seemed to prove this theory. However, he did not publish this **germ theory of infection** until 1878.

Pasteur’s influence in Britain

To begin with, Pasteur’s work had almost no impact on British ideas about the causes of illness and disease. He was not a doctor, and his work focused on decay and spoiled food, not disease.

In Britain, the theory of spontaneous generation continued to be important until the 1870s. It was promoted by Dr Henry Bastian, who was one of the most powerful doctors in the country. Because he was so well respected, few people disagreed with him.

However, some scientists did start to look for a link between the microbes and disease. One of these was Joseph Lister, who read Pasteur’s germ theory and linked it to the infection problems his surgical patients had experienced (see page 82).

Another scientist who promoted the link between microbes and disease was John Tyndall. He had discovered that there were small organic* particles in the air. In January 1870, he gave a lecture, linking his discovery with Pasteur’s germ theory and Lister’s work on wound infection. Tyndall theorised that dust particles carried the germs that caused disease.

Key term

Organic*

Something that is living or that has once been alive.

However, Tyndall was not a doctor: he was a physicist. The medical world trusted Bastian’s beliefs rather than Tyndall’s theory.

Lister’s ideas were also doubted, as he could not prove his theory. Although microscopes meant that microbes were visible, there were lots of them present in the blood or in a wound. Doctors could not yet identify what they were and what role they played. The gut is a good example of why they had problems: when examined under the microscope, scientists saw hundreds of microbes, even in healthy people. It seemed impossible to people that these microbes caused disease, too (see Source A).

Source A

In this extract from an article published in the *British Medical Journal* in 1875, Dr Henry Bastian explains why he does not believe microbes (bacteria) cause disease.

Bacteria... habitually exist in so many parts of the body in every human being... as to make it almost inconceivable that these organisms can be causes of disease. In support of this statement I have only to say, that even in a healthy person they may be found in myriads... the whole alimentary tract from mouth to anus; they exist throughout the air-passages, and may be found in mucus coming from the nasal cavities... They exist... within the skin, not only in the face, but in other parts of the body. Fresh legions of them are being introduced... with almost every meal that is taken.

Therefore, Pasteur’s theory had limited impact in Britain, because **attitudes among doctors** meant people refused to recognise the link between germs and disease – even though the link was correct.

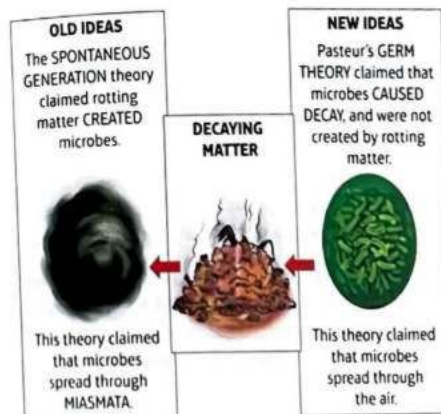


Figure 3.2 Germ Theory vs. spontaneous generation.

Robert Koch’s work on microbes

Pasteur had been the first scientist to identify microbes and their role in decay. However, it was the German scientist Robert Koch who successfully identified that different germs cause many common diseases.

Koch discovered the bacteria that caused tuberculosis, in 1882. He then published his ideas on the methods that could be used to identify disease-causing microbes.

- 1 The microbe is present in every case of the disease.
- 2 Once taken from the body, the microbe can be reproduced into a pure culture*.
- 3 The disease can be reproduced in test animals using that culture.
- 4 The microbe can be taken out of the test animals and used to start a fresh culture.

Key term

Culture*

Bacteria grown under controlled conditions.

Koch continued to look for the microbes causing different diseases. In 1883, he discovered cholera, and in 1884 he proved that it was spread in water supplies when he found it in the drinking water in India, where a cholera epidemic had broken out. This also provided proof for John Snow’s theory (see page 92).

Koch made it easier for future scientists to study bacteria by developing a new method of growing them, using agar jelly in a petri dish. This made it easier to study the bacteria under a microscope. Later, Koch also developed a method for staining them with industrial dyes, to make them easier to see.

Koch’s research inspired other scientists. Over the next two decades, they went on to discover the microbes responsible for other diseases, such as diphtheria, pneumonia, meningitis, the plague, tetanus and various other infections. Koch received the Nobel Prize for Medicine in 1905. He is considered to be the father of bacteriology* and his methods are still used when seeking out the microbes responsible for disease today.

Key term

Bacteriology*

The study of bacteria.

Source B

In this cartoon from 1880, Robert Koch is likened to St George. The saddle is labelled 'Investigation' and he is using a microscope as a weapon to slay tuberculosis, the snake.



Koch's influence in Britain

The identification of microbes that caused particular diseases was an enormous breakthrough in the diagnosis of disease. Whereas before, doctors had studied and treated symptoms, now they studied the disease itself. The medical profession had begun to recognise that the microbe created the symptoms of the disease, and it was the microbe that needed to be removed.

For example, in 1883, the microbe that caused diphtheria was found. Diphtheria was a horrible disease that mostly affected children. It caused a painful cough and a fever. A leathery skin would grow over the tonsils and the back of the throat, which meant that the sufferer could not breathe. By studying the microbe, scientists were able to observe that it produced a poison. The poison, when breathed, stayed in the throat and caused the painful symptoms. Since the microbe had been identified,

scientists were able to seek ways of attacking it directly rather than just treating the symptoms.

Extend your knowledge

Sceptics of the germ theory of disease

Many doctors remained sceptical about the link between germs and disease. One Munich chemist, Max von Pettenkofer, asked Koch for a sample of the cholera microbes, which he then drank. Amazingly, he did not develop cholera – it is likely that the germs were killed off by his stomach acid. Stunts like this did nothing to help convince people that cholera was spread through dirty water.

Summary: the impact of Germ Theory in Britain

Progress in treatment and prevention using Germ Theory was slow. Once Pasteur, Koch and other scientists had found the specific disease-causing microbes, cures and vaccines could be tested. Only after this did Germ Theory begin to have a direct impact on medical treatment.

Even the British government rejected the Germ Theory of disease at first. When Koch travelled to Calcutta in 1884 to study an outbreak of cholera, he proved that it was caused by microbes in the supply of drinking water. However, this was ignored by the British government. Instead, they kept to the idea that the disease was present in the soil, and the miasma was brought out by the weather. This seemed to make sense, since there was more cholera around during the rainy season.

Despite these setbacks, in the 20th century Germ Theory and the new study of bacteriology had an enormous impact on our understanding of the causes of disease and illness. It is now recognised that many diseases are caused by a microorganism – bacteria, virus or fungus. When diagnosing disease, doctors now look for symptoms and try to match them to a disease caused by a specific microbe.

By the end of the 19th century, the mystery around what caused illness and disease had been solved. It took a little more time for this to be accepted by the medical profession, but the evidence had been found. Now it was time to start looking for new treatments based on this new science.

Interpretation 1

In *Disease, Class and Social Change* (2012), Marc Arnold explains that Germ Theory only had a limited impact in Britain during the 19th century.

Many historians have attributed the growing concern with public health from the 1880s as being solely due to the advent of bacteriology, and particularly to Robert Koch's discovery of the tuberculosis bacteria in 1882. However, the impact of germ theory in England during the 19th and early 20th centuries has probably been overstated. In 1897 the physician James Lindsay, reviewing the impact of Koch's discovery in Britain, noted that most of the respondents to an investigation by the British Medical Association did not view tuberculosis as an infectious disease.

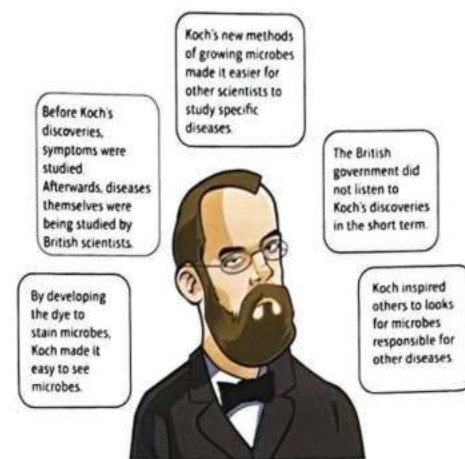


Figure 3.3 Koch's influence in Britain.

THINKING HISTORICALLY

Change and continuity (3a)

Significant change

Things change all the time. Every second something in the world is changing. If historians weighted every change as equal, they would never be able to write anything about the past that was useful. The decisions that historians make about what is historically significant and what is not is a very important part of their work.

The development of the Germ Theory of Disease

In 1861, Pasteur identified that microbes were causing decay in wine.	Dr Bastien, a very well-respected doctor, promoted the theory of spontaneous generation until his death in 1915.	In 1854, John Snow used observation to show that cholera was water-borne.	In 1882, Koch identified the microbe that caused tuberculosis.
Koch developed a new method for growing and observing bacteria.	In 1870, physicist John Tyndall theorised that disease was spread through dust particles.	Joseph Lister read Pasteur's work and linked it with infection problems in surgery.	Leeuwenhoek observed microbes and wrote to the Royal Society about them.

Imagine you are investigating how scientists' understanding of the causes of disease developed over time.

- 1 With a partner, discuss what might make a change significant for this investigation. E.g. how long the change lasted, or the wider consequences of the change, could affect how significant you think a change was. Write down your top three criteria.
- 2 Discuss your choice of criteria as a class, writing down all the suggestions. Assess your own three criteria again and decide if you want to change any of them.
- 3 Using your three criteria for deciding significance, put the eight changes above into order of significance, with the most significant change at the top.
- 4 Compare your list with that of another pair in your class. Is the order similar?
- 5 Compare your list of criteria affecting significance with the other pair. Does this explain why you put the changes into different orders?
- 6 Write a short paragraph explaining what you think makes a change significant.

Activities

- 1 Create a timeline to show the development of Germ Theory. Include events from the work of both Pasteur and Koch.
- 2 Create flash cards to help you revise the new theories. Create one card each for Pasteur and Koch, showing the impact they had in Britain.
- 3 Did Pasteur or Koch do more to improve understanding of disease? Have a debate with a partner to help you better understand the role of each scientist.

Factors affecting the understanding of the causes of illness and disease

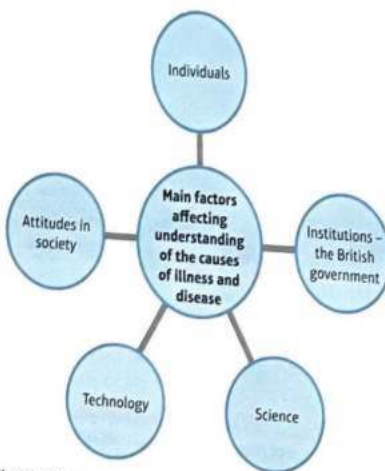


Figure 3.4 Main factors affecting understanding of the causes of illness and disease.

Individuals

The most important individuals in the development of Germ Theory were Pasteur and Koch. Without them, it might have taken a lot longer for the Germ Theory of Disease to be developed. There were also dozens of other scientists whose individual contributions helped put pieces of the puzzle into place, including John Tyndall and Joseph Lister.

Institutions: the British government

The government in Britain did not help to improve understanding of disease. For much of the 19th century, they were not interested in getting involved with everyday life. However, once more people gained the ability to vote, the government was more willing to intervene (although they were more interested in practical solutions to the problems of epidemics like cholera and typhoid). Since Germ Theory offered no practical solution to the problem of disease, they did not promote it.

Science

The second part of the Scientific Revolution focused on finding answers to the big questions of science. There was a strong desire to prove new theories and provide practical solutions to scientific problems.

Improved communication enabled scientists to share their work with each other. Scientists were able to read the work of their peers and to draw their own conclusions and theories. For example, Tyndall's 1870 theory about germs causing disease was based on the work of Pasteur and Lister. Ideas were shared across different branches of science. For example, Pasteur's work on animal diseases inspired other scientists to use his methods to diagnose human diseases.

Technology

The microscope was arguably the most important piece of technology that made the development of Germ Theory possible. Clearer images and higher magnification made it possible to spot most microorganisms, although not all – viruses, for example, are so small that they can only be seen with a very powerful microscope.

Other technological developments also helped. Koch developed a way to grow microbes. His colleague, Joseph Petri, developed the petri dish. By the end of the century, experiments with dyes had helped scientists to better observe bacteria.

Attitudes in society

Attitudes in society were both a help and a hindrance to the widespread acceptance of Germ Theory. People were more interested in finding the reasons behind disease than they had been in previous centuries.

Due to the Enlightenment, they were more interested in rational explanations for disease. Also, overcrowded cities and poor living conditions led to dangerous outbreaks of disease. Many people were horrified by the sights they saw on the street and the impact bad health had on the poor. Furthermore, an unhealthy population could not work. Religious reformers and hard-nosed businessmen agreed: something needed to be done to solve the problem of epidemics.

However, people's reluctance to change their minds slowed the spread of Germ Theory. It took a long time to show that specific microbes were always present when people were suffering from a particular disease. Until this proof was provided, from the 1880s onwards, Germ Theory did not become an accepted fact.

Exam-style question, Section B

'There was rapid change in ideas about the causes of illness and disease in the period c1700–c1900'.

How far do you agree with this statement? You may use the following information in your answer:

- Spontaneous generation
- Louis Pasteur.

You must also use information of your own. **16 marks**

Exam tip

Two hundred years is a long period of time. It is unlikely that rapid change would ever go on for so long. In your answer, try to pinpoint where exactly the rapid change occurred. Make sure you explain what criteria you have used to make your judgement.

Summary

- In the 18th and 19th centuries, scientists started to theorise about germs being produced by decaying matter, a theory named spontaneous generation.
- In 1861, Louis Pasteur, a French chemist, published Germ Theory. This proved that microbes in the air caused decay in substances such as wine and vinegar.
- Pasteur's work was picked up by some medical professions quite quickly, particularly in Britain where Joseph Lister began attempting to remove microbes from his operating theatre. However, many doctors resisted the ideas.
- Robert Koch, a German scientist, began to look for specific microbes that caused disease. He identified lots of these, including the microbe that caused cholera.
- By c1900, the mystery of what caused many illnesses and diseases had been solved – it was just that not everybody believed the solution yet.

Checkpoint

Strengthen

- 51 Describe in detail the roles of Pasteur and Koch in developing the Germ Theory of Disease.
- 52 List the four basic principles of Germ Theory.
- 53 Why didn't scientists always believe Koch's ideas about microbes?
- 54 How much impact did the Germ Theory have on Britain by c1900?

Challenge

- C1 To begin with, Pasteur's work did not have many practical applications. Name some similar discoveries from previous periods.
- C2 Which factor do you think had the biggest impact on the development of understanding about the causes of disease and illness? Explain why you think this.

If you are struggling with these questions, ask your teacher for some hints.

Learning outcomes

- Understand how approaches to prevention and treatment changed and stayed the same, c1700–c1900.

The extent of change in care and treatment

By 1900, the way that sick people were treated and cared for had changed almost completely since 1700. However, there was some continuity in treatment because it took a while for medical science to catch up to the new ideas about the causes of illness and disease. By 1900, most people accepted that germs caused disease and that treatment needed to be focused on removing the germ. Unfortunately, there was still not a great deal of understanding about how best to do this. Scientists were working incredibly hard to come up with treatments for everyday diseases such as syphilis and tuberculosis, but it was not until after 1900 that these were successfully developed. Therefore, old herbal remedies continued to be popular.

Similarly, by 1900, the old belief that prevention was the most important aspect became even more widespread. People began to realise that infection was everywhere: on dirty clothes, medical instruments and unwashed hands, and in the air and water. The old idea of avoiding disease by keeping clean and following a sanitary regime suddenly made sense, because people really did need to protect themselves against something invisible.

Perhaps the biggest change by 1900 was in the **willingness of the government and the population to take steps to prevent disease from spreading.**

Activities

- Create a spider diagram to show changes and continuities in treatments by c1900.
 - Next to each label on your spider diagram, write a sentence to explain why this change/continuity occurred.
- How much had treatment changed by c1900? Write a paragraph to explain the extent of change that occurred by this point.

Improvements in hospitals and the influence of Florence Nightingale

Hospitals in the 18th century

As we discovered in Chapter 2, most of England's hospitals had closed down when Henry VIII dissolved the monasteries in the 1530s. By 1700, there were only five hospitals left in the country – and they were all in London. The country did not invest in new hospitals.

However, new hospitals did begin to appear in other cities in the 18th century, founded using donations from wealthy people and from members of the new middle classes, such as lawyers and businessmen. Some doctors offered their services free of charge to these new hospitals so that they could practise their skills.

Attitudes towards the role of hospitals were changing, too. Hospitals increasingly became places where sick people were treated, as opposed to places where people could rest and pray. Doctors visited patients regularly and there was a surgeon or apothecary on site for daily treatments. A small staff of untrained nurses cared for the patients.

However, hospitals were still not places that people often chose to be treated. The rich received medical treatment, and even surgery, in their own homes, which was much safer.

Hospitals were still quite particular about who they treated. Generally, patients admitted were 'the deserving poor' – respectable, working class people who could not afford to pay their medical bills. This change in the types of people hospitals were willing to treat was very important in the history of medical treatment, because it gave poor people access to trained doctors for the first time.

Unfortunately, as more people started to attend, hospitals became less sanitary. They became less strict about turning away infectious patients.

Although they often had separate wards for infectious patients. However, doctors went from patient to patient and ward to ward without washing their hands or changing their clothes. Diseases spread quickly. People did not understand that germs caused disease, so they did not take steps to avoid spreading the germs.

By the middle of the 19th century, there were a lot more hospitals. However, hospital conditions were very poor.

Florence Nightingale

Source A

A photograph of Florence Nightingale, taken in 1855.



It was in these dirty hospital conditions that Florence Nightingale found herself when she began nursing. Nightingale was born into a wealthy family in 1820. When she was 17, she experienced a religious vision

telling her that her mission was to serve mankind. She eventually convinced her parents to allow her to train as a nurse, first in Germany and then in Paris. In 1853, she became the superintendent of nurses at King's College Hospital in London.

In 1854, Britain went to war with Russia in the Crimea. News reports said that the hospitals there were not fit for the soldiers to be treated in, and there was a national outcry at the rumours that there were no nurses or even bandages available to the soldiers. Because of this outcry, and her position in society, Nightingale was able to convince the government to send her to improve the hospitals in the Crimea, along with 38 other nurses.

While Nightingale was in the Crimea, she made changes to the care of the wounded soldiers in many different ways.

- Nightingale and her nurses demanded 300 scrubbing brushes to get rid of any dirt near patients being treated.
- Nurses were organised to treat nearly 2,000 wounded soldiers.
- Clean bedding and good meals were provided.

Nightingale's efforts had a very positive effect on the mortality rate – within six months, it had dropped from 40% to only 2%.

By the time Nightingale returned to Britain in 1856, she was a national hero. There had been a great deal of bad publicity about the conditions in war hospitals, and Nightingale was famous for having made a big difference. This gave her credibility and helped her to make changes to hospitals in Britain, too.

Extend your knowledge

Statistics

Florence Nightingale was also one of the first people to use statistical diagrams to persuade people that there was a need for change. One of her diagrams was called 'Cause of Mortality in the Army in the East'.

Source B

A copy of a ballad that was written about Nightingale in 1855. Ballads like this, set to popular tunes from the time, would sell for a penny on street corners.



The Nightingale In the East.

TUNE,—"THE COTTAGE AND WATER MILL."

Ryle & Co., Printers, 2 & 3, Monmouth Court, Seven Dials, London.

ON a dark lonely night on the Crimea's dread shore,
There had been bloodshed and strife on the morning before,
The dead and the dying lay bleeding around,
Some crying for help—there was none to be found
Now God in his mercy he pity'd their cries,
And the soldier so cheerful in the morning doth rise
So forward my lads, may your heart never fail,
You are cheer'd by the presence of a sweet Nightingale.

Now God sent this woman to succour the brave,
Some thousands she's sav'd from an untimely grave
Her eyes beam with pleasure, she's bounteous and good,
The wants of the wounded are by her understood
With fever some brought in, with life almost gone
Some with dismantled limbs, some to fragments is torn,
But they keep up their spirits, their hearts never fail
Now they're cheer'd by the presence of a sweet Nightingale.

Her heart it means good—for no bounty she'll take
She'd lay down her life for the poor soldier's sake
She prays for the dying, she gives peace to the brave,
She feels that a soldier has a soul to be saved.
The wounded they love her, as it has been seen,
She's the soldier's preserver, they call her their queen,

May God give her strength, & her heart never fail,
One of Heaven's best gifts is Miss Nightingale.
The wives of the wounded how thankful are they,
Their husbands are car'd for, how happy are they.
Whate'er her country, this gift God has given.
The soldiers they say she's an angel from Heaven
Sing praise to this woman, and deny it who can I
And all women was sent for the comfort of man,
Let's hope no more against them you'll rail,
Treat them well, and they'll prove like Miss Nightingale.

Source C

In this extract from *Notes on Hospitals*, published in 1859, Nightingale explains how important it is for hospitals to be well-ventilated.

To build a hospital with one closed court with high walls, or what is worse, with two closed courts, is to stagnate the air even before it reaches the wards.

This defect is one of the most serious that can be committed in hospital architecture; and it exists, nevertheless, in some form or other in nearly all the older hospitals, and in many even of recent constructions.

The air outside the hospital cannot be maintained in a state sufficiently pure to be used for internal ventilation, unless there be entire freedom of movement. Anything which interferes with this is injurious. Neighbouring high walls, smoking chimneys, trees, high ground, are all more or less hurtful; but worse than all is bad construction of the hospital itself.

The impact of Florence Nightingale in British hospitals

Following her return from the Crimea, Nightingale's experience and popularity meant that she was able to have a big impact on hospital care in Britain in two different ways: the way hospitals were designed and the training of nurses.

For example, Nightingale preferred hospitals to follow the pavilion plan. This meant they were built with improved ventilation, with more windows, larger rooms and separate isolation wards to stop diseases spreading (see Source D).

Also, Nightingale established a nursing school at St Thomas' Hospital in London called the Nightingale School for Nurses in 1860. Figure 3.5 gives more detail on the impact of Nightingale's work.

Extend your knowledge

Nursing now

Now known as the Florence Nightingale Faculty of Nursing and Midwifery, Nightingale's school has been connected to St Thomas' Hospital ever since it was founded by Nightingale in 1860.

3.2 Approaches to prevention and treatment



Figure 3.5 The impact of Florence Nightingale's work.

Hospitals by 1900

Hospitals by 1900 looked very different from the few in Britain in 1700. Many different wards split up infectious patients from those requiring surgery. Operating theatres and specialist departments for new medical equipment provided separate spaces for certain procedures.

Cleanliness was now of the utmost importance: hospitals first focused on cleaning up germs using antiseptics, and by 1900 they were focused on preventing the germs from getting in to begin with. Doctors were a common sight, particularly junior doctors who were training and getting more hands-on experience. Trained nurses lived in nearby houses provided for them.

New ideas about hospitals were adopted quickly. Everybody wanted to have the most modern hospital designs, to help them attract donations and new student doctors.

The function of hospitals had completely changed. Instead of being places for the sick to rest, hospitals had become a place where the sick were treated. This fundamental change in the role of hospitals had forced a change in the way they were built and run.

Source D

This picture and plan show the Birmingham hospital that was opened in 1888. It was built on open ground with separate isolation wards, modelled on the pavilion plan favoured by Florence Nightingale.



Activities

- 1 List the changes that had occurred in hospitals by c1900.
- 2 Explain the impact of more people attending hospitals.
- 3 Write an obituary for Florence Nightingale, explaining the impact she had on nursing and hospital design.

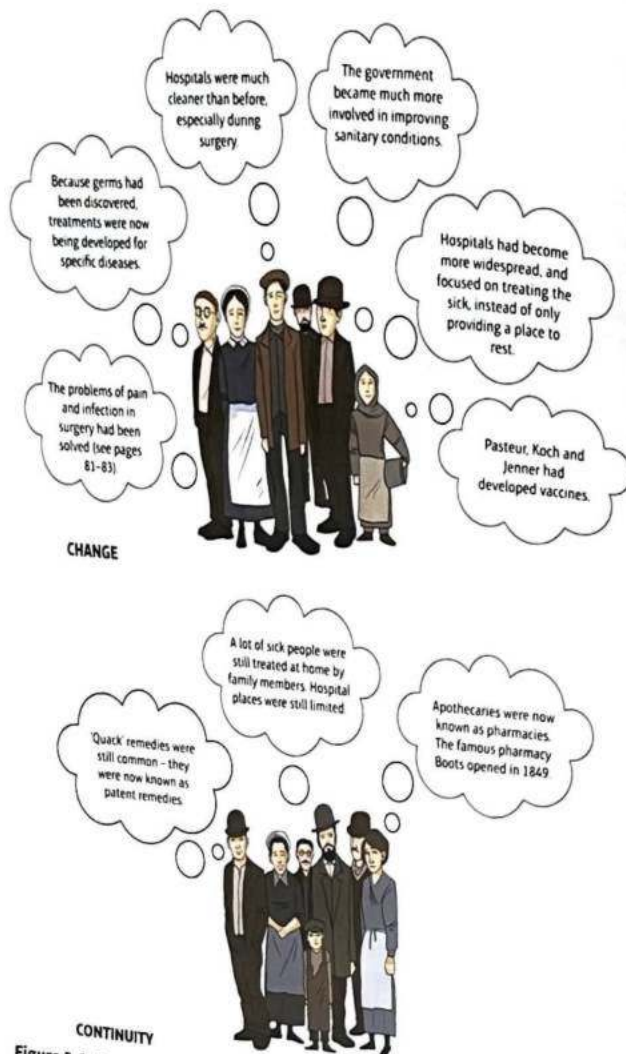


Figure 3.6 Extent of change in care and treatment.

Improvements in surgical treatment

In the 18th century, surgery was a dangerous and usually fatal business. The three big problems that surgeons faced were:

- bleeding
- pain
- infection.

Although substances like opium had been used for some time to calm patients with severe injuries, without **anaesthetic** there was no way of preventing the excruciating pain that they went through – which sometimes sent them into shock. Surgeries had to be performed quickly, before the patient bled to death on the operating table, as blood transfusions had not yet been developed. **Bleeding** continued to be a problem during surgery throughout the 18th and 19th centuries.

The most talented surgeons were able to operate extremely quickly, which improved their patients' chances of survival. However, even if the patient survived, often infection set in as surgeries were not performed in germ-free environments. In fact, they were usually performed in the patient's home, with the surgeon wearing the same clothes he had arrived in.

For these reasons, surgical treatments were quite limited. The most common type of surgery was amputation, which was often necessary due to accidents or complications relating to tuberculosis. Other types of surgeries were rare because the danger of death was so great. However, surgeons were becoming more respected.

In the 19th century, significant developments occurred that tackled two of the three problems of surgery. Firstly, anaesthetics were developed to enable surgeons to put patients to sleep before operating on them – which helped with the **pain**. Secondly, the development of Germ Theory led to an understanding of the importance of cleanliness in the operating room, and **antiseptic surgery** was developed – this helped to stop infection.

Tackling pain: the development of the anaesthetic

Doctors had been experimenting with pain relief for their patients for many centuries, in an attempt to keep them still and quiet for long enough to perform operations. Early experiments with laughing gas proved quite successful for small operations such as pulling teeth. When the chemical ether was discovered and used in America, it caused great excitement among surgeons. However, there were some problems with ether. It often made patients vomit, and the gas irritated the lungs which caused coughing, even while the patient was unconscious. Worse still, it was very flammable, which meant it was a dangerous chemical to keep around, given that operating rooms were lit with candles or gas lamps.

James Simpson and chloroform

James Simpson, a young surgeon from Edinburgh, was convinced that there were better anaesthetics than laughing gas to be discovered. He gathered a group of friends together and they inhaled the vapours of

Timeline

The development of the anaesthetic

1795 Humphry Davy, a dentist's assistant, tried inhaling nitrous oxide, or 'laughing gas'. He discovered that it numbed pain.

1846 Robert Liston, a famous London surgeon, successfully anaesthetised a patient with ether and then amputated his leg.

1853 Queen Victoria used chloroform during the birth of Prince Leopold and spoke favourably of it.

1842 William E. Clarke, an American chemist and doctor, successfully used ether to anaesthetise a patient to remove a tooth.

1847 James Simpson, a surgical professor in Edinburgh, discovered chloroform.

various chemicals to see what might work. After sniffing chloroform, the entire party passed out and were discovered some time later by Mrs Simpson. Clearly, chloroform was an effective anaesthetic.

Source E

An artist's impression of the night when James Simpson and his friends discovered the power of chloroform, c1860.



Although they weren't detected during Simpson's experiments, chloroform did have some negative side effects.

- The dose had to be carefully controlled, as it was easy to overdose a patient and kill them.
- The chemical sometimes affected the heart, which caused some healthy and fit young people to die shortly after inhaling it.

Extend your knowledge

The risks of chloroform

Hannah Greener, a 14-year-old girl who was having an infected toenail removed in 1848, became one of the first patients to die from an overdose of chloroform.

In spite of this, however, chloroform began to be used as a solution to the one of the problems of surgery: pain. After it was administered to Queen Victoria during the birth of her son in 1853, chloroform became even more popular in Britain.

James Simpson was the first person to be knighted for services to medicine. This was because of the positive impact that regular use of anaesthesia had on surgery. More surgeries took place and deeper, more complex surgeries became possible.

Some historians suggest that the use of anaesthetics made it possible for doctors to attempt lengthier and more-complex operations. However, because anaesthetics allowed for deeper surgery to be attempted, infection and bleeding became even bigger problems.

Tackling infection: the development of antiseptic surgery

Anaesthetics had solved the problem of pain. Now, infection had to be tackled.

Historically, due to a lack of understanding about germs, surgeons did not make an effort to keep their surroundings, or even themselves, clean when they operated on patients. In fact, many would wear their most stained doctor's coat to show how much experience they had. Instruments were not washed, and a large number of people would be present during operations – including medical students and 'dressers', whose job it was to hold the patient still during the

operation. As a result of this, many patients survived operations but then died shortly afterwards from infections such as gangrene or sepsis.

Joseph Lister and carbolic acid

Joseph Lister was an English surgeon. By studying infected wounds, he realised that the flesh was rotting. Lister compared his results with the recently-published work of Pasteur, who had identified germs as being responsible for decay. Lister theorised that, if microbes in the air caused wine and vinegar to go bad, perhaps microbes also caused flesh to rot.

Lister started to look for a chemical that would clear bacteria from wounds. He was aware of the use of carbolic acid in sewage treatments. So, in 1865, he operated on a patient with a broken leg and added a bandage soaked in carbolic acid. The wound healed cleanly.

From this, Lister developed a series of steps to ensure that wounds did not become infected. These included spraying carbolic acid in the air during operations.

Source F

This famous picture is taken from William Watson Cheyne's book *Antiseptic Surgery* which was published in 1882.



Lister published his results in *The Lancet*, a medical journal. He detailed 11 different cases where carbolic acid had been used successfully in surgery.

In spite of its success, antiseptic surgery did not catch on quickly.

- News of Lister's success spread more quickly than Germ Theory. However, this meant that the science behind the new method wasn't fully understood. Consequently, not all surgeons were willing to use the

carbolic spray. They still did not believe that the air was full of germs.

- Carbolic spray dried out the skin and left behind an odd smell. Some surgeons argued that, since it made their hands sore, it certainly could not be doing the patient any good.
- Lister focused on encouraging his colleagues to use the carbolic spray instead of scientifically proving his theory. He was a 'doer' rather than a 'thinker'.

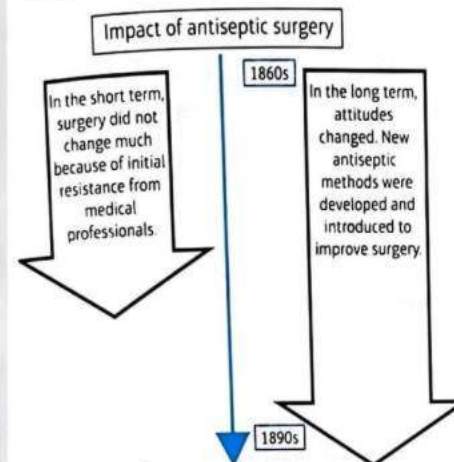


Figure 3.7 The impact of antiseptic surgery.

The key change here wasn't that doctors started using carbolic acid – although it was a great improvement, carbolic spray was only popular for a short amount of time, and even Lister himself stopped using it in 1890. What is important is that the attitude of surgeons towards antiseptic and aseptic surgery* changed. Surgeons finally understood that performing safe surgery was not only possible – it was their duty.

Key term

Aseptic surgery*

Surgery where microbes are prevented from getting into a wound in the first place, as opposed to being killed off with an antiseptic.

Other surgeons began to look for different methods of preventing infection. By 1900, instruments were steam cleaned, operating theatres were scrubbed spotless, rubber gloves and surgical gowns were introduced and surgeons used face masks during operations.

Opposition to change

Although developments in anaesthetics and antiseptics helped to improve the effectiveness and availability of surgery, not everybody welcomed the changes.

- Anaesthetics allowed for deeper surgeries to be attempted. Before the introduction of carbolic spray, infection and bleeding became even bigger problems. The death rate actually increased, which seemed to suggest that anaesthetics were bad. People did not trust the technique.
- The Victorians believed that pain relief was interfering with God's plan, particularly in childbirth, which was meant to be painful.
- Some doctors believed that patients were more likely to die if they were unconscious during the operation, rather than awake and screaming.
- It took a long time for doctors to accept that germs caused infection. Surgeons did not want to believe that they might have been responsible for the infections that killed their patients.

Activities

- 1 Create an annotated timeline to show the changes that took place in the field of surgery during the 19th century.
- 2 Make flashcards for Simpson and Lister, explaining what they did and what impact this had on surgical techniques.
- 3 Have a debate with a partner about which key individual had the biggest impact. Use the flashcards you made to help you.

Exam-style question, Section B

Explain why there was rapid change in surgical treatments in the period c1700–c1900.

You may use the following in your answer:

- chloroform
- Joseph Lister.

You **must** also use information of your own. **12 marks**

Exam tip

Make sure that you focus on the reasons why development of surgical treatment was rapid – avoid simply describing the use of anaesthetics and antiseptics.

Case study – Jenner and the development of the vaccination

Smallpox in 18th-century Britain

At the start of this period, smallpox was a terrible threat to the health of the population of Britain. There were nationwide epidemics in 1722, 1723 and 1740–42. The problem was particularly bad in London, where there were 11 epidemics in the 18th century. The worst of these occurred in 1796, when 3,548 people died. By this time, the population of the city was approaching one million, so the disease spread quickly and easily from person to person.

At this time, people were still unaware of the cause of the disease, but they did have some ideas about how to avoid catching it. It had been noticed that people who caught a mild form of smallpox and then recovered from it did not catch it again. This would later form the basis for vaccination, which works in the same way. However, in the 18th century there was not enough scientific knowledge for people to understand how this worked.

Therefore, some people attempted to inoculate* themselves against smallpox by catching a mild dose of the disease, so that they would avoid catching a more severe form of it later on. Pus from a smallpox scab would be rubbed into a cut on the patient being inoculated by a doctor. Unfortunately, this did not always work: some patients died of the smallpox they were given, as the disease did not affect everyone in the same way.

Key term

Inoculate*

Deliberately infecting oneself with a disease, in order to avoid a more severe case of it later on.

In spite of this, inoculation was seen by many as the best chance of surviving smallpox. However, the procedure was very expensive and so only the very rich could afford it. Many doctors made a fortune carrying out inoculations for wealthy people. One doctor, Thomas Dimmsdale, was made a baron, paid £10,000 and awarded an annual salary of £500 after he inoculated Catherine the Great and her children in 1768. He was one of the most successful inoculators of his time.

Jenner discovers the vaccination for smallpox

Edward Jenner was a Gloucestershire doctor in the late 18th century. He had trained as an apprentice to a surgeon-apothecary and then practised medicine at St George's Hospital in London, before returning to Gloucestershire, where he became a general practitioner (GP). Jenner was particularly interested in inoculations. He gathered evidence of over 1,000 cases where smallpox inoculation had failed.

There were a lot of dairy farms in the area where Jenner worked. He regularly treated dairy maids for cowpox* and noticed that, when there was a smallpox epidemic, those who had previously suffered from cowpox did not catch smallpox. He decided the two must be somehow connected.

Key term

Cowpox*

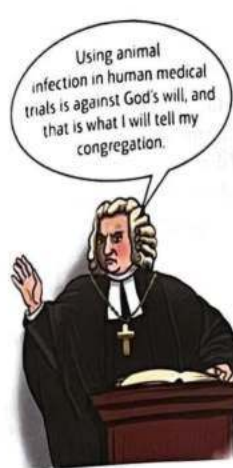
A disease causing red blisters on the skin, similar to smallpox. It can be transmitted from cows to humans.

Jenner needed to test his theory and so, in 1796, he infected a local boy, James Phipps, with cowpox. Six weeks later he attempted to infect James with smallpox, but James did not catch it. Jenner infected more local people with cowpox to further test his theory. In 1798, he wrote up his findings in *An Enquiry into the Causes and Effects of the Variola Vaccinae*. He named the technique 'vaccination' after the Latin word for cow, vacca.

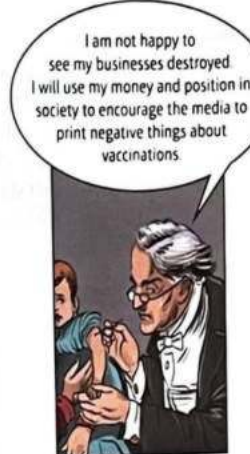
Jenner made sure that the instructions for his new method were very detailed, so that other doctors would be able to follow them. He wanted other people to use the vaccination to prevent smallpox from spreading.

Reactions to the new vaccination

As with so many medical discoveries, it took some time for people to accept vaccinations. Although Jenner knew that the system worked, he was not able to explain how or why it worked – and this made people suspicious. The idea of infecting someone with an animal disease was seen as extremely strange, and a lot of people were against it.



THE CHURCH



INOCULATORS



THE ROYAL SOCIETY

Figure 3.8 Differing opinions about vaccinations in the 19th century.

Although certain groups of people were against vaccinations, there was another, very powerful group that were in favour of them: parliament. As you can see from the timeline below, the British government favoured the new method of vaccination from the first half of the 19th century. This was because it was a safer and more reliable alternative to inoculation. It was also cheaper, because recipients of vaccines did not need to be put into quarantine, whereas those receiving the inoculation were in danger of spreading smallpox to other people.

Timeline

Smallpox and government intervention, 1840–72

1837–40 A smallpox epidemic breaks out. 35,000 people die. *The Lancet*, a medical journal, blames inoculation for the outbreak.

1852 The government makes smallpox vaccination compulsory.

1872 The British government begin to enforce the compulsory vaccination.

1840 The government makes inoculation a crime.

1840 The government agrees to provide children with vaccinations at the taxpayer's expense.

1871 Public Vaccinators* are appointed.

Key term

Public Vaccinators*

Doctors paid by the government to vaccinate people against smallpox.

Key individual: Edward Jenner

Jenner was a trained doctor who had worked as a surgeon and apothecary as well as in a hospital. He used careful scientific method to test and prove his vaccination.

Key institution: Government

The British government played an important role in promoting the vaccine. They provided funding and set up a society to promote vaccination. Later, they made it compulsory for everybody.

Observation and experimentation

Jenner observed the pattern of cowpox and smallpox in dairy maids. He planned his experiment carefully and then repeated it several times so that he could be certain it had not been just a fluke.

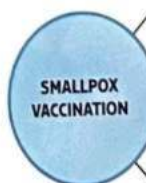


Figure 3.9 Factors assisting Jenner in developing the smallpox vaccine.

Activity

Create a six-picture storyboard to show how Edward Jenner developed the smallpox vaccine.

The impact of the smallpox vaccine

Short term

In the short term, the smallpox vaccine saved many lives. It quickly became very popular overseas and by 1800, 100,000 people around the world had been vaccinated. The French commander Napoleon had his entire army vaccinated in 1805.

The vaccination was slower to become popular in Britain, thanks to the anti-Jenner propaganda promoted by inoculators. Sometimes people still contracted smallpox or died of infection, because the doctors carrying out the procedure mixed up smallpox and cowpox samples or reused needles, and this discouraged people as well. However, after the Royal Jennerian Society had been founded in 1803, 12,000 British people were vaccinated in the space of two years.

Source G

This cartoon was drawn by Cruikshank in 1812. It is titled, 'The cowpox tragedy: scene the last.' The year before, people vaccinated with cowpox had developed smallpox, because of an error with the samples used.



THE COWPOX TRAGEDY
— Scene the Last. —

Long term

By the end of the 19th century, vaccination against smallpox had become normal. Opposition continued throughout the century, but the number of people saved made it clear that the method worked. The number of smallpox cases fell dramatically from 1872, when the government started to enforce compulsory vaccinations – this meant that everyone had to be vaccinated for smallpox.

Extend your knowledge

Smallpox today

In the very long term, it could be argued that Jenner was responsible for the end of smallpox. In 1979, The World Health Organisation announced that the disease had been completely wiped out. This would not have been possible without Jenner's early work.

Jenner had shown that a vaccine could be used to stop smallpox from spreading. His work inspired other scientists, like Pasteur and Koch, to search for vaccinations for other diseases. However, there were no other vaccinations discovered that worked in the same way as the smallpox vaccine. This was a one-off, so scientists were unable to develop other vaccines based on Jenner's method.

Exam-style question, Section B

Explain why there was rapid change in the prevention of smallpox after 1798.

You may use the following information in your answer:

- inoculation
- the government.

You **must** also use information of your own.

12 marks

Exam tip

As well as explaining medical breakthroughs in your answer, in order to show what changed you should compare with the situation before Jenner's research.

New approaches to prevention: the development and use of vaccinations

People still generally believed that the best way to avoid dying from a disease was not catching it at all. People had begun to have new ideas about cures – however, most of these were still not effective on patients who were suffering from a disease. Therefore, scientists continued to focus on prevention and developed the idea of the **vaccination**.

Pasteur presented his case for the germ theory of infection in 1878, after publishing his Germ Theory in 1861. He theorised that microorganisms were responsible for disease. He admired the work of Jenner and started to look for vaccines that would tackle lots of diseases. However, Jenner's work had been the result of observations and experiments, rather than tackling the specific microbe. It would not be possible for other vaccines to work the same way. Pasteur realised that vaccines could only be developed once the germs causing that specific disease had been identified.

Pasteur's first effort at a vaccine was for **chicken cholera**. He identified the germ causing the disease and set about developing a vaccination against it.

In 1879 Pasteur proved that a weakened strain of the disease worked to vaccinate the chickens. He continued with his work, also creating a vaccine for anthrax, another devastating disease affecting animals. He then developed a vaccine for rabies.

Pasteur's work on vaccines involved producing a weakened version of the culture and then treating patients with it. This created an **immune response**, where the body fought off the weakened disease and, in doing so, created antibodies* that prevented the individual from suffering from that disease if the microorganism was encountered again. Pasteur did not know that this was why the vaccine worked, because science had not progressed far enough for him to investigate the method properly yet. However, his methods were clearly effective.

Key term

Antibodies*

Particles inside the body that identify and help to remove germs. The body creates them when first encounters the germ, so that it can fight off the same disease more easily if it comes back.

Until this point, Pasteur's focus had been on animal diseases that caused problems for farmers. His work on vaccinations had little direct impact on disease in humans. However, it inspired research among other scientists, who wanted to find other vaccines for a wide range of human diseases.

Koch's work isolating the microbes that caused specific diseases (see page 71) was very important in developing new vaccines. For example, in 1890, Emil von Behring developed a vaccine against tetanus and diphtheria.

By 1900, scientists all over the world were busy isolating microbes and developing vaccines, thanks to the work of Pasteur and Koch. This was a significant breakthrough in the **prevention** of disease.

Activities

- 1 Draw a storyboard to explain how vaccines were developed.
- 2 As a method of prevention, vaccines were more effective than anything that had been tried in previous centuries. What factors made the development of new vaccinations possible?

The Public Health Act, 1875

Alongside the new scientific methods of prevention, a great deal was also being done to improve living conditions in Britain, particularly in the larger cities. In c1700, the government had little interest in improving conditions in cities. They had a **laissez-faire*** attitude and believed that it was not their responsibility or right to interfere in the way that people lived.

Key word

Laissez-faire*

This French term means 'leave be'. It is used to describe governments who do not get involved in the day-to-day lives of their population.

During the 1800s, this attitude began to change. More men now had the right to vote, so the government began passing laws that appealed to the masses. The government knew that if they appealed to normal people, they would be voted into power in future elections.

As well as this, cholera arrived in Britain. The epidemic led to the deaths of thousands of people. The work of John Snow (see page 92) led people to believe that cholera was spread in dirty water, and this theory was backed up by Pasteur's discovery of microorganisms in 1861.

Extend your knowledge

Edwin Chadwick

In 1842, Edwin Chadwick published his *Report on the Sanitary Conditions of the Labouring Classes*. He had spent some time researching among the urban poor, and this book detailed the results of his research. He showed that people living in cities had a much lower life expectancy than people living in the countryside.

Chadwick concluded that this was down to the filthy living conditions in cities. He campaigned for all cities to set up boards of health, who would be responsible for supplying clean water and disposing of sewage.

Chadwick's work did not have much impact on conditions at the time, but it was only one piece of the puzzle. After more evidence emerged, supporting the theory that clean water was vital for a healthy population, the government was more willing to act.

From the 1860s, the government began to take more action to improve living conditions for people in cities.

- In London, 1,300 miles of sewers were built by 1865.
- In Birmingham, slums were demolished.
- In Leeds, a local business obtained a court order to prevent sewage from being drained into the river from which they got their water.

There had been a change in the way people felt about public health. More people began to recognise that it was now everybody's responsibility.

Extend your knowledge

The first Public Health Act, 1848

The aim of the first Public Health Act was to improve the sanitary condition of towns in England and Wales by encouraging cities to set up boards of health and provide clean water supplies. However, it was not compulsory, so it did not have much impact on the health of the people. It was not until 1875 that rules were put in place to improve sanitary conditions that were compulsory – they had to be followed.

Developments in understanding...	Factors, c1700–c1900	Factor
CAUSE	• Germ theory. • The development of work on identifying microbes.	• Role of technology (microscopes). • Role of science of chemistry. • Role of individuals.
TREATMENT	• Better hospitals and nursing thanks to the work of Florence Nightingale. • Improvements in surgical treatment, because of anaesthetics and antiseptic surgery.	• Role of individuals. • Role of science of chemistry.
PREVENTION	• Development of vaccinations begun by Edward Jenner. • Improved water supply and drainage, with two Public Health Acts in 1848 and 1875.	• Role of individuals. • Role of Government.

In response to this change in attitude, the government passed the **second Public Health Act** in 1875.

City authorities had to follow the rules it set out. The responsibilities included:

- providing clean water to stop diseases that were spread in dirty water
- disposing of sewage to prevent drinking and washing water from becoming polluted
- building public toilets to avoid pollution
- employing a public officer of health to monitor outbreaks of diseases
- ensuring new houses were of better quality, to stop damp and overcrowding

- providing public parks for exercise
- inspecting lodging houses to make sure they were clean and healthy
- creating street lighting to prevent accidents
- checking the quality of the food in shops to make sure that it didn't contain anything that could cause somebody harm. For example, some bakers mixed chalk into flour to make bread whiter.

The government had taken solid steps to prevent the spread of disease – and it worked. The last cholera epidemic in Britain was in 1866–67, and it had a lower mortality rate than previous epidemics, due to some of the new measures that had been put in place.

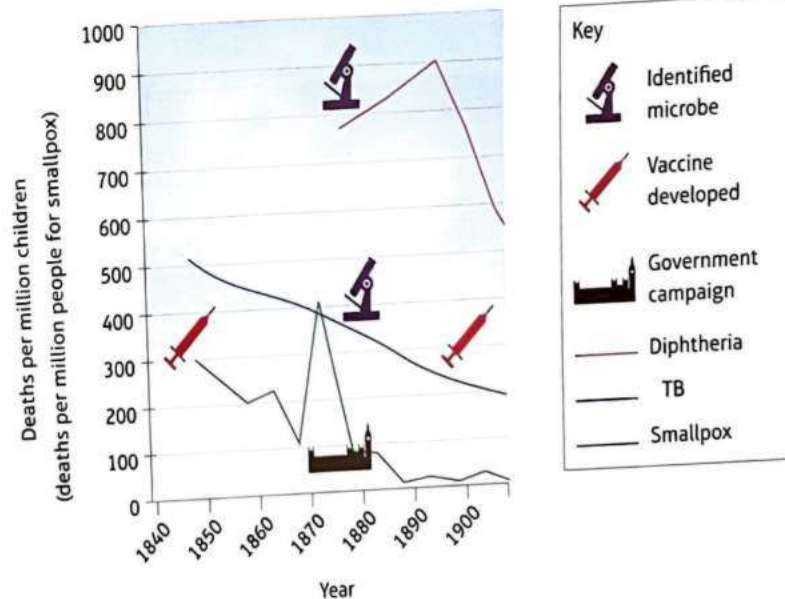


Figure 3.10 Mortality rates of major diseases, 1840–1900.

Summary

- By c1900, the treatment and prevention of disease had changed. This was due to an improved understanding of the cause of disease.
- More hospitals were built, making treatments more widely available.
- Hospitals were cleaner and built to provide a space for recovery, thanks to the work of Florence Nightingale. Nursing had become a respectable profession.
- Herbal and patent remedies were still popular for common illnesses, because few new treatments had been developed.
- Surgical procedures had become a more effective method of treatment as a result of the development of anaesthetics and antiseptics. However, blood loss was still a problem, so surgeons still had to work quickly.
- Scientists had developed a method for vaccinating people against diseases and had begun to develop vaccines for particular illnesses.
- One of these illnesses was smallpox – Edward Jenner proved that catching cowpox prevented people from catching smallpox.
- In the 19th century, the government began to take action to improve public health in cities. This was a result of a better understanding of the link between dirty conditions and disease and it led to a healthier population.

Checkpoint

Strengthen

- S1 What changes and continuities had there been in the treatment of disease by c1900?
- S2 Describe the actions that Florence Nightingale took to improve hospital conditions in Britain.
- S3 Identify the steps that led to improvements in surgical treatment during the 19th century.
- S4 Create a flow diagram to show the different stages in the way people dealt with smallpox, in the period c1700–c1900.
- S5 What were the short-term and long-term impacts of Jenner's new method of preventing smallpox?

Challenge

- C1 As in the previous period (c1500–c1700), attitudes in society had a big impact on change in treatment. Can you list three ways that people's attitudes encouraged developments in treatment and prevention, and three ways that they held them back?
- C2 Compare this list to the one you made at the end of Section 2.2. Had people's attitudes changed or stayed the same between these two periods? Explain your answer, using examples from the text where possible. If you are not confident about any of these questions, form a group with other students, discuss the answers and then record your conclusions. Your teacher can give you some hints.

3.3 Fighting cholera in London, 1854

Learning outcomes

- Understand how the government tackled the cholera epidemics of the 19th century.

Source A

This cartoon was drawn in 1852, for the magazine *Punch*.



A COURT FOR KING CHOLERA.

Fighting cholera

Cholera was a terrible disease. It caused diarrhoea and sickness that became so bad, the victim would become dehydrated*. It was usually fatal: sufferers would die between two and six days after falling sick. As the sufferer became dehydrated, their blood would become thicker, rupturing blood vessels under the skin. This turned the skin blue, so cholera was nicknamed 'the blue death'. It was spread through person-to-person contact, or water contaminated with the faeces of a sufferer.

Cholera did not arrive in Britain until 1831. It spread quickly across the country. It arrived in London in February 1832 and there were 5,275 deaths in the city by the end of the year. Cholera mainly affected the poorest people. There were lots of cases in slum dwellings, as well as in workhouses, prisons and asylums. However, wealthier districts were not immune. As had been the case with the plague two centuries earlier, doctors found it impossible to treat. There were three further severe epidemics across the country in the following decades.

Key term

Dehydrated*

When the body does not have enough water to keep the organs working properly.

Activity

Look carefully at the cartoon. List all the threats to health you can see in the picture.

Year of epidemic	Total cholera-related deaths in England and Wales
1831–32	21,882
1848–49	53,293
1853–54	20,097
1865–66	14,378

Attempts to prevent the spread of cholera

Some steps were taken to try to clean up the filthiest areas of the cities and so prevent the spread of cholera. The belief that miasma and rotting material caused disease was still widespread, so local councils and populations turned their attentions to the mess in which they were living. The government encouraged cities to set up boards of health and provide clean water supplies. However, this did not have a great effect on people's living conditions.

Source B

This letter was printed in *The Times* in 1849, during the second cholera outbreak. It was written by a group of residents of Soho, London.

Sir, May we be and beseech your protection and power. We are Sir, as it may be, living in a wilderness, so far as the rest of London knows anything of us, or as the rich and great people care about. We live in muck and filth. We ain't got no privies [toilets], no dust bins, no drains, no water-supplies, and no drain or sewer in the whole place. The Sewer Company, in Greek Street, Soho Square, all great, rich powerful men take no notice whatsoever of our complaints. The stench of a gully-hole is disgusting. We all of us suffer, and numbers are ill, and if the cholera comes Lord help us all.

John Snow

John Snow was a surgeon who had moved to Soho in 1836 and had become London's leading anaesthetist. It was Snow who gave Queen Victoria chloroform during the birth of Prince Leopold in 1851. He was popular and well-respected.

Source C

This cartoon was published in *Punch* in 1858. In it, the River Thames is offering his 'children' to London – the diseases diphtheria, scrofula (a type of tuberculosis) and cholera.



Snow observed cholera during the epidemic of 1848–49. He wrote up his theories in *On the Mode of Communication of Cholera*. In it, he suggested that:

- cholera could not be transmitted by a miasma, because it affected the guts, not the lungs
- drinking water was being contaminated by the cholera-ridden faeces being disposed of in the city's drains.

Snow concluded that cholera was transmitted by dirty drinking water.

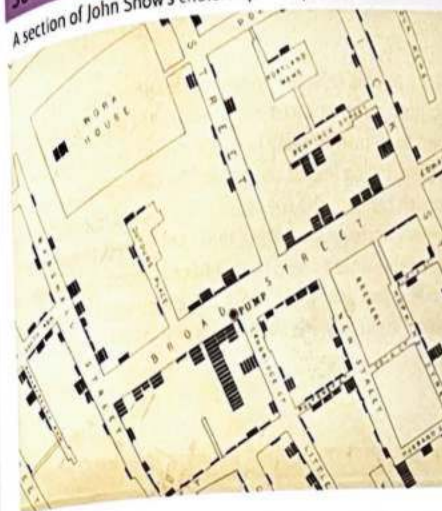
The 1854 epidemic

In August of 1854, cholera broke out in Soho, where Snow lived. This prompted Snow to investigate the 93 deaths in his local area.

Snow created a spot map to show where the deaths had occurred in the area around Golden Square and Broad Street. He took a street map and drew spots onto it to represent the deaths that had taken place (see Source D).

Source D

A section of John Snow's cholera spot map, 1854.



After looking at the map, John Snow realised that there was a pattern: the number of deaths seemed to be centred around the water pump on Broad Street.

To Snow, it was clear that the water pump was the source of the infection. He removed the handle from the pump, preventing locals from pumping water, and the cholera outbreak went away as quickly as it had arrived.

Later inspections of the well underneath the water pump revealed that it was extremely close to a cesspit* – less than one metre away. Although the cesspit had a brick lining, it had cracked, meaning waste from the cesspit was seeping into the well and spreading cholera.

Key term

Cesspit*

A pit for storing sewage or waste.

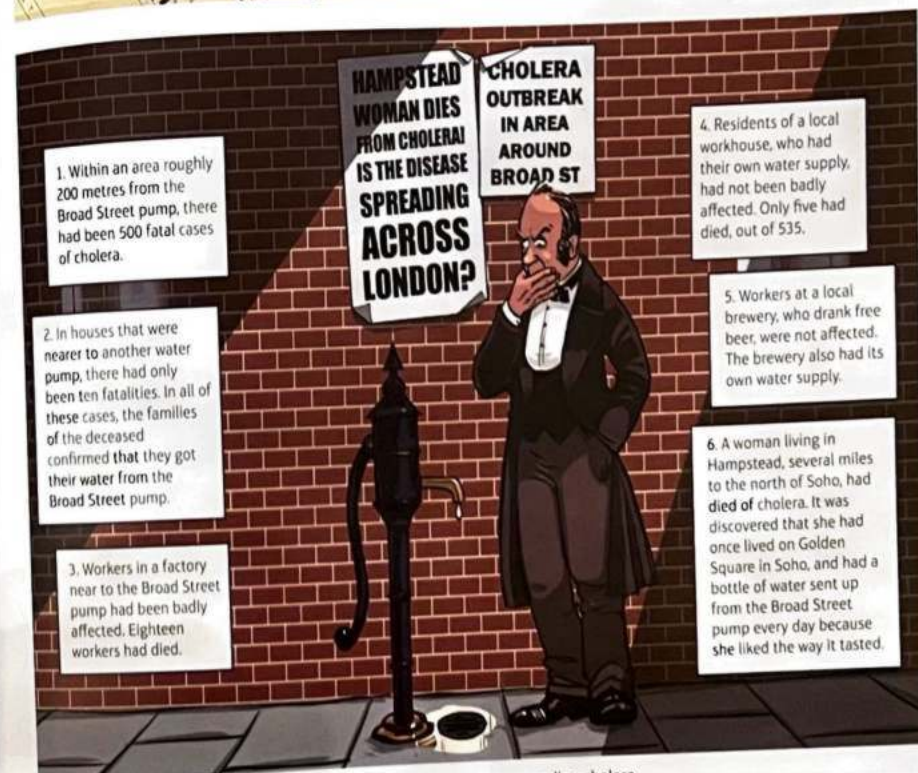


Figure 3.11 How John Snow proved that the Broad Street pump was spreading cholera.

The impact and significance of John Snow and the Broad Street pump

In 1855, Snow presented his findings to a House of Commons committee. He showed the evidence that he had gathered, which proved that cholera was transmitted by dirty water. He recommended that the government start making massive improvements in the sewer systems of London. By doing this, Snow argued, another cholera outbreak could be avoided.

The government did eventually agree to invest in a new sewer system, which was planned by Joseph Bazalgette and completed in 1875. However, this was not just due to the work of John Snow. An unusually hot and dry summer in 1858 had caused 'The Great Stink'. The Thames was low and the stench of the exposed sewage

on the riverbanks heating up and steaming gently in the sun became terrible. This nudged the government into action and the work on the new sewers was begun in 1860.

Many people rejected Snow's work. Other scientists pointed out that cases would still occur among people who lived further away from the pump, even if they were drinking less of the water. The General Board of Health clung to the theory of miasma and rejected Snow's findings. Admitting that cholera was present in the water would mean having to take steps to provide clean water, which was going to be very costly – and, the Board argued, there was no scientific proof that it would work.

Although Snow had plenty of practical evidence to show that cholera was spread in water, he had no scientific evidence to show what caused the disease. It would be another seven years before Pasteur published his Germ Theory – three years after Snow had died – and another 30 years before Koch finally isolated the bacterium that caused the disease.

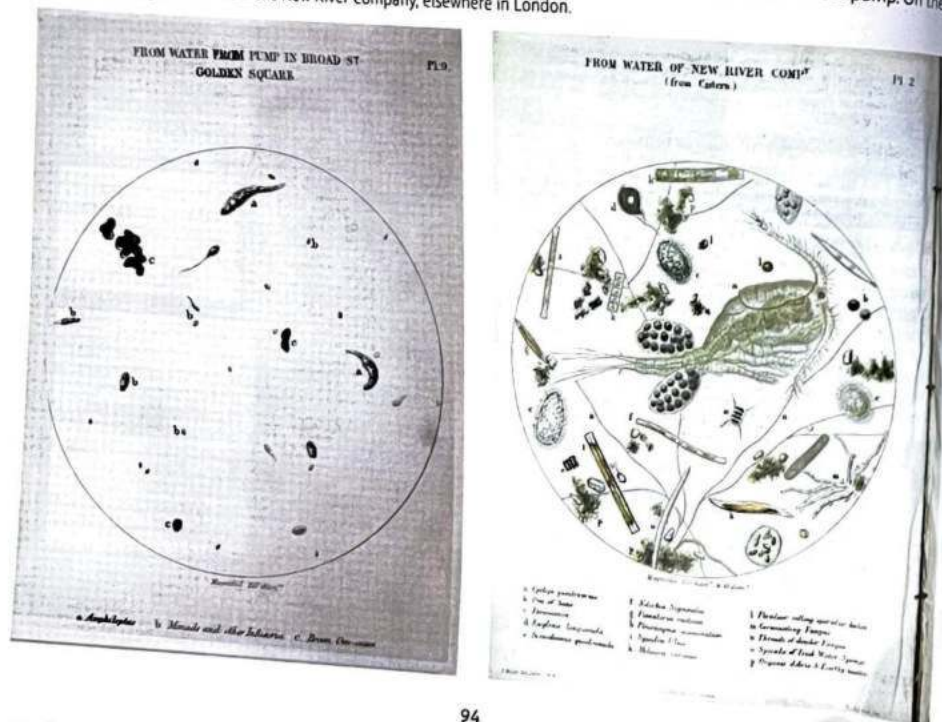
Therefore, in the short term, Snow's work had an immediate impact on the residents of Soho Square, many of whom avoided cholera thanks to his removal of the pump handle. However, his impact outside of this area was very limited. It was not until much later that the importance of clean water was accepted.

Preventing cholera: the role of individuals and institutions

Role of the government	Role of the individual: John Snow
Encouraged local councils to clean up their cities and provide clean water.	Observed the pattern of cholera cases.
Listened to John Snow's evidence about cholera.	Designed an experiment to prove that cholera was caused by dirty water.
Arranged for a new sewer system to be built in London.	Prevented residents from drinking the water.
Eventually passed the 1875 Public Health Act to force other cities to clean up.	Presented his findings to the government.

Source E

These two drawings accompanied the General Board of Health's report on the cholera epidemic, which was published in 1855. The Board had rejected Snow's theory that cholera was spread through water. Here, they attempt to prove their theory by offering two drawings of water magnified 200 times. On the left, a drawing of water taken from the Broad Street pump. On the right, a drawing of water from the New River Company, elsewhere in London.



THINKING HISTORICALLY Change and continuity (3b)

Significant to whom?

Historians are interested in different aspects of the past, and ask different questions. The interest of the historian is a very important factor in their decision about what is significant and what is not.

Historian's focus	Social historian	British historian	Scientific historian
Title of investigation	How did medical developments change living standards in Britain during the 19th century?	How did British scientists change our understanding of the causes of disease?	What role did science and technology play in the development of medicine?

Changes and events during the 19th century

In 1861, Pasteur identified that germs caused decay.	The British government passed laws to make cities cleaner and protect people's health.	In 1865, Lister developed a carbolic spray to make surgery safer.
In 1854, John Snow made a link between cholera and dirty water.	Better microscopes enabled scientists to see microbes and link them with diseases.	In 1797, Edward Jenner developed a vaccination against smallpox.
Robert Koch developed methods to allow bacteria to be grown and observed more easily.	The Enlightenment meant that people were more interested in looking for rational explanations for disease.	In 1870, John Tyndall theorised that dust particles carried germs that caused disease.

For each historian, make a diagram to show the relative significance of the changes and events that would interest them. Write the historian in the middle of the page and then add the events and changes that would interest them: the more significant, the closer to the historian.

- 1 Look at the change or event of most interest to the social historian. How important is it to the other historians?
- 2 Why would the work of Joseph Lister be of interest to the social historian and the scientific historian?
- 3 Would the British historian include the work of Pasteur or ignore it? Explain your answer.
- 4 How important is it for the reader to understand the interest and focus of the historian when reading their work?

Learning outcomes

- Understand how advanced understanding of genetics and lifestyle choices affect health.
- Understand how improvements in technology have helped to improve diagnosis.

By 1900, Germ Theory had been around for nearly 40 years, and microbes had been clearly linked to outbreaks of disease. People finally understood what caused common diseases, such as cholera and diphtheria.

Because of this, throughout the 20th century, doctors no longer referred to miasmata, the Four Humours or the supernatural when diagnosing illness. This is the first period in which doctors were working solely with solid scientific discoveries. For the first time, they had more than just ideas about what caused diseases and illnesses – they had solid, evidence-based, knowledge.

At the start of the 20th century, diagnosis was something that happened between a doctor and a patient. The doctor would observe the patient and consider the symptoms. He would consult medical textbooks and diagnose the disease based on this knowledge.

During the 20th century, there was a move towards laboratory medicine, with more examination of **samples**. These samples might include skin or blood, or more detailed samples of flesh gathered from the patient in a procedure called a **biopsy**. The samples would be examined by medical scientists in a laboratory, using microscopes and other technology. In addition, the patient might be x-rayed to allow doctors to see what was going on inside the body, and the pictures produced would be examined by doctors looking for the cause of the disease or illness. All of this additional information meant that diagnosis was, and is, now more accurate. The exact microbe causing a disease can be identified and targeted.

Therefore, the biggest change in diagnosis in the 20th century was that, instead of being based on symptoms, and the experience and knowledge of the doctor, it was now based on medical testing.

Advances in understanding: the influence of genetic and lifestyle factors on health

The science of genetics

In 1900, it was clear to scientists that microbes did not cause all illnesses and diseases. For example, some babies were born with conditions that appeared to have developed in the womb.

This fact continued to stump doctors for the first half of the 20th century. It was clear that there was still a piece of the puzzle missing, related to the way that children inherited certain traits from their parents and how that related to hereditary diseases*.

Key term

Hereditary diseases*

Hereditary diseases are caused by genetic factors. This means that they can be passed on from parents to their children, or other descendants.

Early research into genetics

By 1900, a German scientist, Mendel, had theorised that genes come in pairs, and one is inherited from each parent. These were known as the **fundamental laws of inheritance**. Unfortunately, he did not have scientific proof that his laws were correct. Microscopes were not yet powerful enough to be able to identify gene pairs.

Timeline

Early work on genetics

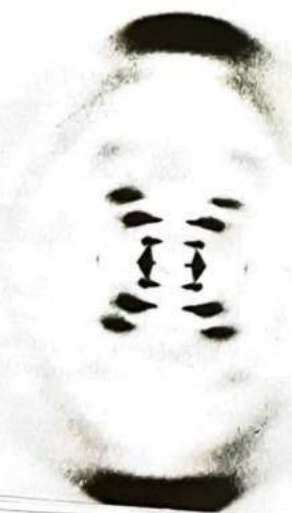
1902 Archibald Garrod, an English doctor, theorises that hereditary diseases are caused by missing information in the body's chemical pathways.

1951 At King's College in London, Rosalind Franklin and Maurice Wilkins create images of DNA using x-rays (see Source A).

1941 US scientists George Beadle and Edward Tatum prove Garrod's theory.

Source A

Rosalind Franklin took this x-ray photograph of DNA in 1951 whilst working alongside Maurice Wilkins.



By 1951, scientists knew that characteristics were passed down from parents to children, as children often look like their parents. They theorised that a substance in human cells passed on this information from one person to the next. This substance also passed on a variety of hereditary diseases. However, it was not until 1953 that technology finally made it possible for scientists to find the missing piece of the puzzle: DNA*.

Key term

DNA*

Short for deoxyribonucleic acid, DNA carries genetic information from one living thing to another. DNA information determines characteristics like hair and eye colour.

Watson, Crick and the discovery of the human gene

James Watson was an American biologist. Francis Crick was an English physicist. In 1953, they were both working at Cambridge University, where they shared an office. Even though neither men were investigating DNA, they both had a strong interest in researching and finding out more about human biology.

Crick and Watson saw the x-rays provided by Franklin and Wilkins (see Source A). They built their model of DNA and shared it with Franklin, who made a correction based on her x-rays. Wilkins also shared clearer photographs that they had managed to take of the DNA. Due to this additional input, Crick and Watson were able to solve the puzzle of the structure of DNA. They discovered that it was shaped as a double helix, which could 'unzip' itself to make copies.

Watson and Crick published their paper in April 1953. Crick suggested that they had discovered the secret of life. Understanding the shape of DNA meant that they could now begin to look at its structure and identify the parts that caused hereditary diseases.

The mapping of the human genome*

Once the structure of DNA was understood, teams of scientists began to break it apart to understand how it worked. All the information that builds a person is stored in their DNA.

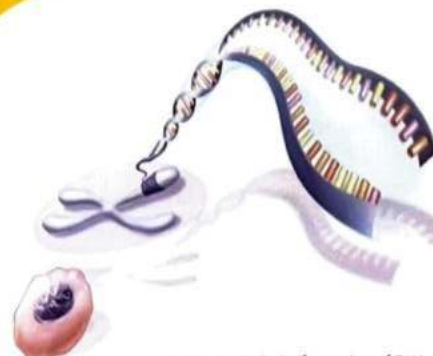


Figure 4.1 An image of the double helix formation of DNA. DNA is stored in every human cell.

Understanding that information – mapping the DNA's code – was vital to helping scientists understand the cause of genetic diseases, such as haemophilia*.

The **Human Genome Project** was launched in 1990. It was originally led by James Watson himself. For a decade, 18 teams of scientists all over the world worked together to decode and map the human genome. Even though hundreds of scientists were working towards this goal, they did not complete the first draft until 2000.

Once the human genome was mapped, it then became possible for scientists to use this blueprint of human DNA to look for mistakes or mismatches in the DNA of people suffering from hereditary diseases.

Source B

In this 2013 news article, Angelina Jolie explains why she chose to have a mastectomy.

Angelina Jolie bravely reveals she has had a preventive double mastectomy after tests showed an 87% chance of contracting breast cancer.

The Hollywood actress is healthy and made the decision to undergo the procedure after discovering she carries the BRCA1 cancer gene.

Angelina said: "My doctors estimated that I had an 87% risk of breast cancer and a 50% risk of ovarian cancer. I made a decision to have a preventive double mastectomy."

The surgery was successful and doctors believe Angelina's chances of developing breast cancer have reduced to less than 5%.

For example, scientists have now been able to identify a gene that is sometimes present in women who suffer from breast cancer. Although they cannot use this knowledge to **treat** breast cancer, women now have the opportunity to **prevent** this disease by identifying their risk of developing the disease and then having a mastectomy*. A famous example of this is the actress Angelina Jolie, who had herself tested for the gene because her mother had died of breast cancer.

Key terms

Genome*

The complete set of DNA containing all the information needed to build a particular organism. In humans, this is more than three billion DNA pairs. It is unique for every human being, except identical twins.

Haemophilia*

A genetic disease passed from parent to child that stops blood clotting. Sufferers from haemophilia must be careful, as an open wound will not heal correctly. Famously, many of Queen Victoria's ancestors suffered from haemophilia.

Mastectomy*

Surgery during which a person has one or both of their breasts removed.

Factors helping the development of genetics

Technology

Discovering the shape of DNA, understanding how it works and then mapping the individual genes has been made possible through improvements in technology. Advances in microscopes and the ability to produce higher-powered images enabled scientists to identify the DNA and then start to examine how it is formed.

The electron microscope was first developed in 1931 by a German physicist, Ernst Ruska, and electrical engineer, Max Knoll. Within two years, they had built a model that was able to magnify more than any of the optical microscopes that scientists had been using up to that point.

Electron microscopes work by using a beam of electrons to illuminate the sample being examined, instead of regular light. This allows for a much more powerful magnification. An optical microscope that uses visible light can clearly magnify a sample up to 2,000 times; an electron microscope can produce a clear image up to 10,000,000 times magnified.

Science

Understanding DNA required a lot of collaboration on the part of the scientific community. The Human Genome Project was an example of a new kind of 'big science' – thousands of scientists from all over the world collaborating to solve the same puzzle. All the data produced from the study was made public, so that it could benefit as many people as possible.

The impact of the science of genetics

A better understanding of DNA and how each part of the genome affects the body has helped scientists to recognise genetic disorders, such as Huntington's and Down's Syndrome. These disorders are caused by missing information in the genome: if that information can be put back in by scientists, this could theoretically lead to a treatment in some cases.

However, this is **not a current treatment**. A good understanding of genetics has helped doctors to better understand what causes diseases and illnesses, but the science is not yet at the stage where treatments of this nature are widely available for many diseases.

Activities

- 1 Create a chain of paper people. Each person should be a different figure that was important in the discovery of DNA. How many people can you make in your chain?
- 2 Explain why the discovery of the shape of DNA was so important for scientists.
- 3 Which factor was more important in the development of our understanding of genetics: science or technology? Write a short paragraph explaining your opinion.

Lifestyle and health

During the 20th century, we have gained a better understanding of the impact of lifestyle choices on the body and how these are linked with diseases and illnesses.

Smoking

Smoking became more popular from the 1920s. It was associated with being young and free. By the 1950s, doctors had started to notice a worrying rise in the number of men suffering from lung cancer, and this was linked with smoking (see page 122).

Doctors now recognise that smoking is associated with an enormous variety of diseases, including high blood pressure, a wide variety of cancers (including lung, throat and mouth), heart disease, and even gum disease and tooth decay. Smoking is the biggest cause of preventable diseases in the world. It is even dangerous to people who inhale the smoke second-hand. Studies show that children exposed to second-hand smoke are more likely to develop asthma than those who do not.

Diet

Due to the Theory of the Four Humours, our medieval ancestors believed that what we ate had a huge impact on our health. Although nobody believes this theory anymore, we now recognise that what you eat (and how much of it) has a huge impact on your health – but in very different ways to what was suggested in the Middle Ages.

Most people are probably familiar with the usual advice about a healthy diet: plenty of fresh fruits and vegetables, and most other things in moderation. Two particularly important substances when it comes to health are **sugar** and **fat**. Too much sugar can cause the health to develop type 2 diabetes. This is an incurable condition where the body is not able to process sugar in the bloodstream. Too much fat can lead to heart disease.

Not getting the right amount of nutrients or not eating enough at all can also cause health problems.

The influence of other lifestyle factors

- **Drinking too much alcohol**, either through binge drinking or drinking a lot over a long period of time, can lead to liver diseases and kidney problems.
- People now recognise that sharing bodily fluids with other people, either through **intravenous drug taking** or **unprotected sex**, can lead to the spread of certain diseases.
- The fashion for **tanning**, either naturally or using sun beds, has led to a rise in the number of cases of skin cancer worldwide.

Activities

- 1 Create flashcards to show different lifestyle factors and the disease that they may cause. Write the factor on one side and the associated illnesses on the other.
- 2 Using your flashcards, write an advice leaflet to explain threats to health.
- 3 How is your leaflet different from a leaflet that might have been made for people in the 15th century?

Improvements in diagnosis: the impact of the availability of blood tests, scans and monitors

New methods of diagnosis

The development of machines and computers has enabled doctors to have a better understanding of a patient's symptoms than in any previous time. For example, x-rays and CT scans (see Figure 4.2) mean that doctors no longer have to use surgery to diagnose all diseases.

The impact of technology

The enormous leap forward in technology since 1900 has made diagnosing disease much more accurate. This has, in turn, had a huge impact on doctors' ability to treat patients.

New methods of diagnosing disease are being developed all the time.

Source C

This article was published on the *Medical News Today* website in November 2015. It describes the use of a special type of dressing to help diagnose infection in burn patients.

Infections are the primary cause of complications in burn injuries, especially in children. Even a relatively mild hot water scald can readily become infected. Many deaths from burn injuries are due to sepsis [blood poisoning].

... Researchers at the University of Bath, in conjunction with the Healing Foundation Children's Burns Research Centre and the University of Brighton, have created a ground breaking solution to diagnosing wound infection. The team has developed a prototype dressing that changes colour when a wound becomes infected. The wound dressing on an uninfected area displays a discrete circular design. Within four hours of an infection, the colour and pattern change.

Exam-style question, Section B

Explain **one** way in which understanding of the causes of disease and illness was different in c1750 from the present day. **4 marks**

Exam tip

It is helpful to have a 'go-to' connective word to use here, to make sure you clearly demonstrate the change. 'Whereas' and 'however' are both good choices.

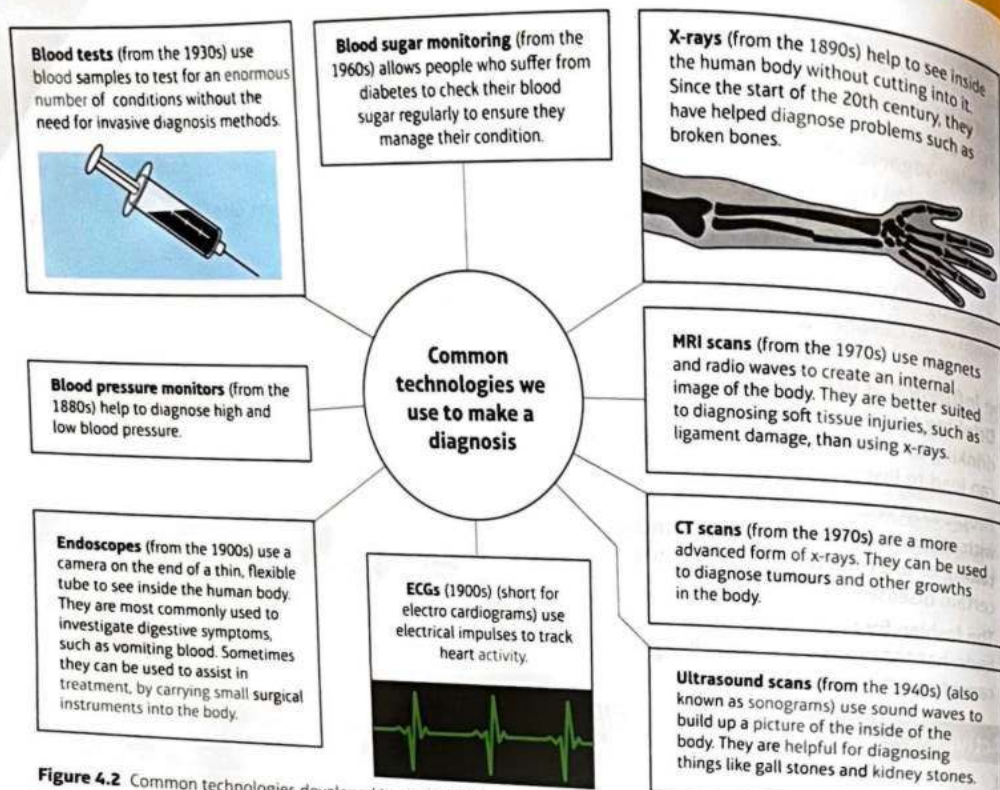


Figure 4.2 Common technologies developed in period 1900 – present to help diagnosis.

Summary

- After 1900, understanding about the causes of disease and illness progressed rapidly.
- By the end of the 20th century, doctors recognised that many factors caused disease.
- The structure of the gene was discovered by Watson and Crick in 1953 and the human genome was mapped by 2000.
- Lifestyle factors such as smoking, a poor diet and drinking alcohol can all contribute to illness and disease.
- New technology has enabled doctors to carry out more detailed diagnoses of their patients.

Checkpoint

Strengthen

- 1 Explain how the mapping of the human genome helped doctors to better understand the causes of disease and illness.
- 2 Describe three ways that technology has led to a better understanding of the causes of disease and illness.

Challenge

- 1 How did changes in the way scientists work lead to an improved understanding of the causes of disease?
 - 2 Explain five examples of the impact of technology on diagnosing the causes of illness and disease.
- If you are struggling, discuss the answers to these questions in groups. Your teacher can give you some hints.

4.2 Approaches to prevention and treatment

Learning outcomes

- Understand the impact of the development of magic bullets, including antibiotics.
- Understand the impact of the NHS on public health.
- Understand how the government tried to prevent certain diseases.

Medical treatments

The first chemical cures: magic bullets

The term 'magic bullet' was used to describe a chemical cure that would **attack the microbes in the body causing disease**, whilst at the same time **leaving the body unharmed**.

In the late 19th century, more microbes responsible for specific diseases were being discovered. This meant that scientists could begin to search for substances to attack and destroy these microbes.

Doctors now understood that the body produced **antibodies** to fight diseases that had previously infected it – this is how vaccines work. The hunt was on for artificial or chemical antibodies that would work in the same way, attacking the infection without harming the body.

The first big breakthrough was made in the treatment of syphilis. Syphilis continued to be a problem throughout the 19th and 20th centuries, and treatment of it had not really improved since mercury treatments of the 16th century. There had been some success with arsenic compounds*. However, it was very difficult to find a form of arsenic that attacked the disease and not the body, as arsenic is poisonous.

Key term

Compound*

A compound is a mixture two or more different elements.

Extend your knowledge

Prontosil

Prontosil was used as a cure for puerperal fever at Queen Charlotte's maternity hospital in London. The death rate for puerperal fever dropped from 20% to 4.7%.

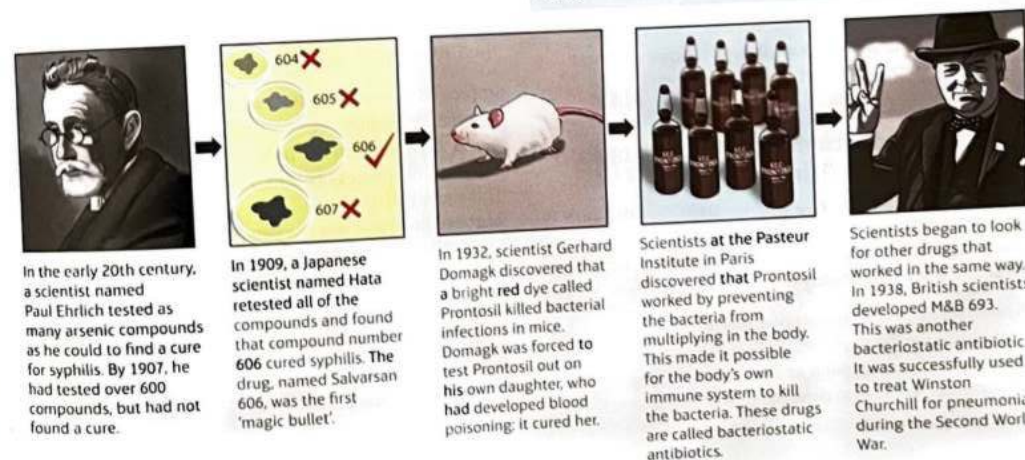


Figure 4.3 The discovery of the first two magic bullets.

The development of antibiotics

The term **antibiotic** is used to describe any treatment that destroys or limits the growth of bacteria in the human body. The first true antibiotic was **penicillin**. Penicillin was different to Salvarsan 606 and Prontosil as it was created using microorganisms, not chemicals. Penicillin was isolated from a mould sample by Alexander Fleming in 1928 and developed into a usable treatment by Florey and Chain in 1940 (see pages 119–121).

Inspired by the discovery of penicillin, other scientists investigated moulds and fungi in the search for more antibiotics. Streptomycin was discovered by American scientist Selman Wakston in 1943. This antibiotic was so powerful that it was even effective against tuberculosis, which had previously been thought to be incurable. During the 1950s and 1960s, even more antibiotics were discovered.

Research into the development of new antibiotics has not stopped. In the 21st century, pharmaceutical companies continue to test substances to develop new antibiotics. This is because some bacteria have developed a resistance to the antibiotics we already have. If new treatments are not developed, scientists fear that the old antibiotics will become totally ineffective against diseases that we think we have beaten, such as septicaemia (blood poisoning).

Therefore, in the short term, antibiotics have been a miracle cure for a variety of diseases. However, their long-term impact has yet to be measured.

The impact of science and technology on advances in medicines

As with diagnosis, the way that we treat diseases now is almost unrecognisable from the way that people treated them before 1900. This is largely due to huge **advances in science and technology**.

Scientists have now developed medicines that pinpoint and treat specific diseases. Even if they are unable to cure some diseases, such as diabetes and lung cancer, treatments have been developed to help patients

Source A

This article, entitled 'Too much of a good thing', was published in *The Telegraph* in 2013, by Joe Shute.

Antibiotics are no longer effective. The drugs that have transformed life and longevity and saved countless millions since penicillin was discovered by Sir Alexander Fleming in 1928 now saturate [fill] every corner of our environment. We stuff them into ourselves and our animals; we spray them on crops, dump them in rivers, and even – as emerged at a meeting of science ministers from the G8 last year – paint them on the hulls of boats to keep off barnacles.

As a result, an invisible army of super-resistant bacteria has evolved, one that is increasingly claiming lives – currently more than 25,000 a year in Europe alone...

Many leading scientists and doctors and politicians are freely adopting the language of global catastrophe. Infections such as tuberculosis and septicaemia – the scourge of earlier centuries – are once again killing us at frightening rates. We have used, or are using, our so-called drugs of last resort.

manage their illness. Scientists are now able to identify the causes of disease in most cases, because they know what they are looking for – for example, a microbe, a tumour or an unusual gene. This is a huge change from the 19th century.

Improved scientific understanding has also led to better testing and trialling of new treatments before they are given to patients. In the past, drugs did not have to go through this process before being used to treat disease. This meant that mistakes were made. The most famous of these mistakes was the use of the drug thalidomide in the 1960s to treat pregnant women suffering from morning sickness. The drug caused birth defects.

Now, it takes several years for a new drug to be trialled thoroughly before being used. This slows down progress but ensures drugs are safe for everybody.



Change and continuity (4b&c)

The bird's eye view

Development	Example of immediate changes	Example of change in the medium term	Example of change in the long term
The development of antibiotics		Scientists developed a way of making antibiotics, which meant they could be modified to attack particular diseases.	Antibiotics are still used widely to treat diseases and infections. However, there are an increasing number of diseases that are resistant to them.

Imagine you are looking at the whole of history using a zoomed-out interactive map like Google Maps™. You have a general view of the sweep of developments and their consequences, but you cannot see much detail. If you zoom in to the time when antibiotics were first developed, you can see the event in detail but will know nothing of its consequences in the medium or long term.

Look at the table above and answer the following:

- 1 What were the immediate changes brought about by the development of antibiotics? Write down at least two changes that could complete that column in the table.
- 2 Look at the medium-term changes and the long-term changes. How are they similar? How are they different?
- 3 Work in groups of three. Each take the role of the teacher for one of the above (the immediate changes, the medium-term changes or the long-term changes). Give a short presentation to the other two students in your group, explaining the key changes over your timescale. They may comment and ask questions.

Answer the following individually:

- 4 How was your explanation of change different to the other explanations? Write a short paragraph using examples from what you and the rest of your group said.
- 5 What happens to the detail, particularly of the medium- and short-term changes, when you zoom out to look at the long-term changes?
- 6 What are the advantages and disadvantages of zooming in to look at a specific time in detail?

New technology has made it easier to create and provide drugs to treat diseases.

- **Mass production of pills** has made the distribution of drugs much easier.
- **The development of capsules**, which dissolve in the stomach to release the drug, means taking drugs to treat disease is easier.
- **Hypodermic needles** allow the precise dose to be introduced directly into the bloodstream.
- **Insulin pumps** for young people suffering from diabetes deliver insulin without the need for injections.

Activities

- 1 Draw a timeline to show when the different drugs described in this section were developed. Label each one with details, such as who was responsible for its development and which diseases the drug fights.
- 2 Describe how science and technology assisted in the development of new chemical treatments.
- 3 Individuals like Ehrlich and Domagk inspired British scientists to look for new treatments, while science and technology made them possible. Which factor do you think has been the most important in the development of treatments post-1900?

Medical care: impact of the NHS

Phase one: Improved access to care

The National Health Service (NHS) was launched in 1948 by the government. Its aim was to provide medical care for the entire population of Britain. It was paid for by National Insurance contributions, taken from wages in the same way as tax. It was the largest government intervention in medical care. The new NHS took over existing hospitals and medical services.

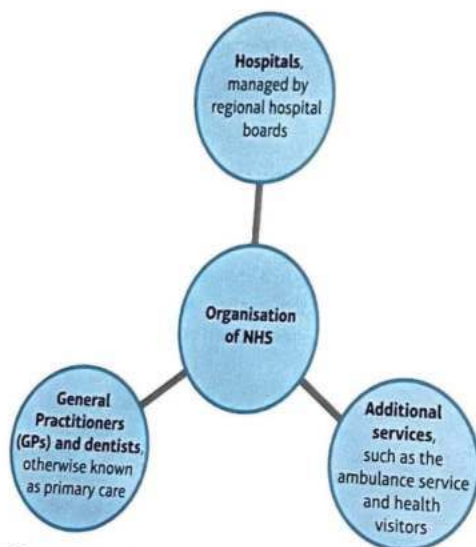


Figure 4.4 The three parts of the NHS in 1948.

The government aimed to provide the same level of service for everybody in the country, no matter how rich or poor they were.

For example, workers earning under a certain amount were already entitled to medical care through the 1911 National Insurance Act. However, this did not extend to women who were at home raising their families. After 1948, women were able to get treatment for painful conditions like varicose veins, which might previously have been left untreated. Similarly, children could be treated for minor problems before there was any lasting damage.

To begin with, hospitals were not much changed by the launch of the NHS. Post-war Britain did not have a lot of money to spend on medical care. The government was now responsible for 1,143 voluntary hospitals and 1,545 city hospitals, which was a huge undertaking. Many of the hospitals had been built in the 19th century and desperately needed updating. There were also more hospitals in London and the South East than there were across the rest of the country.

Similarly, many GP surgeries were in need of modernisation, as well as the standards of the GPs themselves. Studies in the 1950s suggested that up to a quarter of GPs were not satisfactory. With little time or opportunity to keep up-to-date with medical developments, many GPs were behind the times. The problem was made worse by the NHS, because more and more people began visiting GPs. Waiting times increased and appointment times decreased.

Therefore, **access had improved** because the NHS was available to all. However, **provision had not improved** in the short term. During the 1960s, however, the government implemented changes to improve the NHS. Plans were made to ensure that hospitals were evenly spread across the whole country. In 1966, a GP's charter was introduced, which encouraged GPs to work in group practices and gave them incentives to keep up with medical developments. The government had to manage the NHS rather than just fund it. This led to improvements in the standard of care.

Phase two: High-tech medical and surgical treatments in hospitals

Hospital treatments have changed a lot since 1900. Treatments that we consider routine today, like hip replacements and blood transfusions, did not exist before 1900. Patients benefit from high-tech treatments that would have been unimaginable in previous centuries. Once the three major problems of surgery – **pain, infection and blood loss** – had been solved, doctors were able to carry out more daring and intrusive surgeries than ever before. The development of new machinery to treat the body and even replace parts of it that had stopped working also improved treatment in hospitals.

There are hundreds of examples of new high-tech medical and surgical treatments being carried out. The table opposite shows a few of the most famous examples.

	New technology	Treatment made possible
Medical treatments	Advanced x-rays	Doctors can now also use x-rays to target and shrink tumours growing inside the body, using a treatment known as radiotherapy . Combined with chemotherapy, this is an effective treatment for many types of cancer.
	Smaller, cheaper machines	Processes like dialysis , where the blood of patients with kidney failure is 'washed' by a machine, and heart bypasses , where a machine performs the functions of the heart, have become more widely available as machines have become smaller and more portable.
	Robotics	Better prosthetic limbs are now produced. This is partly in response to the number of soldiers surviving bomb attacks in recent wars in Iraq and Afghanistan.
Surgical treatments	Microsurgery	The first successful kidney transplant was performed between identical twins in the USA in 1956. This paved the way for transplants of other organs, including lungs (from 1963), and livers and hearts (from 1967). These were made possible by improved surgical techniques, including the use of microsurgery to reattach tiny nerve endings and blood vessels.
	Laparoscopic (keyhole) surgery	Using tiny cameras and narrow surgical instruments, surgeons can now operate inside the body through tiny incisions some distance away from the area to be operated on. This allows for quicker healing and less trauma to the body.
	Robotic surgery	Surgeons can now use computers to control instruments inside the body, allowing for more precise surgery with smaller cuts. Operations can be performed on a tiny scale where precision is of vital importance – for example, in brain surgery.

Interpretation 1

In this interpretation, taken from *Health and Medicine in Britain since 1860* by Anne Hardy (2001), the short-term impact of the NHS is called into question.

The implementation of the NHS by no means resolved the problem of delivering adequate health care to Britain's people, although it did offer a considerable improvement over the combination of private and insurance medicine that had preceded it. The medical services offered under the 1948 Act were pre-eminently providers of treatment for existing illness rather than agents for preventing its development.

Extend your knowledge

3D printers

At the cutting edge of medical research, scientists are attempting to develop a way of creating new organs using 3D printers. However, this is still decades away from being a usable treatment.

Interpretation 2

Former Labour Prime Minister Tony Blair wrote this as part of a foreword to a document about the modernisation of the NHS. It was published in 1997.

Creating the NHS was the greatest act of modernisation ever achieved by a Labour Government. It banished the fear of becoming ill that had for years blighted the lives of millions of people.

The extent of change in care and treatment

Treatment

Looking back to the 17th century, when Thomas Sydenham imagined a world in which each disease would have its own treatment, the change in the way diseases have been treated is immense.

In the 20th century, medical science made a huge leap forward towards Sydenham's dream. In 1900, 25% of deaths were caused by infectious diseases. By 1990, that number had fallen to less than 1%.

In c1900, most people were still taking herbal remedies or patent medicines, such as Beechams, bought from the chemist to treat their illnesses. Now, thanks to advances in science after 1900, there are a wide variety of specific, effective medicines matched with the diseases that they treat.

However, scientists continue to face problems when developing treatments.

- It is very difficult to develop a vaccine against some viruses. A different flu vaccine is available every year, as scientists develop it in response to the most common strain of the flu virus at that time.
- New diseases keep appearing, which do not respond to any chemical treatments that we currently know about. Scientists have to go back to the lab and continue testing compounds in the hope that they might find one that is effective.
- Lifestyle factors have caused an increase in illnesses such as heart disease and cancer, for which there are no certain cures.
- Microbes are living organisms and they have evolved to beat some of the cures doctors have been using. This has led to drug-resistant bacteria, such as MRSA*. Tuberculosis cases are once again on the rise in the UK.

We may not be facing all the same problems as our ancestors when researching treatments for illness and

Key term

MRSA*

A strain of drug-resistant bacteria that is particularly hardy and resistant to antibiotics.

disease, but we face many new ones. Therefore, alternative remedies, such as herbal medicines, acupuncture and homeopathy, are still popular treatments for disease.

Improved access to care

In c1900, most sick people were still cared for in the home by women. Doctors had to be paid and so were only used for serious illnesses.

The situation improved slowly during the first half of the 20th century. In 1919, the government set up the Ministry of Health to help determine the level of health care across the country.

There was rapid improvement in the availability of care outside the home from 1948 onwards. The NHS made medical services free at point of service. This gave everybody access to medical care and treatment.

However, the introduction of the NHS made it clear, once and for all, that hospitals were just for treating the sick. In earlier periods, hospitals had been places for the elderly to rest. This change left a gap in services. Up until the end of the Second World War, elderly people with no family had often lived out the last days of their life in hospitals. This was no longer possible.

Exam-style question, Section B

'Treatment of diseases and care of the sick completely changed after c1800'.

How far do you agree with this statement?

You may use the following information in your answer:

- magic bullets
- the NHS.

You **must** also use information of your own. **16 marks**

Exam tip

Focus on the word 'completely' in this question. Nobody would argue that treatment and care has stayed the same since c1800, so in your answer you should focus on whether it is entirely different or whether there are some things that are the same as previous centuries.

Activity

Draw a table to show the change and change continuity in care and treatment from earlier periods to the end of the 19th century.

Preventing disease

By c1900, many different approaches to preventing disease had been put in place. The government now took responsibility for providing clean water and removing waste. The public understood the importance of these factors, due to the development of Germ Theory. As more people were given the vote, the government paid more attention to what its citizens wanted.

The government has taken significant action to improve the public's health since the beginning of the 20th century. The *laissez-faire* attitude was now behind them. There are two reasons for this.

1 Increased understanding of cause

Now that we understand what causes disease, the government recognises that its intervention can have an impact. Without this understanding, the British government would not have acted in the same way, because they did not know that their intervention could change things.

2 Increased understanding of methods of prevention

Once the causes of disease and health problems was understood, methods of prevention could be tested and introduced. These have included:

- **Compulsory vaccinations:** inspired by the positive impact of the smallpox vaccination, other campaigns were launched in the 20th century.
- **Passing laws to provide a healthy environment:** these include the Clean Air Acts (see page 114) and adding the chemical fluoride to the water supply to help prevent tooth decay.
- **Communicating health risks:** lifestyle campaigns help people to identify and tackle health risks. During times of global epidemics, such as during the 2014–15 outbreak of ebola in West Africa, the government tracked travellers from affected regions and put quarantine measures in place to stop the spread of disease. Communicating risks to the population has become key in preventing disease.

Charities also contribute to healthy lifestyle campaigns. For example, the British Heart Foundation creates adverts encouraging people to protect their heart by giving up smoking, eating less fat and exercising.

New approaches to prevention: mass vaccinations

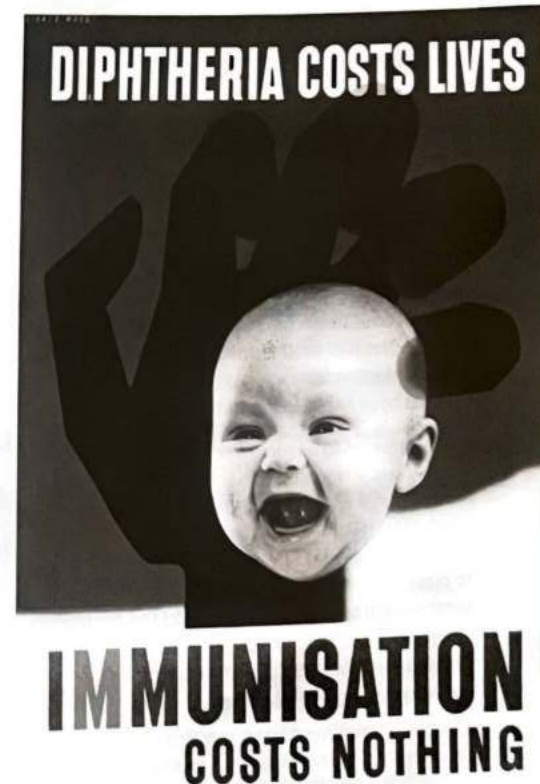
Timeline

Government introduction of vaccinations

1942 Diphtheria	
	1950 Poliomyelitis and whooping cough
1961 Tetanus	
1970 Rubella	1968 Measles

Source B

This poster advertising diphtheria immunisation was published in 1943, one year after the national campaign was launched.



The national vaccination campaign against diphtheria was launched in 1942 – the first of its kind. Before this, local governments were responsible for vaccination campaigns that were not funded by the central government, which meant they were not widespread. Around 3,000 children died each year of diphtheria.

During the Second World War, the government put a national campaign in place to immunise all children against diphtheria. There were fears that the cramped conditions of air-raid shelters during the war might lead to an epidemic. Because of this, infection rates plummeted. By the middle of the century, diphtheria was seen as a disease of the past.

Another significant vaccination campaign was against poliomyelitis (polio). Polio is a very contagious disease that causes paralysis. In the early 1950s, there were as many as 8,000 cases reported every year in Britain. The vaccination was developed in the USA by Jonas Salk and was introduced to the UK in 1956, followed by a more effective vaccination in 1962. As with diphtheria, the number of infections dropped very rapidly. The last case of a person contracting polio in the UK was in 1984.

Some vaccines are aimed at protecting future generations. Rubella, or German measles, is not a life-threatening disease for most people – however, it can be very dangerous if a pregnant woman catches it because it will affect the unborn child.

Other vaccines target diseases that can lead to other diseases. The HPV vaccine, for example, protects women against infection from a sexually transmitted disease that has been linked to cervical cancer.

There continues to be controversy surrounding vaccinations. Many people resent government intervention and choose not to vaccinate their children. A lack of trust in the medical profession has led to fears that vaccines are unsafe. While vaccination is the best way to prevent the spread of dangerous epidemic diseases, there is still freedom of choice to reject this method.

New approaches to prevention: government legislation

The government has passed laws to provide a healthy environment for the population. Examples of these were

the Clean Air Acts of 1956 and 1968. They were triggered by bad episodes of smog in London in 1952.

Smog is very heavy fog caused by air pollution. In an era a time when everybody burned coal to heat their homes, there was a great deal of smoke and soot in the air, particularly in London, where a lot of people lived. Sometimes there was so much pollution that smog would cover the city for days at a time.

Smog is no longer a significant problem in the UK. However, the government continues to pass laws to protect the population from air pollution. An example of this is by limiting car emissions.

Source C

Lavinia Hand was born in London in 1924. During the Great Smog she lived with her husband, George, and their two-year-old son Peter. Here she recollects the conditions.

It was really a very terrible fog. Everybody had coal or wood fires at that time so there was a lot of soot in the air, but it was traffic pollution as well. The air was so dangerous that thousands of people died. We lived in Wood Green and George worked in Finchley. He wore a mask but the smog made our eyes sting as well and the mask didn't help with that. I wouldn't take Peter out in it.

The government have also passed legislation making it illegal to smoke in all enclosed workplaces. It came into force on 1 July 2007 as part of the Health Act of 2006.

Source D

A picture of a London policeman wearing a facemask in the last London smog, in 1962. Pharmacists sold masks like this to help people avoid the negative effects of breathing in the pollution.



New approaches to prevention: government lifestyle campaigns

As well as providing direct intervention to prevent disease, the government also aims to help people prevent disease themselves, by promoting healthier lifestyles. Some examples of their work include:

- advertising campaigns warning against dangers to health, such as smoking, binge drinking, recreational drug use and unprotected sex
- events such as Stoptober, which encourage people to stop smoking for a month
- initiatives encouraging people to eat more healthily and get more exercise, such as the Change4Life campaign (see Source E).

Source E

Change4Life is a Public Health England campaign designed to help families eat well and move more.

Exam-style question, Section B

Explain why there was rapid progress in disease prevention after c1900.

You may use the following in your answer:

- government intervention
- vaccinations.

You **must** also use information of your own. **12 marks**

Exam tip

Remember that all of that knowledge needs to **support** your arguments – avoid simply describing disease prevention.

THINKING HISTORICALLY Cause and Consequence (2a)

The web of multiple causes

Why were doctors able to treat more diseases in the 20th century?

Study these causes that would help historians to explain why doctors were able to treat diseases successfully in the 20th century.

Fleming, Florey and Chain had developed penicillin, which led to the discovery of other antibiotics.	Technology such as x-rays made it easier for doctors to identify the specific causes of disease.	Everybody now understood and accepted that many diseases were caused by microbes.
Scientists had isolated the microbes that caused different diseases.	Chemical cures, such as chemical 'magic bullets', had been developed.	The NHS provided medical care that was free at point of service.

Work in pairs. Take an A3 sheet of paper. You will need to use all of this.

- Write two of the causes on the paper with some space between them.
- Think of anything that connects the causes, draw a line between them and describe the connection by writing along the line.
- Add all the other causes in turn and make as many links as you can with other causes.
- Now add the outcome to the diagram: 'People believed in Germ Theory'. Think carefully about where to place this and how it should be linked into the diagram of causes.

Answer the following:

- Explain why it would be difficult to put the above causes on a timeline.
- Which of the causes have an effect on other causes? Which are linked directly to the outcome?
- Can your diagram help you to work out which causes were more important?
- Write a paragraph explaining why doctors were able to treat diseases successfully in the 20th century. Make sure you mention all the causes in the table. Use the links you have identified between the causes to help you.

Summary

- Penicillin was discovered and then developed into a usable treatment for a wide variety of diseases.
- New technology helped to improve the way that medicines were given to patients.
- The government established the NHS in 1948. This made free medical care available to everybody.
- High-tech treatments, such as organ transplants and radiotherapy, helped doctors tackle diseases.
- Government sponsored campaigns encouraged people to lead healthier lives in order to prevent disease.

Checkpoint**Strengthen**

- Choose three pieces of new technology and explain how they have helped to treat disease.
- What impact did the NHS have on the health of the nation?
- List five methods of preventing disease that have appeared since c1900.

Challenge

- Which factor do you think has had the biggest impact on treatment and prevention since 1900: government or science and technology? Select at least three examples from the text to support your answer.

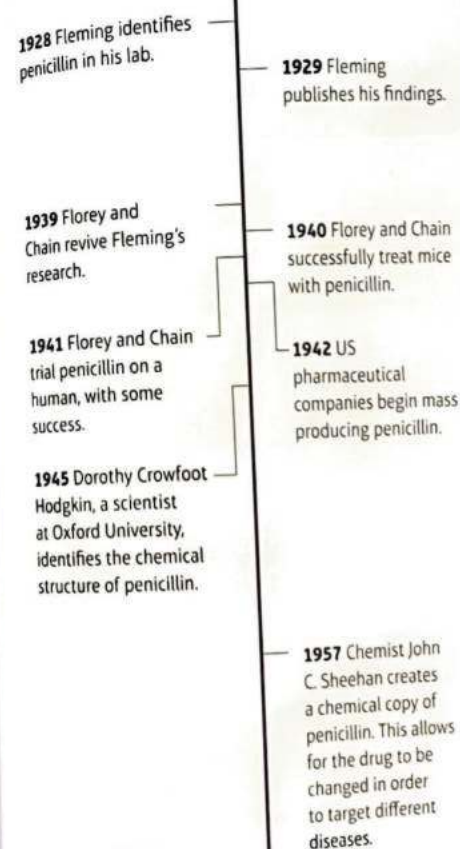
Before writing your answer to question C1, you might find it useful to create a mind map listing your examples.

4.3 Fleming, Florey and Chain's development of penicillin

Learning outcomes

- Understand how Fleming, Florey and Chain discovered and developed penicillin.
- Understand how penicillin is used to treat diseases.

Following the development of the first 'magic bullet' in 1909, the search for effective treatments continued throughout the 20th century. The development of penicillin into a usable drug revolutionised the way that infections were combated and has saved countless lives.

Timeline**The development of penicillin****Alexander Fleming and the discovery of penicillin**

Alexander Fleming was a British doctor working at St Mary's Hospital in London. He had a particular interest in bacteriology and had been one of the first doctors to use the first 'magic bullet' to treat syphilis. During the First World War, Fleming had worked in battlefield hospitals trying to improve treatments for wounded soldiers. He was dismayed at the number of men who died from simple infections.

During the 1920s, Fleming researched substances that might be effective in combating these simple infections. In 1928, he noticed something unusual about his dirty petri dishes: one of them had developed some mould. The mould appeared to have killed off the harmful *staphylococcus* bacteria that had been growing in the dish.

Fleming tested the mould and identified it as penicillin. He was not the first person to notice what it could do: in the Middle Ages, people were aware that mouldy bread had healing properties, and Joseph Lister used it to treat a patient in 1871. However, Fleming published his findings at a time when scientists were actively looking for chemical treatments for disease, and so more notice was taken of it.

Unfortunately, Fleming did not believe that penicillin could work to kill bacteria in living people. His first experiments with the mould showed that it became ineffective when mixed with blood in test tubes in the laboratory, so he did not pursue funding to perform further tests on the mould.

Source A

A photograph of a plaque on the wall of St Mary's Hospital in Paddington, London, commemorating Fleming's achievement.



Source B

A photograph of Alexander Fleming studying mould cultures in his laboratory at the Wright Fleming Institute in London.



Source C

In 1945, Alexander Fleming was awarded the Nobel Prize in Medicine for his discovery of penicillin. He shared this with Howard Florey and Ernst Chain. In this extract from his acceptance speech, Alexander Fleming describes the process of discovering the antibiotic.

In 1928 an accidental contamination of a culture plate by a mould set me off on another track. I was working on a subject having no relation to moulds or antiseptics and if I had been a member of a team engaged on this subject it is likely that I would have had to neglect the accidental happening and work for the team with the result that penicillin would not then have been described and I would not be here today as a Nobel Laureate...

... I isolated the contaminating mould. It made an antibacterial substance which I christened penicillin. I studied it as far as I could as a bacteriologist. I had a clue that here was something good but I could not possibly know how good it was and I had not the team, especially the chemical team, necessary to concentrate and stabilise the penicillin.

Florey and Chain and the development of penicillin

Howard Florey was an Australian pathologist working at Oxford medical school. His colleague, Ernst Chain, had escaped Nazi Germany, where he had been a biochemist.

Florey and Chain were conducting research in the field of antibiotics. As part of this, they were looking for neglected research that might be worth investigating. They came across Fleming's findings and decided that the mould should be tested further. Chain grew the mould in his laboratory and used extracts of it in tests for treatment.

In 1940, Florey and Chain tested their extracted penicillin on infected mice. The results were promising: it looked as though the penicillin was killing the infection. Unfortunately, it was very difficult to produce penicillin in large quantities. The active ingredient in the liquid produced by the mould only represented one part per two million – that meant growing a great deal of mould before it was possible to get started on a human trial.

The scientists set about growing as much penicillin as possible. They used whatever they could to grow the mould: milk churns, bed pans and even a bath tub. By 1941, Florey and Chain had a human patient on whom to try the drug: a local policeman who had developed an infection. He had been scratched by a rose bush and had developed septicaemia – fatal blood poisoning.

Florey and Chain only had a very small amount of penicillin, but they gave it to the policeman anyway. He showed signs of recovery almost straight away. However, there had not been enough penicillin to cure him completely and there was no more available. Florey and Chain collected the patient's urine and extracted leftover penicillin from it. This was then given to the patient and, again, he showed signs of getting better. Unfortunately, this could only be done so many times and the patient eventually died.

Nevertheless, penicillin had proved to be effective in fighting infection in the human body.

THINKING HISTORICALLY

Change and continuity (4a)

Significance

Look at Source C on page 118 and Interpretation 1 above.

- 1 In what ways does Fleming think he was significant in the discovery of penicillin?
- 2 How significant does Fleming feel he was in the discovery of penicillin?
- 3 Compare this to historian Patricia Harding. What significance does Harding credit to Fleming?
- 4 Why do you think these views might differ so greatly?

Interpretation 1

There is some controversy over who should get the credit for the development of penicillin. In this study of the role of science in science education, published in 1996, Patricia Harding sets out an alternative point of view.

Fleming did not develop penicillin. He found it in 1928, extracted it from a culture of [the mould] *Penicillium*, and worked on it for a short time. By 1931 he had abandoned it as an antiseptic for medical use and used it only as an ingredient in culture medium to selectively grow certain organisms. Penicillin was developed therapeutically in 1940 by a group of scientists at Oxford under the leadership of Howard Florey. However, when penicillin made such an impact on the world, Fleming managed to get the credit, and the group at Oxford were ignored by the public. Early accounts propagated [spread] the "myth" of the development of penicillin, but several scientists who knew what had happened later told the real story.

Activity

Discuss Interpretation 1 with a partner. Why does Patricia Harding feel that Fleming should not get the credit for penicillin?

Mass production of penicillin

Florey and Chain proved that penicillin was effective in treating infections, but they were still struggling to mass produce it. They needed a large-scale factory where the penicillin could be grown and extracted on an industrial scale.

Florey first approached British pharmaceutical companies for assistance. Unfortunately, this was during the Second World War and the companies were busy producing materials for the war effort.

However, the USA had not yet joined the war. In July 1941, Florey visited the USA and convinced

pharmaceutical companies to begin penicillin production. The companies started growing the penicillin in beer vats. It was a very slow process – after a year, the US companies only had enough penicillin to treat ten people.

However, once this had begun, the impact of penicillin could be shown. The US government, observing the benefits of the drug, funded 21 pharmaceutical companies to begin mass producing it. British pharmaceutical companies became involved in 1943, when they too started to mass produce the drug. By D-Day, in June 1944, there was enough penicillin available to treat all Allied casualties.

Factors enabling the development of penicillin

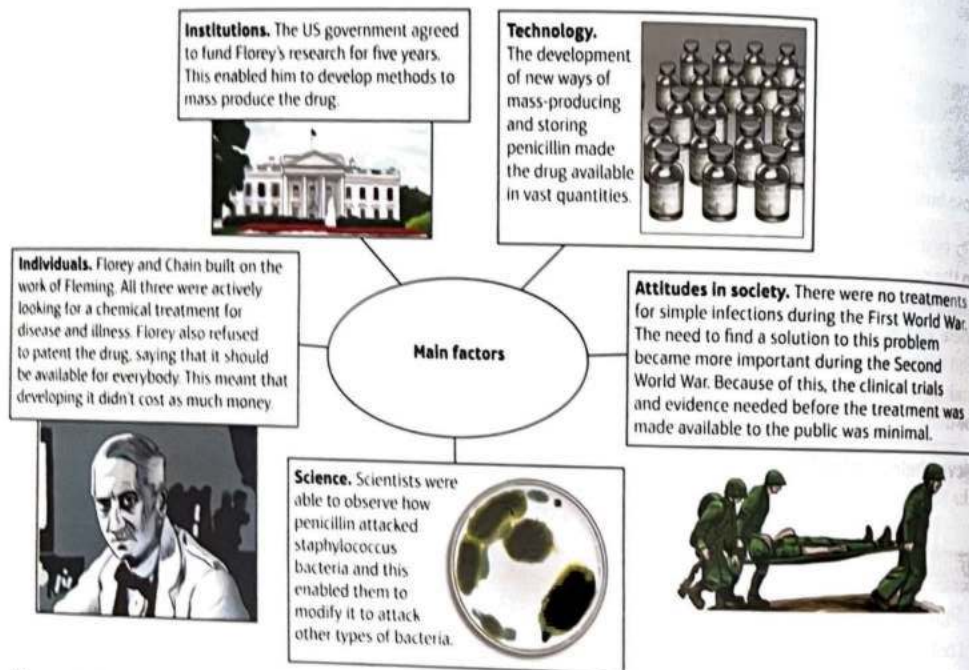


Figure 4.5 The main factors that enabled the development of penicillin.

Use of penicillin

Penicillin is effective in treating diseases caused by a certain family of bacteria.

Penicillin is also used to **prevent** infection, particularly with patients who have had teeth extracted.

The development of penicillin also encouraged scientists to look for other moulds that could be used to fight bacterial infections – such as streptomycin, which was the first drug found to be effective against tuberculosis. Once Dorothy Hodgkin had mapped the chemical structure of penicillin (see the timeline, on page 117), scientists were able to begin working on

synthetic versions of it that were slightly modified to treat specific diseases.

Now that doctors could offer treatments that worked against a wide range of illnesses, confidence in medical treatments began to rise. Patients were more willing to seek out medical treatments from doctors.

Unfortunately, as explained on page 112, some bacteria are now resistant to penicillin. Bacteria can mutate to resist attack from penicillin mould. The first penicillin-resistant strain of bacteria appeared in 1942. Pharmaceutical companies continue to work hard to develop new forms of penicillin and other antibiotics that will kill off the bacteria.

Summary

- Alexander Fleming discovered penicillin by accident in 1928. His research went unnoticed until 1938, when it was developed by Howard Florey and Ernst Chain.
- Florey and Chain were able to create a usable drug from the mould, which they tested on mice and then a human being. However, it was very difficult to mass produce.
- When the Second World War broke out, Florey was able to secure funding from the US government to mass produce the drug.
- Penicillin is effective against a wide range of illnesses.

Checkpoint

Strengthen

- 1 Describe the process by which Fleming discovered penicillin.
- 2 How did Florey and Chain test penicillin when they isolated it?
- 3 How did attitudes in society lead to the development of penicillin?
- 4 How was penicillin developed after the Second World War?

Challenge

- 1 How was the discovery of penicillin linked to:
 - government intervention
 - science and technology?
- 2 With a partner, discuss the roles played by Fleming, Florey and Chain in the discovery and development of penicillin. Was it the right decision to award them the Nobel Prize jointly?

Discuss the answer to C1 in groups if you are struggling. Your teacher can give you some hints.

Learning outcomes

- Understand why lung cancer has become more widespread since the 19th century.
- Understand how science and technology has helped in diagnosing and treating lung cancer.
- Understand how the government has tried to prevent lung cancer.

Lung cancer is the second most common cancer in the UK. It mainly affects people over the age of 40, with diagnoses being highest among people aged 70–74.

Most lung cancers are caused by external factors. Around 85% of cases are people who smoke, or have smoked. Other chemicals in the air, such as radon gas, are also sometimes to blame. However, a small number of people develop lung cancer for no apparent reason.

There were very few cases of lung cancer discovered in the 19th century. Records kept at the University of Dresden in Germany, for example, show that only 1% of all cancers they found at autopsy were caused by lung tumours. However, by 1918, that had increased to 10%, and by 1927 it was more than 14%.

In 1950, the British Medical Research Council published a study that showed conclusively that the rise in lung cancer was linked to cigarette smoking. Aggressive advertising by the tobacco companies since the First World War had led to a huge rise in the number of

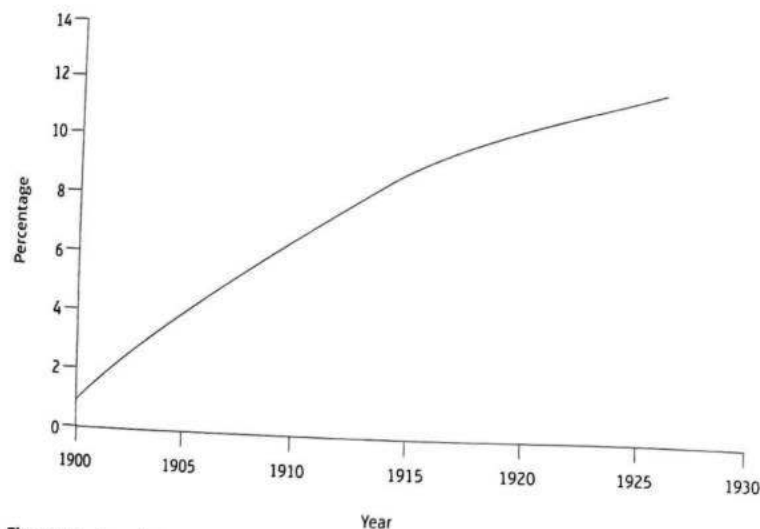


Figure 4.6 A graph showing the increase in lung cancer cases over time.

smokers. In spite of the results of the study, deaths among men from lung cancer continued to rise. They peaked in 1973 when nearly 26,000 deaths occurred due to lung cancer. The death rate among women also continued to rise until the 1990s.

The use of science and technology in diagnosis

One of the reasons why lung cancer is so hard to treat is that usually, by the time the cancer is detected, it is already very advanced. Patients often mistake their symptoms for other diseases. There is no national screening programme for lung cancer, so people are not routinely tested to see if they have it. This is because the tests are not accurate enough to outweigh the negative effects of the screening (for example, exposure to radiation during an x-ray scan).

Diagnosis is also difficult, although technology has led to improvements in this area.

Diagnosing lung cancer

Before more advanced technology had been discovered, lung cancer was diagnosed using an x-ray machine. A doctor would examine the x-ray to look for a tumour.

This way of diagnosing was not ideal. Often other things, like lung abscesses, might be mistaken for cancer –

or, worse, cancer could be mistaken for something less threatening. X-rays were not detailed enough to accurately diagnose cancer.

Figure 4.7 shows how lung cancer is diagnosed today.

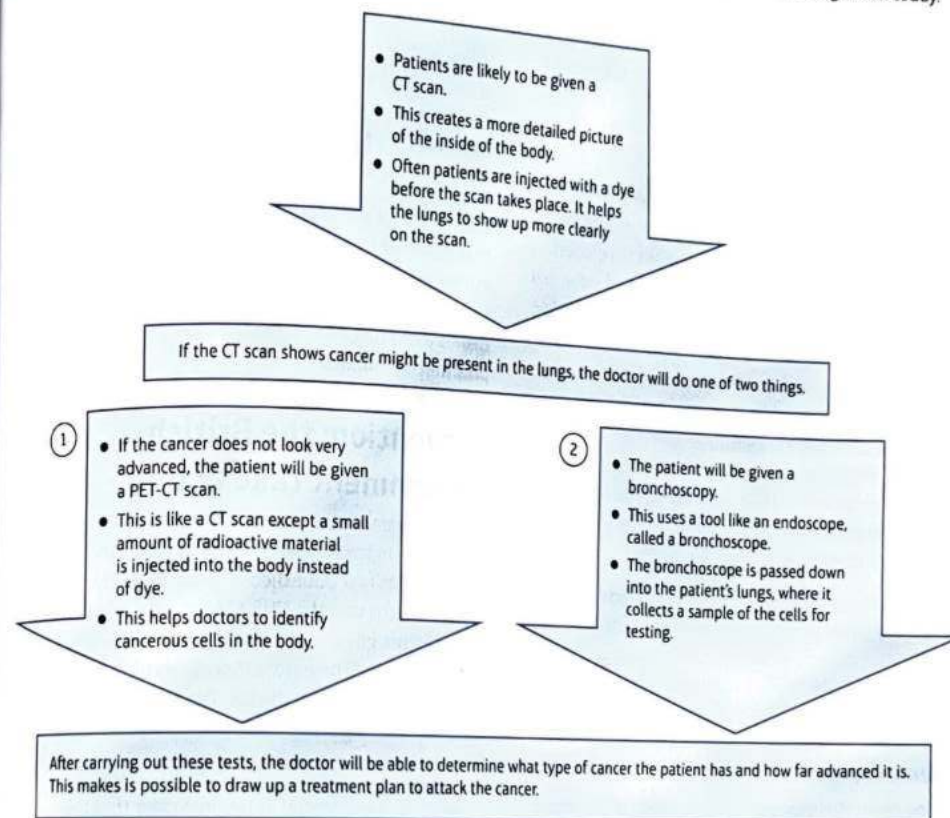


Figure 4.7 The stages of diagnosing lung cancer.

Activities

- 1 Explain the most common causes of lung cancer.
- 2 Explain why lung cancer is difficult to treat.
- 3 How has science and technology helped in the diagnosis of lung cancer?

The use of science and technology in lung cancer treatment

If lung cancer is diagnosed early, doctors can perform an operation to remove the tumour and the infected portion of the lung. This can range from the removal of just a small piece, to the removal of the entire lung. It is possible to breathe normally with only one lung. There are also other treatments to consider.

Transplants

It is possible to replace cancerous lungs with a transplant from a healthy donor. However, this raises a number of ethical problems if the patient developed lung cancer after smoking for a long period of time. Is it fair to give new lungs to somebody because they chose to ruin their own?

Radiotherapy

During radiotherapy, concentrated waves of radiation are aimed at the tumour to try to shrink it. Small tumours can be treated this way instead of with surgery. Larger tumours can be prevented from growing bigger using radiotherapy.

The radiotherapy can either be administered as beams of radiation directed at the tumour from outside the body, or by placing a small piece of radioactive material directly next to the tumour using a very thin tube, called a catheter.

Chemotherapy

During chemotherapy, patients are injected with many different drugs. These either shrink the tumour before surgery, prevent the cancer from reoccurring, or provide relief from the symptoms of lung cancer when surgery is not possible.

Lung cancer patients are now more likely to be treated using a range of these strategies. For example, they might have surgery to remove the tumour and then have radiotherapy and chemotherapy to tackle any remaining cancerous cells.

Genetic research

It is not yet possible to use genetics to treat lung cancer.

However, scientists have been studying the genes of lung cancer sufferers in order to help doctors prescribe more effective treatments. For example, some chemotherapy drugs work better in lung cancer patients whose tumours have a certain genetic mutation.

The reason for this is not clear. However, it does show that treatment can be more effective if it is prescribed for the individual, rather than on a 'one size fits all' basis. This idea of tailoring treatments to a person's DNA is a growing field in medical research known as pharmacogenomics.

Prevention: the British government take action

The government was slow to respond to the evidence that cigarette smoking was linked to lung cancer. This evidence was first published in 1950. By 1985, smoking-related deaths cost the NHS £165 million a year. However, the government earned around £4 billion from the tobacco tax. There were thousands of jobs related to the tobacco industry in the UK. There was also an ethical issue: was it the government's job to limit personal freedoms?

However, as time passed, it became clear that the government needed to intervene. The death rate was too high – they would not normally ignore an epidemic of such magnitude.

This table shows some of the actions that the government took.

Changing behaviour

The government passes laws to **force** people to change behaviour that damages their health.

In 2007, the government **banned smoking in all workplaces**. People were no longer allowed to smoke in pubs, cafés, restaurants or offices.

In 2015, the **ban was extended to cars** carrying children under the age of 18. There is significant evidence to suggest that second-hand smoke has a negative impact on health, particularly among children. Although many still argue that a smoking ban is an attack on personal choice, others argue that it is not the choice of the child to be exposed to the smoke. Therefore, the government stepped in to protect their health.

In 2007, the government **raised the legal age for buying tobacco** from 16 to 18. They did this to try to reduce the number of teenagers who smoke.

Influencing behaviour

The government controls communication to **influence** people to change behaviour that damages health.

The **ban on tobacco advertising** began with a ban on cigarette television advertising in 1965. Over time, the rules governing how and where cigarettes could be advertised were extended, until the government banned cigarette advertising entirely in 2005. This included the sponsorship of major sporting events in the UK, such as the Grand Prix.

The government has produced many **campaigns to advertise the dangers of smoking** over the past decades. These have included highlighting the impact of pregnant women smoking, the number of chemicals included in cigarette smoke and statistics about the health impacts and the diseases caused by regular use. Education to discourage young people from smoking is now included in schools.

Government research in 2012 suggested that it is important to discourage young people from smoking. Now, all cigarette products in shops must be **removed from display**.

Increased taxation on tobacco products was introduced to encourage people to stop smoking.

Source A

A 2012 Department of Health consultation on smoking, which sets out the reasons why the government has taken action to limit visibility of cigarettes.

Evidence shows that cigarette displays in shops can encourage young people to start smoking. The figures for England show that:

- 5% of children aged 11-15 are regular smokers
- more than 300,000 children under 16 try smoking each year
- 39% of smokers say that they were smoking regularly before the age of 16.

Covering tobacco displays will protect children and young people from the promotion of tobacco products in shops, helping them to resist the temptation to start smoking. It will also help and support adults who are trying to quit.

More than 8 million people in England still smoke – it is one of the biggest preventable killers causing more than 80,000 deaths each year. Nearly two-thirds of current and ex-smokers say they started smoking before they were 18.

Government campaigns and legislation have led to a change in attitude among the public. The number of smokers is falling. It is now very unlikely that you will see a television character or a public figure smoking. If you do, it will probably be portrayed negatively.

5.1 The historical context of medicine in the early 20th century

Learning outcomes

- Understand the growth of medical knowledge in the late 19th and early 20th centuries.
- Understand the developments in aseptic surgery, x-rays and blood transfusions that were the basis of further developments during the First World War.

In the years before the outbreak of the First World War, many medical breakthroughs had occurred. These were used as the foundation for medical advancements in the British sector of the Western Front. These included aseptic surgery, x-rays and blood transfusions.

Understanding infection and the move towards aseptic surgery

Joseph Lister first used carbolic acid to prevent infection in surgery in 1865, based on Louis Pasteur's work on Germ Theory.



The air was sterilised by being pumped over the heating system to kill germs. Two German surgeons, Gustav Neuber in the 1880s and Ernst von Bergmann in the 1890s, first developed these methods.

Figure 5.2 The key features of early 20th-century aseptic surgery.

By the late 1890s, Lister's methods had laid the foundations for aseptic surgery. By 1900, most operations were carried out using aseptic methods.

Exam-style question, Section A

Describe **two** features of aseptic surgery in the early 20th century. **4 marks**

Exam tip

Make sure that you develop each feature you identify with some supporting information.

The development of x-rays

The development of x-rays was completely accidental. In 1895, Wilhelm Roentgen, a German physicist, was studying the effects of passing an electrical current through a glass tube covered in black paper. He noticed that although everything in the room was darkened, a screen about a metre from the equipment had begun to glow. He called these rays that could pass through glass 'x'.

Further experiments led him to realise that these rays could penetrate many objects. He held a piece of lead in front of the tube and was able to see his own flesh glowing around his bones. He then placed photographic paper between the tube and his hand and created the world's first x-ray image. Shortly after, he also took an x-ray of his wife's left hand. It was possible to see not just her hand on this x-ray, but also her wedding ring.

The importance of the use of x-rays was understood immediately.

By early 1896, radiology departments* were opening in a number of hospitals, contributing to advancing knowledge and applying the new science in a medical setting. At Glasgow Royal Infirmary, a radiology department headed by Dr John Macintyre produced a number of interesting x-rays including x-rays of a kidney stone, a penny in a child's throat, and even a frog's legs in motion.

At Birmingham General Hospital, Dr John Hall-Edwards was one of the first doctors to make a diagnosis based on information from an x-ray, when he located a needle in a woman's hand. It was this potential for carrying out diagnosis before operations took place that would help medical treatments on the Western Front.

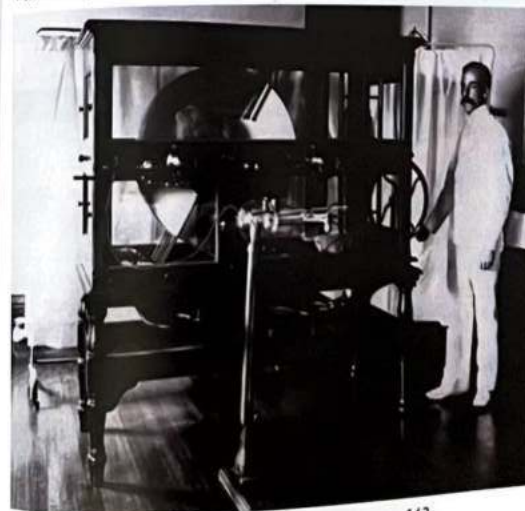
However, many problems emerged with the early use of x-rays.

- The health risks associated with x-rays were not fully understood. The amount of radiation that was released in early x-rays was about 1,500 times the amount that is released today. Exposure to x-rays was harmful. Patients could lose hair or suffer burns.
- Wilhelm Roentgen had used a table-top machine, but the glass tube used with this was very fragile, so could break easily.
- Taking an x-ray of a hand using the table-top machine took about 90 minutes – which was a very long time to keep your hand still!
- Larger x-ray machines were being developed, but were very difficult to move around.

However, these dangers and problems did not prevent the continuing use of x-rays.

Source A

A 19th-century x-ray machine in a hospital.



Key term

Radiology department*

The hospital department where x-rays are carried out.

Activities

Look at Source A.

- 1 Describe the main features of the machine shown.
- 2 What problems might arise from the use of this machine?
- 3 What other types of sources might help you to understand the problems linked to the use of x-rays?

Extend your knowledge

Roentgen rays

Although Roentgen called his discovery x-rays, everyone else knew them as **Roentgen rays**. However, during the First World War, anything with a German name became very unpopular in Britain. King George V, for example, changed his family name from Saxe-Coburg-Gotha to Windsor. This anti-German feeling led to the name Roentgen rays being permanently changed to x-rays in 1915.

Key terms

Blood transfusion*

Blood taken from a healthy person and given to another person.

Universal blood group*

This blood group can be used in a transfusion to a recipient with any other blood group.

The development of blood transfusions and the storage of blood

An average adult body contains about five litres of blood. If somebody loses too much blood, then they are likely to go into shock and die. In the 19th and early 20th centuries, blood loss was often the result of complex surgeries.

With the development of aseptic surgery and x-rays in the late 19th century, it was possible to carry out more complex surgical operations safely. However, if the problem of blood loss could not be solved, then the success of these operations would be irrelevant.

James Blundell did the first experiments in human blood transfusion* in 1818 to help women under his medical care who lost blood when they gave birth. Between 1818 and 1829, Blundell carried out ten transfusions, with up to half of the patients surviving. In these years, Blundell developed many of the techniques and basic equipment which would continue to be used up to the First World War. As blood could not be stored – it had to be used as soon as it was available – transfusions were carried out with the donor (the person giving the blood) being directly connected to the recipient (the person receiving the blood) by a tube.

The table below shows the main problems with blood transfusions, and the attempts that had been made to solve them by the early 20th century.

Activities

- 1 What were the main features of aseptic surgery?
- 2 Do you think that the benefits of x-rays outweighed their dangers?
- 3 With a partner, discuss how successfully the problems associated with blood transfusions had been solved by the early 20th century.

Problem with transfusion	Attempted solution to the problem
Blood coagulates (clots) as soon as it leaves the body. This meant that the tubes which transfused blood from one person to another could become blocked up.	There were attempts to find chemicals, such as sodium bicarbonate, to prevent clotting. In 1894, Professor Almroth Wright, a British scientist, concluded that the soluble solution of certain acids could prevent clotting, but he thought that side effects, such as convulsions, could not be prevented.
Rejection of the transfused blood because the blood of the donor and the blood of the recipient was not compatible.	In 1901, Austrian doctor Karl Landsteiner discovered the existence of three different blood groups – A, B and O. The following year a fourth blood group, AB, was also found. This information was used in 1907 by Reuben Ottenberg, an American doctor, who was the first person to match a donor and a recipient's blood type before a transfusion. He also identified blood group O as a universal blood group.*
Danger of infection from unsterilised equipment.	The introduction of aseptic methods of surgery had largely solved this problem in hospital conditions by the early 20th century.

Summary

- By 1900, most surgery was carried out using aseptic methods.
- X-rays were discovered in 1895. They were used almost immediately for diagnostic purposes in medicine.
- Blood transfusions had to take place on a person-to-person basis because there was no way to store blood.
- The recognition of different blood groups enabled blood transfusions to become more effective.

Checkpoint

Strengthen

S1 Explain how the following were important developments in medicine?

- Aseptic surgery
- X-rays
- Blood transfusions

Challenge

- C1 Although there had been many advances in medicine, there were still many problems remaining. Identify these problems.
- C2 From your list of problems, which ones do you think would be particularly challenging in a war environment? If you are not confident about any of these questions, form a group with other students, discuss the answers and then record your conclusions. Your teacher can give you some hints.

5.2 The context of the British sector of the Western Front

Learning outcomes

- Understand the organisation of the trench system and how it affected medical treatment.
- Know about the key battles on the Western Front in the First World War and how they affected the development of medical treatments.

Flanders and northern France

Britain declared war on Germany on August 4 1914. Germany invaded France through Belgium. The British government sent the British Expeditionary Force (BEF) to northern France to try to stop the German advance through Belgium. The BEF was made up of 70,000 professional soldiers, fighting alongside a larger French army. Near Mons, close to the French-Belgian border, the BEF faced a German army that was more than double their size – 160,000 soldiers. Although they stopped the German advance briefly, they were ordered to retreat to the River Marne in order to prevent Paris falling to the Germans. After the Battle of the Marne, the German forces pulled back to the River Aisne and it was here that trench warfare began.

The communications trench which ran between the other trenches.

At the very rear were artillery emplacements

Holes were dug into the side of the trenches where men could take protective cover when needed. These were known as dugouts

The support trench. This was about 80 metres behind the frontline trench and the troops would retreat here if the frontline trench came under attack.

The reserve trench. This was at least 100 metres behind the support trench and was where reserve troops could be mobilised for a counter-attack if the frontline trench was captured by the enemy

The frontline trench. This was where attacks would be made from

Trenches were dug in a zig-zag pattern

The trench system

By the end of 1914, much of Belgium and northern France had been occupied by the Germans, although they had been stopped from advancing into the heart of France. This meant that battles could not follow along the same lines as previous wars, which had involved much more movement of single armies. Instead, it became a static war based on trenches, which needed to be defended from the enemy, with attempts made to advance from the trenches to seize land from the enemy army. A line of trenches was eventually established all the way from the English Channel in the north, to Switzerland in the south.

Construction and organisation

Although some basic trenches were dug in 1914, a more complex system began to evolve from 1915. The trenches were generally dug to a depth of about 2.5 m. The main elements in the trench system are shown in Figure 5.3.

Trenches were easier to defend than attack. Machine guns could fire rapidly, and barbed wire was placed in no-man's-land (the area between two opposing lines of trenches) to slow down the progress of any attack.

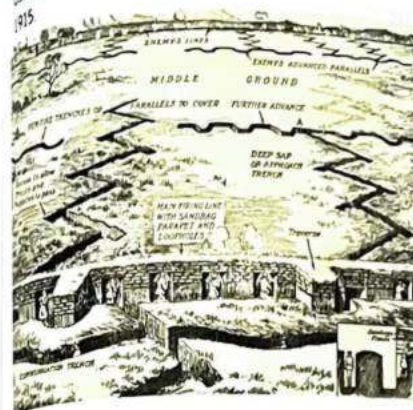
New tactics were developed to try to deal with the advantages held by the defenders. Such tactics included the use of gas.

Figure 5.3 The trench system during the First World War.

5.2 The context of the British sector of the Western Front

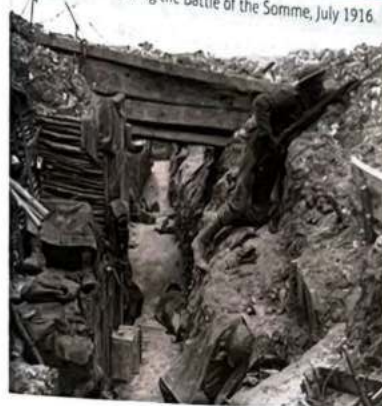
Source A

A typical trench system, sketched during the war. The exact date and artist is not known. It was likely sketched after 1915



Source B

A British trench during the Battle of the Somme, July 1916



Activities

Study Sources A and B.

1. What differences do you note between the sketch diagram of a trench system and the photo of a trench?
2. What are the benefits and drawbacks of each of these types of sources for the historian?
3. Work in pairs. Discuss which of the two sources you find the most helpful and why.

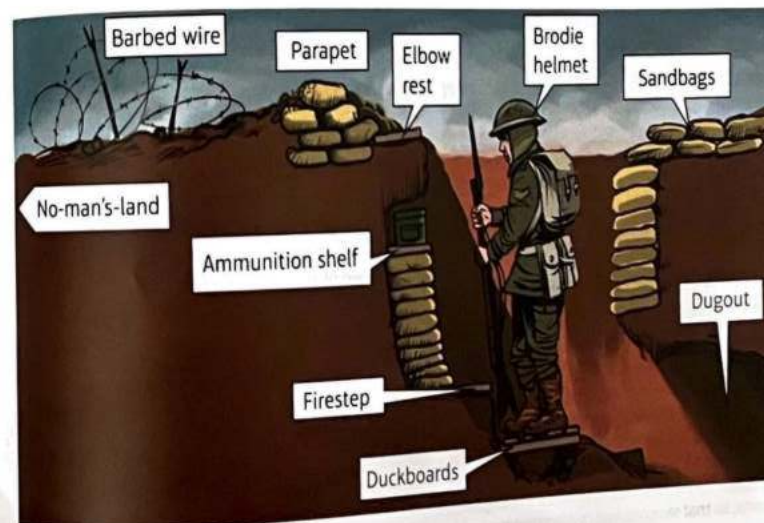


Figure 5.4 Cross section of a trench.

The Ypres Salient*, the Somme, Arras and Cambrai

This section will outline the key battles in the British sector of the Western Front. It will be important to refer back to this section when looking at specific medical advancements on the Western Front.

Key term

Salient*

An area of a battlefield that extends into enemy territory, so that it is surrounded on three sides by the enemy and is therefore in a vulnerable position.



Figure 5.5 Map of the Western Front.

1914: the First Battle of Ypres

During the first months of the war, the BEF had moved to the town of Ypres in western Belgium, in order to prevent the German advance towards the sea. In the autumn of 1914, the Germans launched an attack on the British positions to the east and north-east of Ypres. Although the British lost over 50,000 troops in this battle, which went on from 12 October to 11 November, they held on to Ypres. This meant they controlled the English Channel ports, so that supplies and reinforcements could

be provided. By the end of the battle, the Germans had extended their control around the edge of the Ypres Salient as far as the village of Messines.

The use of mines at Hill 60

Hill 60 was a man-made hill to the south-east of Ypres. The Germans had captured it in December 1914 and its height gave them a strategic advantage in this area. The British used the method of offensive tunnelling to take it back in April 1915. This involved tunnelling into and under the hill. Five mines were placed in the tunnels. When they exploded they blew the top off Hill 60 and the British were able to take this strategically important position.

Extend your knowledge

Digging tunnels on the Western Front

The men who joined the Tunnelling Companies had all worked underground before the war – including, for example, coal miners from Northumberland, sewage drain diggers from Manchester and tube tunnel diggers from London. These men were often shorter than the minimum height requirement for the army. They experienced a very high death rate.

1915: the Second Battle of Ypres

As soon as the battle for Hill 60 was finished, the Second Battle of Ypres began. This took place as a sequence of battles over a period of about one month, from 22 April through to 25 May 1915. It is significant in the history of the First World War as it was the first time that the Germans used chlorine gas on the Western Front (see pages 154–155). British losses during this month totalled about 59,000 men. By the end of the battle, the Germans had moved about two miles closer to the town of Ypres on the eastern side of the salient.

1916: the Battle of the Somme

Source C

A still from the 1916 British film, *The Battle of the Somme*.



Figure 5.6 Map of the Ypres Salient.

Activity

This photograph claimed to show British soldiers advancing at the start of the Battle of the Somme. It is now thought this sequence was staged away from the frontline. Does this affect the way in which you can use the photograph as evidence?

The British attack on the Somme aimed to take ground from the Germans and was launched on 1 July 1916. The casualties on both sides were enormous: on the first day alone, British casualties were over 57,000, with deaths totalling around 20,000 men. The British tried two new strategies, which would both eventually contribute to increased casualty rates:

- the use of the **creeping barrage**, which saw artillery launched from the trenches towards the German lines just ahead of the British infantry as it advanced forwards.
- the **first use of tanks in warfare** – however, the use of tanks had many technical problems and they were not very successful.

By the time the Battle of the Somme ended in November 1916, it is estimated that the British had suffered over 400,000 casualties.

Tunnels, caves and quarries at Arras

The area around Arras is very chalky, and so it is easy to tunnel through. Quarries and tunnels had been dug in this landscape since Roman times. In 1916, the British decided to link the existing tunnels, caves and quarries to create an underground network around Arras to act as shelters against German attacks. They were also built to enable safe underground movement. The work was carried out by Tunnelling Companies from Britain and New Zealand. In total, they dug more than 2.5 miles of tunnels in five months. Up to 25,000 men could be stationed in the tunnels, which contained electric lights, running water, a light railway system and a fully functioning hospital (see page 162).

1917: the Battle of Arras

In April 1917, 24,000 men who had been hiding in tunnels dug near the German trenches attacked. The aim of the offensive was to break through the German lines. In the first few days, it appeared that this aim was achieved, as the British advanced about eight miles. However, as the advance slowed, virtually no further progress was made and by the end of the offensive in May, there was a large number of casualties (nearly 160,000 British and Canadians).

Source D

From 'The General', a poem written by Siegfried Sassoon in 1918. Sassoon served as an officer on the Western Front from 1914. He was treated for shellshock (a psychological condition caused by prolonged exposure to bombardment) in 1917, after which he returned to fight on the Western Front.

"Good morning, good morning," the General said,
When we met him last week on our way to the line.
Now the soldiers he smiled at are most of 'em dead,
And we're cursing his staff for incompetent swine.
"He's a cheery old card," muttered Harry to Jack
As they slogged up to Arras with rifle and pack,
But he did for them both by his plan of attack.

Activities

Study Source D.

- 1 Does the detail in the poem match the events at the Battle of Arras?
- 2 What questions should you ask before using a poem as a piece of historical evidence?

1917: the Third Battle of Ypres

The purpose of the Third Battle of Ypres in 1917 was for the British army to break out of the Ypres Salient. The British wanted to remove the German advantage of having the higher ground. Throughout June, the British had prepared for the main attack in the battle of Messines, where they had driven the Germans off the ridge that formed part of the Ypres Salient and which the Germans had occupied since October 1914. The British launched their main attack on 31 July, marching east from Ypres towards Passchendaele. The army advanced about two miles on the first day. Soon though, the weather turned to rain and the ground became waterlogged – so much so that many men fell in the mud and drowned. This campaign lasted until November. By then, the British had moved the edge of the salient back by about seven miles. The cost of this advance was an estimated 245,000 British casualties.

1917: the Battle of Cambrai

The Battle of Cambrai was launched on 20 October 1917. The artillery barrage was changed so that less warning of the coming attack was given to the Germans. It was accompanied by the first large-scale use of tanks – nearly five hundred were used in this battle. They were

able to move easily across the barbed wire and their machine guns were very effective.

Problems of transport and communications

The constant shelling, and the type of terrain that soldiers were fighting on (similar to the conditions in the Third Battle of Ypres), left the landscape full of craters and holes and destroyed many roads. This led to major problems in transporting injured men away from the frontline. Before the war, this region had been used as farmland, and the use of fertiliser was extensive. This meant there was a lot of bacteria in the soil that could lead to infected wounds.

Men who were injured on the Western Front needed to be moved away from the frontline in stages, as soon as their condition was stable enough. Stretcher bearers, like Edward Munro (see Source E), would carry away the large numbers of wounded from the frontline, both during the day and at night. This meant they often had to expose themselves to shelling and gunfire. Further away from the frontline, it was possible to carry out more advanced medical procedures, and also to provide some protection against shelling. The faster an injury could be treated, the more likely a person was to survive.

Source E

From Edward Munro's *Diaries of a Stretcher Bearer*. This entry comes from 7 November 1916, when Munro was in the Somme.

We commenced to carry down the wounded of whom there were a considerable number. The 7th Brigade had made an attack on the German line the previous night and had suffered many casualties. The country over which we have to carry is most difficult to traverse [walk across], being pitted with shell holes, mostly waterlogged. Fritz [a reference to the Germans] keeps up a fairly constant shelling. Yesterday he caught some of the 6th Ambulance bearers, killing two. In this area was started the system of carrying the stretchers shoulder high – four to a stretcher, this being much less fatiguing than the old method of two carrying with slings... The carrying at night is very trying as there are no clearly defined tracks. The landmarks that serve to guide one in the daytime are not visible at night.

Source F

Stretcher bearers carrying a wounded man to safety at the third battle of Ypres in August 1917. This photograph was taken by Lieutenant John Brooke, an official photographer for the British army on the Western Front.



Horse-drawn and motor ambulances

When the BEF was first sent to France in August 1914, the military leadership decided not to send any motor ambulances with them. It was soon realised that this was a mistake, as the horse-drawn ambulance wagons could not cope with the large number of casualties. Men who were transported in these wagons were shaken about, which often made their injuries even worse. This lack of transport actually led to soldiers being left to die or being taken prisoner by the Germans.

When news of this reached Britain, *The Times* newspaper ran a public appeal for donations. By October 1914, after only three weeks, the appeal had raised enough money to buy 512 ambulance wagons, which would make transporting wounded soldiers much easier, and would prevent injuries being made worse from the move. The first motor ambulances were sent to the Western Front in October 1914, as a result of work by the Red Cross. However, motor vehicles could not operate in much of the muddy terrain of the frontline, so horse-drawn

Activities

Study Source F.

- 1 Describe the photo Source F – what is happening?
- 2 How could a historian use this source for an enquiry into the problems that faced stretcher bearers?
- 3 What other sources would be helpful in considering the work and problems faced by stretcher bearers on the Western Front?

wagons continued to be used throughout the war. In worse terrain, six horses, rather than the usual two, pulled the ambulance wagons.

Train, barge and ship ambulances

Wounded men might also be transported by train or by canal in the final stage of their evacuation to the Base Hospitals on the French coast (see page 162). In the first few months of the war on the Western Front, the Royal Army Medical Corps (RAMC), which was responsible for medical care in the army (see page 158), had to use French goods trains. The first ambulance train designed for carrying wounded soldiers arrived in France in November 1914. It had spaces for stretchers fitted down both sides of the carriage.

Later, some trains sent to France even contained operating theatres. There were concerns that the numbers of wounded being moved on the railways damaged the war effort because they contributed to too many trains moving around on the rail network of northern France and Belgium. This led to the decision to also make use of canal barges as transport for the wounded to Base Hospitals. Although the journey was slow, it was comfortable, and some of the wounded bypassed the Base Hospitals to be transferred directly onto the ships that were transporting wounded men back to Britain.

Source G

From a speech made by Walter Roch in Parliament, 23 June 1915. Roch was a Liberal MP and was taking part in a debate on how the government should spend its money.

I want to bring to the notice of the House information in connection with the treatment of the wounded in Flanders. The information is not my own personal knowledge, but from several very close personal friends who have been connected with this, although I cannot give their names. I am told that it is of the utmost importance that the men who are wounded should be treated as quickly as possible, and that their wounds should have the best possible attention as soon as may be. The suggestion I have to make is that there should be many more of these evacuation hospitals than there are in France at the present moment, that they should be much better equipped with operating theatres and other appliances, and that they should be more sanitary and hygienic in their nature.

The message and the messenger

There are many sources of information about the past. Historians use these sources to help them **draw conclusions**. When information is used to help you form a conclusion, it is used as **evidence**.

Read Source G. In this source, Walter Roch, a Liberal MP, says a number of things. Roch was making a public speech that he knew would be:

- a heard in London by his audience
- b reported in parliamentary records and possibly in the press

What **information** does the source contain? What was Roch **saying**? Answer the following questions to find out.

- 1 What does Roch say is important for wounded men?
- 2 What does he say the conditions in the hospitals should be like?
- 3 What impression do you think other MPs would have had about Roch's sources of information?

Historians are not usually interested in information for its own sake. Historians are interested in **using** information to work out the **answers** to questions about the past. Use the information you have just extracted from the source and the information about its **context** to try and work out answers to the following questions about the way in which Parliament was helping the wounded in 1915.

- 4 What does Roch's speech suggest about the conditions in hospitals at this date?
- 5 What do you think Roch wanted Parliament to spend some of its money on?

Summary

- Trench warfare had begun on the Western Front by the end of 1914.
- As the trench system developed, a complex network of trenches was created in which men could live and fight.
- Tunnels and caves at Arras were used as part of the defensive system.
- Chlorine gas was first used by the Germans at the Second Battle of Ypres in 1915.
- The first motorised ambulances sent to France were provided by public donations and organisations such as the British Red Cross.
- Wounded men were moved away from the frontline by trains and canal boats.

Checkpoint

Strengthen

- S1 In what ways were the problems of transporting wounded men dealt with?

Challenge

Think about the different types of source used in this section on the context of the British sector of the Western Front.

- C1 Using the types of sources from Figure 5.2 (page 142) make a table with one page for each type of source. Divide each page into three columns: 'Example', 'Strengths', 'Weaknesses.'

- C2 Fill in this table for Sources A, C and D.

If you are not confident about any of these questions, your teacher can give you some hints.

Learning outcomes

- Understand the main medical problems that were faced on the Western Front and how they were dealt with.
- Understand the main types of wounds and injuries that were experienced by soldiers fighting on the Western Front.

Main medical problems on the Western Front

Conditions in the trenches were very unpleasant. Sanitary, or hygienic, conditions for such large numbers of people posed a problem. In summer, the combination of sewage and dead bodies made the smell dreadful, whilst in winter,

bad weather gave rise to both flooding and frostbite. In November and December 1914, there were over 6,000 cases of frostbite. Rat infestation was normal. A number of medical problems were caused by these conditions.

The nature of wounds

Rifles and explosives

In a case study of over 200,000 wounded men admitted to Casualty Clearing Stations (CCS) on the Western Front, it was discovered that high-explosive shells and shrapnel were responsible for 58% of wounds. When a shell exploded, it could kill or injure a soldier immediately. A shell explosion also scattered shrapnel (fragments of metal in the casing), which travelled at fast speeds over a wide area. This meant that anyone who was in the way of the shrapnel was likely to be wounded. About 60% of injuries were to arms and legs.

Extend your knowledge

Famous fighters

Three young men who went on to become well-known authors all suffered from trench fever: A. A. Milne (*Winnie the Pooh*), J. R. R. Tolkien (*Lord of the Rings*) and C. S. Lewis (*Chronicles of Narnia*). Milne and Tolkien did not return to frontline fighting after their bout of trench fever in 1916.

	Main symptoms	Attempted solutions to deal with the problem
Trench foot	Painful swelling of the feet, caused by standing in cold mud and water. In the second stage of trench foot, gangrene set in. Gangrene is the decomposition of body tissue due to a loss of blood supply.	• Prevention was key. • Rubbing whale oil into feet to protect them. • Keeping feet dry and regularly changing socks. • If gangrene developed, then amputation was the only solution to stop it spreading along the leg.
Trench fever	Flu-like symptoms with high temperature, headache and aching muscles. This condition was a major problem because it affected an estimated half a million men on the Western Front.	• By 1918, the cause of trench fever had been identified as contact with lice. • Delousing stations were set up. After this, there was a decline in the numbers experiencing the condition.
Shellshock	Symptoms included tiredness, headaches, nightmares, loss of speech, uncontrollable shaking and complete mental breakdown. It has been suggested that about 80,000 British troops experienced shellshock.	• The condition was not well understood at the time. • In some cases, such as Siegfried Sassoon and Wilfred Owen, this involved treatment back in Britain. The Craiglockhart Hospital in Edinburgh treated 2,000 men for shellshock. • However, some soldiers who experienced shellshock were accused of cowardice. Many were punished for this – some were even shot.

The case study also found that bullets were responsible for 39% of wounds. Machine guns could fire 450 rounds a minute, and their bullets could fracture bones or pierce organs. Rifles could fire accurately at up to 500 m, but lacked the speed of machine guns.

Shrapnel, wound infection and head injuries

When men were injured, either by shrapnel or by bullets, the metal would penetrate their body, taking with it the fabric of the uniform from the area surrounding the wound. The soil in the region, which had been intensively farmed with large quantities of fertiliser before the war, contained the bacteria for both tetanus and gas gangrene. Gangrene is an infection caused by a lack of blood to an area in the body. Gas gangrene is an infection that produces gas in gangrenous wounds. When wounds were exposed to the soil the presence of these bacteria made infection much more likely. The impact of tetanus was reduced by the use of anti-tetanus injections from the end of 1914. However, there was no cure for gas gangrene. The bacteria for gas gangrene spread through the body quickly and could kill a person within a day.

At the start of the war, the headgear worn by soldiers was a soft cap. To protect against head injuries, a trial using the Brodie helmet was carried out in 1915. This was a steel helmet with a strap that prevented it being thrown off the head in an explosion. It was estimated that it reduced fatal

Source A

From an interview with Captain Maberly Esler in 1974. He was a medical officer. Here he is recalling events at Hooge, in the Ypres salient, in June 1915.

We'd never attempt any major surgery or anything like that in the trenches – one couldn't do it. The only thing you could do was to cover a wound to keep it from getting infected, or stop a haemorrhage by compression if they were bleeding to death. Several people got tetanus afterwards from an infection in the ground which was carried in shelled areas. The ground had been shelled for such a long time it was in rather a septic sort of condition.

head wounds by 80%, so the helmet was then provided to all soldiers fighting on the Western Front.

The effects of gas attacks

Gas attacks caused great panic and fear, as is shown in Wilfred Owen's poem *Dulce et Decorum Est* in Source B. It was not, however, a major cause of death, with only about 6,000 British soldiers dying as a result of gas attacks. The British army gave troops on the Western Front gas masks from 1915, which became more sophisticated over time. Still, gas attacks were greatly feared by soldiers on the Western Front.

Chlorine

First used by the Germans in 1915 at the second battle of Ypres.

It led to death by suffocation.

The medical services had no experience in dealing with gas attacks, and so had to experiment with treatments.

Gas masks were given to all British troops in July 1915.

Before this, soldiers developed their own system of gas masks. They soaked cotton pads with urine and pressed them to their faces to help stop the gas entering their lungs.

The British retaliated with their own chlorine attack later in 1915 at the Battle of Loos, but the wind changed direction and the gas blew back on the British lines.

Phosgene

First used at the end of 1915 near Ypres.

Its effects were similar to those of chlorine but it was faster acting, killing an exposed person within two days.

Mustard

First used in 1917 by the Germans.

It was an odourless gas that worked within 12 hours, causing both internal and external blisters and could pass through clothing to burn the skin.



Figure 5.7 Three types of gas attacks on the Western Front.

Source B

From '*Dulce et Decorum Est*', a poem written by Wilfred Owen in 1917 whilst he was being treated for shellshock. He served on the Western Front in 1916–17 and returned in 1918, where he was killed in action shortly before the end of the war. The text in the title and the end of the poem is in Latin and means 'it is sweet and fitting to die for one's country'.

Gas! Gas! Quick, boys! – An ecstasy of
flumbling,
Fitting the clumsy helmets just in time;
But someone still was yelling out and
stumbling,
And hound'ring like a man in fire or lime...
Dim, through the misty panes and thick green
light,
As under a green sea, I saw him drowning.
In all my dreams, before my helpless sight,
He plunges at me, guttering, choking, drowning.
If in some smothering dreams you too could
pace

Behind the wagon that we flung him in,
And watch the white eyes writhing in his face,
His hanging face, like a devil's sick of sin;
If you could hear, at every jolt, the blood
Come gargling from the froth-corrupted lungs,
Obscene as cancer, bitter as the cud
Of vile, incurable sores on innocent tongues,

My friend, you would not tell with such high
zeal

To children ardent for some desperate glory,
The old Lie; *Dulce et Decorum est*
Pro patria mori.

Source C

From the notebook of Lance Sergeant Elmer Cotton, who served in the 5th Northumberland Fusiliers in 1915. He is describing the effects of a chlorine gas attack.

It produces a flooding of the lungs. It is the equivalent to drowning, only on dry land. The effects are these – a splitting headache and a terrific thirst (but to drink water is instant death), a knife-edge pain in the lungs and the coughing up of a greenish froth off the stomach and the lungs, finally resulting in death. It is a fiendish death to die.

Activities

- 1 Study Sources B, C and D.
- 2 Draw up a table to show the similarities and differences between the three sources when presenting the effects of a gas attack.
- 3 How much weight do you give to each source separately as a description of the effects of a gas attack?
- 4 What other sources could you use to find out more about the effects of a gas attack?

Source D

From a 1919 painting by John Singer Sargent. Sargent was commissioned by the British War Memorials Committee to paint this in 1918 and researched the painting by visiting both Arras and Ypres before the end of the war. These soldiers have experienced a mustard gas attack.



Asking questions: dealing with gas attacks

You need to think about the way sources require historians to ask themselves three sorts of questions. Look at Source E.

Content questions

What can you learn from the content? You can see that the source is a soldier wearing a mask.

Provenance questions

Now you need the caption! How does provenance affect the usefulness of the source? Remember, for provenance, you need to break things down into nature, origin, and purpose.

- 1 Nature – it is a photograph.
- 2 Origin comes from the caption – this was taken in the same month that the Second Battle of Ypres began.
- 3 Purpose can be difficult to determine for a photograph. Might it have been for use as propaganda? In this case, it is hard to think of a propaganda purpose. Does it look like it was set up?

Source E

Photograph of a man wearing a cotton wool pad respirator, April 1915. This was a simple form of gas mask. The Second Battle of Ypres also began in April 1915.



Perhaps it was to inform soldiers about the best way that they could protect themselves against gas attacks.

Context questions

You know two things that are relevant to Source E.

- 1 The first use of chlorine gas by the Germans took place in April 1915 at the Second Battle of Ypres.
- 2 The British were not prepared for gas attacks and so they had to experiment to find the best way to protect soldiers.

So, if you were asked how useful Source E is, you could say something like this [key: **content**, **provenance**, **context**]:

'Source E is useful because it shows the first attempts to make a gas mask. Although we don't know why the photo was taken, it might have been set up so that soldiers could learn how best to protect themselves. We know that proper gas masks were not provided at the time because the gas attacks were unexpected.'

What this does (and what you can do) is explain why the source is useful, using criteria you have explained based on the three types of question.

Source F

From the autobiography of Geoffrey Keynes, *The Gates of Memory* (1981). This account was originally published in 1968. Keynes was important in developing blood transfusions during the war.

On 6 February 1915 I was detailed for duty on an ambulance train. The numbers of patients carried on each journey varied between 100 and 400. During my turn of duty in the train we carried nearly 19,000 patients. Medical duties were usually restricted to ensuring that the wounded men, who had already been attended in a Dressing Station or Casualty Clearing Station travelled as comfortably as possible with the help of sedative and pain-killing drugs. Frequently they had to suffer violent jolts during shunting operations.

Usually the patients had been fully cared for before being sent on by train to the base hospitals, but on one occasion (12 March 1915), the train was ordered to go close to the frontline and take on casualties who had barely received first aid.

Now look at Sources F and G. The content of Source F describes the work done by doctors on an ambulance train. Although the provenance tells us that this was written many years later, the detailed references to dates and figures in the source suggests that the writer kept a diary and that he is using this to help him remember what happened. Source G, on the other hand, is clearly a historical photograph, preserving a moment in time. We don't know exactly what date this photograph was taken – only the month and year. Its purpose was to record the awful conditions that

Source G

A photograph of a wagon belonging to the Field Ambulance service, in use in the Somme region, September 1916.



faced men on the Western Front. Both sources tell you something about the different types of transport that were used to carry the wounded to safety and treatment.

Activities

- 1 Explain the differences between questions for comprehension, questions for context, and prompted questions.
- 2 Pick a source that could be used by a historian who was studying day-to-day life in your school now.
 - a Describe the source.
 - b Give an example of a comprehension question that could be asked about the source, and explain what you would learn from it.
 - c Give an example of a provenance question (nature, origin, purpose) that could be asked about the source and explain what you would learn from it.
 - d Give an example of a context question that could be asked about the source, and explain what you would learn from it.
- 3 Give an example of a question or enquiry prompted by the source, and explain why this would be an interesting enquiry to follow.

Summary

- Common medical problems that faced men fighting on the Western Front were trench foot, trench fever and shellshock.
- The introduction of the Brodie helmet saved many lives by protecting the head against shrapnel injuries.
- Gas attacks caused burning skin and suffocation.

Checkpoint

Strengthen

- 51 Explain all the medical problems and possible injuries that soldiers faced in the trenches that are referred to in this section. Then note down all the solutions to the problems.

Challenge

- C1 How useful are Sources B, C and D on the effects of gas attacks on the Western Front? If you are not confident about any of these questions, form a group with other students, discuss the answers and then record your conclusions. Your teacher can give you some hints.

Learning outcomes

- Understand the main stages of the chain of evacuation and what happened to wounded soldiers at each stage.
- Know about the role played by the RAMC* and FANY* in dealing with wounded soldiers.

Key terms

RAMC*

Royal Army Medical Corps. This branch of the army was responsible for medical care and was formally founded in 1898.

FANY*

First Aid Nursing Yeomanry. Founded in 1907, this was the first women's voluntary organisation to send volunteers to the Western Front. It provided frontline support for the medical services, for example by driving ambulances and engaging in emergency first aid.

To deal with the large numbers of casualties in the First World War, the number of medical professionals needed to be increased dramatically.

The table below shows the number of medical professionals in 1914 and 1918.

	1914	1918
Medical officers	3,168	13,063
Other ranks (e.g. private)	16,331	131,099

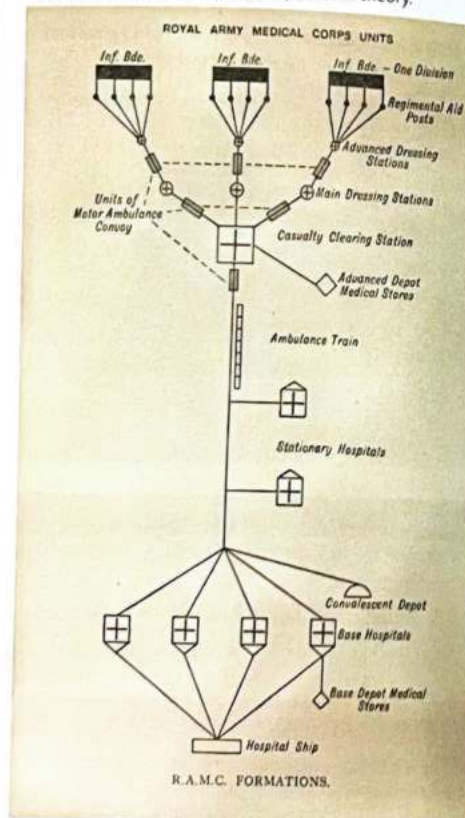
More than half of Britain's doctors were serving with the armed forces and most of these were deployed to the Western Front.

The system of transport and the stages of treatment

Because of the large numbers of casualties, it was essential that there was an efficient system to get the wounded from the frontline to a safe area where they could be treated. This system became known as the **chain of evacuation**.

Source A

From F. S. Brereton, *The Great War and the RAMC*, published in 1919. Brereton served as a Lieutenant-Colonel in the RAMC on the Western Front. This is a diagram showing how the chain of evacuation might operate in theory.



The main stages in the chain of evacuation were **Regimental Aid Posts (RAP)**, **Dressing Stations (ADS and MDS)**, **Casualty Clearing Stations** and **Base Hospitals**. Remember – these were the main stages, but were not always followed in the same order for every casualty.

Source B

From Ward Muir's *Observations of an Orderly*, published in 1917. Muir was a Lance Corporal in the RAMC and worked in a hospital in London that received patients from the Western Front at the end of the chain of evacuation.

The orderlies meet each convoy at the front door of the hospital. The walking-cases are the first to arrive – men who are either not ill enough, or not badly enough wounded, to need to be put on stretchers in ambulances. They come from the station in motor-cars supplied by the London Ambulance Column. The few minutes which the walking-case spends in the receiving hall are occupied in drinking a cup of cocoa, and in 'having his particulars taken'. Poor soul! – he is weary of giving his particulars. He has had to give them half-a-dozen times at least, perhaps more, since he left the front. At the field dressing-station they wanted his particulars, at the clearing-station, on the train, at the base hospital, on another train, on the steamer, on the next train, and now in this English hospital.

Activities

Study Source A and Source B. When Source B refers to 'particulars' it means personal details.

- Why do you think the first thing that happened to the new arrivals was that they were given a cup of cocoa and had their 'particulars' taken?
- How useful are these two sources in helping to establish the key stages in the chain of evacuation?
- What other sources would give you useful information on the chain of evacuation? Think of as many as possible. Now compare your list with a partner. Decide which would be the most useful.

Regimental Aid Post (RAP)

The RAP was generally located within 200 m of the frontline, in communication trenches or deserted buildings. It was made up of a Regimental Medical Officer, with some help from stretcher bearers with first-aid knowledge. Wounded men would either walk in themselves or be carried in by other soldiers. The purpose of the RAP was to give immediate first aid and to get as many men back to the fighting as possible. It could not deal with serious injuries. These had to be moved to the next stage in the chain of evacuation.

Dressing Stations (ADS and MDS)

In theory, there should have been an Advanced Dressing Station (ADS) about 400 m from the RAP and a Main Dressing Station (MDS) a further half a mile back. In practice, this was often not the case – and there may only have been one Dressing Station. Where possible, the Dressing Stations were located in abandoned buildings, dug-outs or bunkers, in order to offer protection from enemy shelling. Where these were not available, tents would be used. Each dressing station would be staffed by ten medical officers, plus medical orderlies and stretcher bearers of the RAMC. From 1915, there were also some nurses available for this part of the chain of evacuation. To get to the Dressing Station, men would either walk, if they were able to do so, or be carried in by stretcher in stages.

Source C

From the diary of E. S. B. Hamilton, 19 August 1916. Hamilton had been in France for over a year at this time, as part of the Field Ambulance. At the time of this diary entry, he was working at an Advanced Dressing Station on the Somme.

The dugout [of the ADS] is awfully overcrowded both night and day and it is impossible to get it cleaned or aired. [There were] something like 800 people through here in about thirty hours the day before yesterday. This is far too much work for the personnel [of] three officers and about 115 men. Result [is] a lot of the men are done up and the officers seedy and depressed.

Extend your knowledge

John McCrae

John McCrae was a Canadian medical officer serving on the Western Front. He treated many wounded soldiers during the Second Battle of Ypres. The famous war poem *In Flanders Fields*, written by McCrae, was composed at the Essex Farm Advanced Dressing Station in May, 1915. The remains of this dressing station can still be seen today.

Source D

A photograph of an Advanced Dressing Station. This was taken in August 1916 at Pozieres Ridge, which was part of the Somme campaign.



Those working at the Dressing Stations belonged to a unit of the RAMC called the **Field Ambulance**. This should not be confused with the vehicles that carried the sick and wounded, which were known as **ambulance wagons**. In theory, each Field Ambulance unit could deal with 150 wounded men, but when major battles were taking place, they would have to deal with many more. The Field Ambulance at Hooze in the Ypres Salient dealt with about 1,000 casualties on 10–11 August 1917. The Field Ambulance units did not have the facilities to tend to wounded men for more than a week. Men who had been treated would either be returned to their units if they were fit enough to fight again, or they would be moved on to the next phase of the chain of evacuation by horse or motor ambulance.

Casualty Clearing Stations (CCS)

Casualty Clearing Stations were located a sufficient distance from the frontline to provide some safety against attack, but close enough to be accessible by ambulance wagons. Often the CCS closest to the frontline would specialise in operating on the most critical injuries, such as those to the chest. They were set up in buildings such as factories or schools and were often located near to a railway line to allow the next stage of the chain of evacuation to take place quickly.

When wounded soldiers arrived here, they were divided into three groups. This system was called **triage**, from the French word for sorting or selecting. Triage helped medical staff make decisions about treatment.

The three categories the wounded were divided into are:

- 1 **The walking wounded.** These were the men who could be patched up and returned to the fighting.
- 2 **Those in need of hospital treatment.** These men would need to be transported to a Base Hospital once they had been treated for any immediate life-threatening injuries.
- 3 **Those who were so severely wounded that there was no chance of recovery.** These men would be made comfortable, but the medical resources available were given to those who were more likely to survive their wounds.

Here are some useful statistics about the CCS during the Third Battle of Ypres in 1917.

- There were 24 CCS in the Ypres Salient.
- 379 doctors and 502 nurses treated more than 200,000 casualties.
- The medical staff operated on 30% of the men who were admitted.
- In total, 3.7% of the men admitted died.

Exam-style question, Section A

How useful are Sources C and D for an enquiry into the treatment of the wounded at ADSs on the Western Front?

Explain your answer, using Sources C and D and your knowledge of the historical context. **8 marks**

Exam tip

Go through these steps when preparing your answer.

- Always concentrate on the enquiry – in this question, the treatment of the wounded at ADSs, and then identify points from the content.
- Think about the provenance (nature, origin, purpose) of both sources.
- Think about what you know about the context – the Advanced Dressing Stations.
- Link together the content, provenance and context in your answer and include how the provenance and context affects the usefulness of the content.

Source E

Photograph of a ward in the Casualty Clearing Station at Hazebrouck in 1915. Hazebrouck was close to Ypres.



Activities

In pairs, study the statistics about the Casualty Clearing Stations during the Third Battle of Ypres and Source E.

- 1 What do you learn from the statistics?
- 2 Why can statistics be useful to historians?
- 3 What problems should historians be aware of when using statistics?
- 4 Describe the main features of the Casualty Clearing Station in the photograph.
- 5 What evidence does Source E provide that medical treatment in the Casualty Clearing Stations was effective?

The role of FANY

The first six FANYs arrived in France on 27 October 1914. However, the British would not make use of them so they devoted their energies to helping French and Belgian troops.

Finally, in January 1916, the British army decided to allow FANYs to drive ambulances. They became the first women to carry out this role, replacing British

Red Cross male ambulance drivers. They were used to transport wounded troops by ambulance in the Calais region. Although there were never more than 450 FANYs in France, they did open the way for other women who were attached to other organisations, such as the Voluntary Aid Detachments (VADs), to participate in the frontline.

Extend your knowledge

What else did FANY do?

FANY did things other than driving ambulances to support the soldiers on the Western Front.

- They drove supplies such as food and clothes to the frontline.
- They had a mobile bath unit which provided baths to the soldiers in water heated by the power from the van's engine.
- They set up cinemas to help the morale of soldiers.

Source F

From Pat Beauchamp's autobiography, *Fanny Goes to War*, published in 1919. Beauchamp first worked as a nurse, bringing in the wounded from the trenches, and from 1916 as an ambulance driver. Here she is writing about an account of FANYs from an English newspaper.

The following is an extract from an account by Mr. Beach Thomas in a leading daily: "Our Yeomanry nurses who, among other work, drive, clean, and manage their own ambulance cars, have done prodigies [wonders] along the Belgian front. One of their latest activities has been to devise and work a peripatetic [travelling] bath. Ten collapsible baths are packed into a motor car which circulates behind the lines. The water is heated by the engine in a cistern in the interior of the car and offers the luxury of a hot bath to several score men."

Base Hospitals

Base Hospitals on the Western Front were located near the French and Belgian coast, so that the wounded men who were treated there would be close to the ports, from which they could be transported home to Britain. At the start of the war there were two types of Base Hospital – the Stationary Hospital and the General Hospital. However, in practice, they worked in very similar ways. Men were treated in both types of hospital until they could be returned to Britain for further treatment or were fit enough to return to the fighting.

As the war progressed, Casualty Clearing Stations played an increasingly important role in dealing with wounds, instead of Base Hospitals. It had become clear that if contaminated wounds were not dealt with quickly, wounded men were more likely to develop gangrene. This meant that the CCSs started doing operations that it was originally believed would be done in the Base Hospitals.

By May 1916, for example, at Number 26 General Hospital at Étaples, most head and chest patients had been operated upon before their arrival at the Base Hospital there. In turn, the Base Hospitals became increasingly responsible for continuing treatment that was begun in the CCSs, before men were either returned to frontline fighting or transported back to Britain. The size of the Base Hospitals increased, especially after a major offensive had taken place. In 1917, three new Base Hospitals with a total of 2,500 beds were available.

As the Base Hospitals were not carrying out their intended role, other important roles emerged for them. They experimented with new techniques which, once successful, were used in the CCSs. For example, by dividing patients up into different wards according to their wounds, such as amputees, head wounds, chest wounds, and by allocating doctors to a specialised ward, it was possible for doctors to become expert in the treatment of particular wounds.

The Casualty Clearing Stations retained their role as the most important place for operations until the spring of 1918. The static nature of trench warfare had meant that the CCSs had been relatively safe early on in the war, but in March 1918, the Germans launched the Spring Offensive. This was a last-ditch attempt to win the war before American troops arrived in Europe and joined the British. It resulted in many CCSs having to move back, so

much of the surgery that was required for the wounded was now undertaken again in the Base Hospitals.

Extend your knowledge

'Blighty' wounds

A 'Blighty' wound, was a wound serious enough to get soldiers away from the fighting and back to Britain via the chain of evacuation, but would not result in permanent medical problems.

Source G

A photograph of a British Base Hospital in 1916. The nurses are preparing for a visit from King George V.



The underground hospital at Arras

In November 1916, tunnelling began under the town of Arras. In 800 m of tunnels, a fully working hospital was created so close to the frontline that it was, in reality, a Dressing Station. From here, wounded soldiers would move through the chain of evacuation. It was sometimes called **Thompson's Cave** after the RAMC officer who was responsible for equipping it. There were waiting rooms for the wounded, 700 spaces where stretchers could be placed as beds, an operating theatre, rest stations for stretcher bearers, and a mortuary to lay out the dead. Electricity and piped water were supplied to the hospital. The hospital was abandoned during the Battle of Arras in 1917, when it was hit by a shell which destroyed the water supply, but luckily did not injure any people.

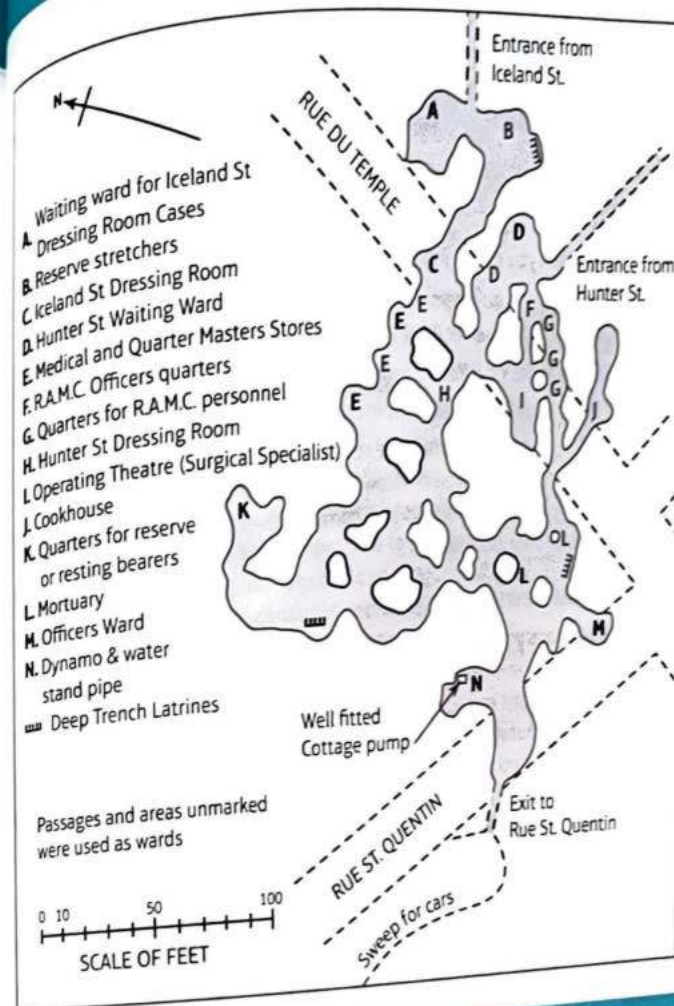


Figure 5.8 Plan of the Arras Underground hospital, also known as Thompson's Cave, based on a drawing from April 1917.

Source H

From Major-General Sir W. G. Macpherson, *Medical Services General History*, published in 1924. Macpherson was on the Western Front from 1914. From 1916–18, he was in charge of the RAMC. He wrote this history based on official records to which he had access. Here he is writing about the underground hospital at Arras.

Dressing Stations were established in caves, cellars and basements of buildings, protected as strongly as possible with sandbags on the outskirts of the town. The chief of these was in a large subterranean cave, from which stone had been excavated for building the town in the 16th century. It was close to the 3rd Division trenches and only 800 yards from the frontline. Two entrances for stretchers were tunnelled into it from the communication trenches, and an exit tunnelled out from the back into Rue St. Quentin, where an approach was constructed for ambulance cars. This cave was fitted with electric light and a piped water supply and was able to accommodate 700 wounded on stretchers in two tiers.

Source I

From *The Daily Telegraph*, a British newspaper, 29 April 1915.

POISON BOMBS: CANADIAN'S HEROIC CONDUCT

There appears little doubt that the material used by the Germans in the "poison bombs" is chlorine. This is the only conclusion one can arrive at after hearing the graphic narration of a Canadian who was enveloped in the fumes near Ypres.

The Canadian said, "Directly we opened fire the Germans rained shrapnel over us. We kept the guns going, wounded as some of us were. That we could stand. We had no complaints, because it was honest warfare. Then came the surprise. We saw bombs burst in the air and throw off a greenish-yellow vapour..."

At yesterday's meeting of the London Education Committee, the chairman (Mr. Gilbert) called attention to the request of the Government for respirators [gas masks] for the troops.

Using the range of sources

Following up an enquiry

In the examination, you are asked to suggest a possible question and a type of source that you could use to follow up another source. The framework of the question helps you through the four-stage process.

- 1 The detail you might follow up is the request for respirators.
- 2 The question you might ask is 'How many gas masks were given to Canadian troops after April 1915?'
- 3 There are lots of different types of source in this unit. You could suggest private diaries, local newspapers or official records.
- 4 Lastly, you have to explain the reason for your choice. For example, you could say that you would follow up Source I with a private diary because gas attacks would be a common event soldiers would write about in their diaries.

National newspapers

Source I shows you some of the strengths and weaknesses of national newspapers as sources. The report refers to events that are going on in both the Western Front and at home in response to these events. The chairman of the London Education Committee is named, but the soldier is anonymous. He is only referred to as 'the Canadian'. The article appears to give valid information, but it is also a form of propaganda.

Activities

Read Source I in small groups.

- a What can you find about the German attack at Ypres, the types of weapon used and the response at home?
- b Which of the things you discovered would you also expect to find in a local newspaper?
- c How useful is this national newspaper for studying gas attacks?

Summary

- The Royal Army Medical Corps (RAMC) provided both doctors and other staff to support the medical services.
- The First Aid Nursing Yeomanry (FANY) provided additional support to the RAMC.
- Moving the wounded from the frontline to the appropriate medical facility was known as the chain of evacuation.
- The main stages for the most severely wounded were the Regimental Aid Post, the Dressing Station, Casualty Clearing Station and Base Hospitals.

Checkpoint

Strengthen

- S1** Create a flow diagram that shows the main stages in the chain of evacuation.

Challenge

- C1** How useful are Sources C on page 149 and I on page 164 for a study of gas attacks?

5.5 The significance of the Western Front for experiments in surgery and medicine

Learning outcomes

- Understand how the experience of war on the Western Front gave rise to new techniques in medical treatment.
- Understand how new methods of surgery developed to treat the large number of head injuries that were a result of trench warfare.

New techniques in the treatment of wounds and infection

A major problem that faced the RAMC at the start of the war on the Western Front was dealing with infections caused by gas gangrene. It was not possible to perform aseptic surgery in Dressing Stations and Casualty Clearing Stations, due to the contaminated conditions and because of the large numbers of wounded men needing treatment.

Because of this, other methods of treatment had to be found. This led to much disagreement in the medical profession between those medics who were facing the frontline conditions on a daily basis, and those who were back in Britain, unfamiliar with what medics in the trenches were facing.

Extend your knowledge

Amputations

Queen Mary's Hospital at Roehampton, just outside London, became the leading centre for fitting new limbs for men who had experienced amputations. Opening in 1915, it had fitted artificial limbs to more than 26,000 men by the end of the war in 1918.

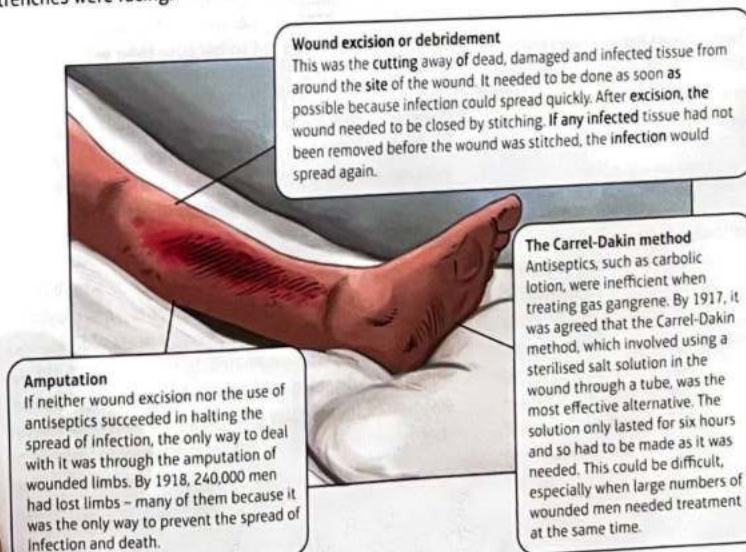


Figure 5.9 The main techniques used to prevent infections from spreading.

Source A

From the diary of B. C. Jones, 1915–16. Jones served with the Royal Field Artillery in France from the start of the war until he was wounded in 1915.

7 December. A German shell hit the dugout of our telephone pit. I remembered no more until I woke up in Bethune Casualty Clearing Station Number 33, where I find I have been severely wounded. Left hand blown off, left arm ripped up 12 inches. Scalp wound 6 inches, wound on over side of knee (left) 5 inches.

9 December. Operation on upper arm for gangrene (successful).

12 December. I remain here for 8 days then removed to St Omer by hospital barge, very comfortable. I am then removed by train to Étaples. I am sent to England on the Hospital Ship. Return to Nottingham where I am in bed until the end of February.

3 June 1916. I am eventually transferred to Brighton where I am operated on and re-amputated. Awaiting Roehampton for artificial limb.

Source B

From Ward Muir's *Observations of an Orderly*, published in 1917. Muir was a Lance Corporal in the RAMC and worked in a hospital in London that received patients from the Western Front at the end of the chain of evacuation.

The majority of stretcher-cases reach us in a by no means desperate state, for, as I say, they seldom come to England without having been treated previously at a base abroad (except during the periods of heavy fighting). And it is remarkable how often the patient refuses help in getting off the stretcher on to the bed. He may be a cocoon of bandages, but he will courageously heave himself overboard, from stretcher to bed, with a wallop which would be deemed rash even in a person in perfect health.

Activities

Study Sources A and B.

- 1 What conclusions can you draw from the fact that the soldier in Source A needed to have a limb re-amputated when he returned to England?
- 2 With a partner, make a list of inquiries you think Source A and B would be useful for.

The Thomas splint

In 1914 and 1915, men with a gunshot or shrapnel wound to the leg only had a 20% chance of survival. This was because these wounds created a compound fracture where the broken bone pierced the skin. It was particularly serious if the femur (thigh bone) was fractured, because a large amount of muscle would be damaged. This meant there was likely to be major bleeding into the thigh.

The splint that was in use as the wounded man was transferred from the frontline did not keep the leg rigid. By the time the wounded man arrived at the Casualty Clearing Station, where he could be operated on, he would have lost a great deal of blood, was likely to be in shock and might already have developed gas gangrene in the wound. This combination of factors reduced his chances of surviving an operation to the wound. Many of those who survived did so because their wounded leg was amputated.

It was clear that a way of improving the survival rate for men with this type of injury was needed. In fact, the solution, which only came into use in 1916, had been available since well before the start of the war.

In the late 19th century, Robert Jones worked with his uncle, Hugh Thomas, in his medical practice, where his uncle had designed a splint to stop joints from moving. When the war broke out, Jones was 57 years old. He offered his services immediately to the war effort. He worked with disabled soldiers in a hospital in London and started to make use of his uncle's **Thomas splint**.

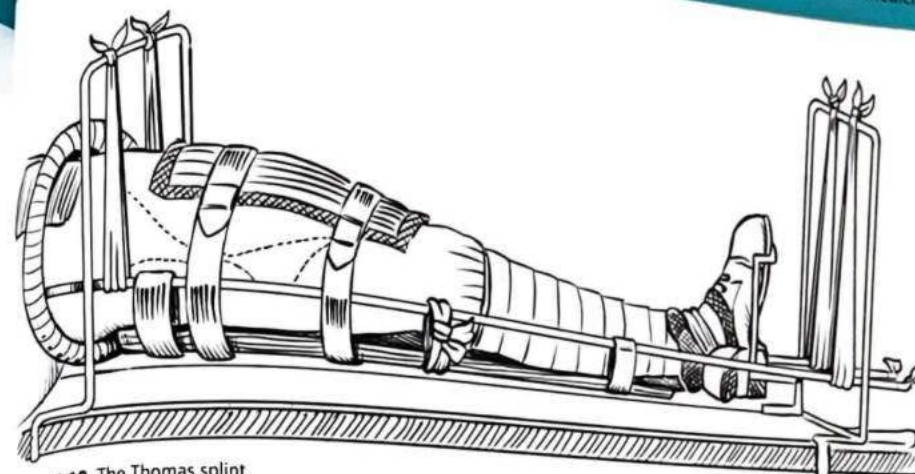


Figure 5.10 The Thomas splint.

As a result of this, in December 1915, he was sent to Boulogne to instruct medical practitioners on how to use the Thomas splint. The introduction of its use from this time increased the survival rate for this type of wound from 20% to 82%.

The use of mobile x-ray units

Extend your knowledge

Marie Curie

Marie Curie was a leading Polish-French scientist who did ground-breaking work on radioactivity. When the war broke out, she immediately understood the importance of x-rays for detecting shrapnel and was responsible for equipping 20 mobile x-ray vans to work in the French sector of the Western Front. These vans were known as 'petites Curies' – little Curies.

X-rays were used from the start of the war. Their main use was to identify shell fragments and bullets in wounds, which, if not removed when the person was wounded, could cause infection. Two x-rays would be taken from different angles and this helped the surgeon to identify quite accurately the location of shrapnel and bullets in the body.

Although a relative success, there were some problems with the use of x-rays for medical practitioners on the Western Front.

- X-rays could not detect all objects in the body. For example, fragments of clothing that were driven into wounds with shrapnel would not show up on an x-ray. However, doctors soon realised that they needed to look for these fragments in the same area as shrapnel when operating on the wounded.
- The length of time that a wounded man had to remain still whilst the x-ray was taken was still several minutes, which could cause problems depending on the wound.
- The tubes used in x-ray machines were fragile and overheated quite quickly. This meant that x-ray machines could only be used for about one hour at a time and then had to be left to cool down. This posed a problem when there was a major offensive going on and large numbers of wounded men were being brought in. The solution was to have three machines which would be used in rotation. When a machine became too hot to continue working, it would be replaced by another one. There had been an advance made in the technology of tubes in the USA by William Coolidge in 1913, but this was not available to the RAMC on the Western Front until the USA entered the war in 1917.

Source C

From *Radiography and Radiotherapeutics*, by Robert Knox, published in 1917. This was a textbook on the use of x-rays written by a British doctor.

The need for portable outfits in connection with the war has led to a great development in the provision of motor wagons containing complete x-ray apparatus with all accessories. The mechanism used for driving the wagon i.e. the motor is coupled with a powerful dynamo which delivers a continuous current.

Exam-style question, Section A

How could you follow up Source C to find out more about x-rays on the Western Front?

In your answer, you must give the question you would ask and the type of source you could use.

Copy out and complete the table below.

Detail in Source C that I would follow up	4 marks
Question I would ask	
What type of source I could use	
How this might help answer my question	

Exam tip

This question involves a four-stage process. The example shows you what to do at each stage.

- **Pick a detail.** For example, 'a great development in the provision of motor wagons.'
- **Question I would ask.** How many x-ray vehicles were used in 1917 on the Western Front?
- **Type of source I would use.** Military records showing the number of x-ray vehicles being used in 1917 on the Western Front.
- **How this might help answer my question.** The data would show whether the numbers increased and quantify the words 'great development' in the statement in Source C.

Source D

A photograph of a mobile x-ray unit, taken in 1917. Notice how the equipment has been laid out.



The Base Hospitals and some of the larger Casualty Clearing Stations had static (unmoving) x-ray machines as part of their equipment. Those that did not have them could call on a mobile unit. There were six mobile x-ray units operating in the British sector on the Western Front. Setting up the equipment from the mobile unit took some time. A tent was attached to the back of the van with a table where stretchers could be placed. The x-ray machine was set up next to this table and linked to the engine of the van, which was used to power the x-ray machine. The equipment for processing the x-ray films was set up inside the van. Although the quality of the x-rays taken by the mobile units was not quite as good as that taken by static units, it was sufficient to identify shrapnel and bullets and prevent infection for many of the wounded soldiers (see Source D).

Blood transfusions

The use of blood transfusions from 1915 in the British sector of the Western Front was pioneered by a Canadian doctor, Lawrence Bruce Robertson, in the Base Hospital at Boulogne. He used the indirect method, where a syringe and tube was used to transfer the donor blood to the patient. The purpose of this was to stop the patient going into shock through blood loss before surgery. Even where a wound was relatively minor, shock could kill a soldier. Those who did not experience a negative reaction to the blood transfusion generally recovered. As blood transfusions proved so successful at the Base Hospital, it was decided to extend their use. Therefore, by 1917, blood transfusions were being administered in the Casualty Clearing Stations as a routine measure in the treatment of shock.

Geoffrey Keynes, a British doctor and lieutenant in the RAMC, designed a portable blood transfusion kit that was used to provide blood transfusions close to the frontline. Despite Robertson's pioneering work, this kit did not use stored blood because of the difficulties in keeping the blood fresh when there was no refrigeration available. Keynes added a device to the blood bottle to regulate the flow of the blood which helped prevent clotting. In 1915, Keynes used the new method in a Casualty Clearing Station on the Western Front. By his own accounts, it saved countless lives.

The blood bank at Cambrai

The identification of blood groups and the use of blood type O as a universal donor blood type meant that the risk of being transfused with the wrong blood group was reduced (see page 144). The problem of clotting remained, and there was never enough blood on hand to meet demand. However, as the war continued, some advances were made in the storage of blood.

- In 1915, American doctor Richard Lewisohn discovered that by adding sodium citrate to blood, the need for donor-to-donor transfusion was removed. Blood transfusions could be done indirectly, with patients not needing to be in the same room.
- In the same year, Richard Weil discovered that blood with sodium citrate could be refrigerated and stored for up to two days.
- In 1916, Francis Rous and James Turner found that by adding a citrate glucose solution to blood, it could be stored for a much longer period – up to four weeks.

The use of stored blood was clearly demonstrated in 1917 at the Battle of Cambrai. Before the battle, Oswald Hope Robertson, a British-born American doctor, stored 22 units of universal donor blood in glass bottles. He built a carrying case for the bottles in ammunition boxes which he packed with ice and sawdust. He called this a 'blood depot'. During the battle, he treated 20 severely wounded Canadian soldiers with the 22 units of blood, some of which had been collected 26 days before use. They were so badly affected by shock that none of them were expected to survive. In fact, of the 20 wounded men, 11 survived.

Robertson's work at Cambrai was the first time stored blood was used to treat soldiers in shock, and, although it was only on a small scale, demonstrated its potential to save lives. This was important, because during times of heavy fighting, only the most severely wounded were taken to the Casualty Clearing Stations. The less severely wounded, who were normally the men who gave blood for transfusions, would not be taken there. Therefore, the availability of blood stored in a number of blood depots made a huge difference to men's chances of survival.

The attempts to deal with increased numbers of head injuries

About 20% of all wounds in the British sector of the Western Front were to the head, face and neck. This was the part of the body that was most exposed in the trench warfare of the Western Front. Injuries of this nature could be caused by both bullets and shrapnel.

Brain Surgery

Injuries to the brain were very likely to prove fatal at the start of the war because:

- the issue of infection applied just as much to head injuries as it did to wounds to other parts of the body
- there were difficulties involved in moving men with head injuries through the chain of evacuation, as they were often unconscious or confused
- there were very few doctors who had experience of neurosurgery* before the war.

Key term

Neurosurgery*

Surgery carried out on the nervous system, especially the brain and the spine.

Despite the inexperience of doctors in dealing with head wounds, observation quickly led to improvements in methods of treatment.

Harvey Cushing, an American neurosurgeon, developed new techniques in brain surgery on the Western Front. He experimented, for example, with the use of a magnet to remove metal fragments from the brain. He also used a local anaesthetic* rather than a general anaesthetic* when operating. The reason for this was that the brain swelled as a result of general anaesthetics and this increased the risks of the operation. His methods

Observation	New method of treatment
Men who were operated on quickly were more likely to survive.	Specific Casualty Clearing Stations became chosen as centres for brain surgery. For example, during the Third Battle of Ypres, all head injuries were moved to the Casualty Clearing Station at Mendinghem.
It was dangerous to move men too soon after an operation.	Patients remained at the Casualty Clearing Station for three weeks after surgery.
Injuries that looked fairly minor could be hiding more severe injuries.	All head wounds were always carefully examined.

became more effective as he learned more through observation. He operated on 45 patients in 1917 with an operation survival rate of 71%, compared to the general survival rate of 50% for brain surgery.

Key terms

Local anaesthetic*

Keeping a patient awake during an operation, with the area being operated on numbed to prevent pain.

General anaesthetic*

Putting a patient to sleep during an operation.

Source E

From *A Surgeon's Journal 1915-18*, by Harvey Cushing, published in 1936. Here he is describing the conditions under which he is working during the battle of Passchendaele on 19 August 1917.

My prize patient, Baker, with the shrapnel ball removed from his brain, after doing well for three days suddenly shot up a temperature to 104 last night about midnight. I took him to the operating theatre, reopened the perfectly healed external wound, and found to my dismay a massive gas infection of the brain. I bribed two orderlies to stay up with him in the operating room, where he could have constant thorough irrigation over the brain and through the track of the missile [passing a warm saline solution along the path taken by the shrapnel to prevent infection]. No light except candles was permitted last night.

Plastic Surgery

The development of plastic surgery was largely the work of a New Zealand doctor called Harold Gillies. In civilian life, he was an ENT (ear, nose and throat) surgeon. He was sent to the Western Front in January 1915. There he met Charles Valadier, a French man who had been working for the British Red Cross as a dentist since October 1914. Head injuries that might not kill, could cause severe disfigurement. This led Gillies to become interested in facial reconstruction – how to replace and restore those parts of the face that had been destroyed. As he had no background in this type of surgery, he devised new operations to deal with problems as they confronted him.

Source F

Four photographs documenting the facial reconstruction of a soldier whose cheek was extensively wounded during the Battle of the Somme (July 1916).



Activities

Study Source E and answer the following questions:

- 1 Why do you think Baker was Cushing's 'prize patient'?
- 2 What happened to make Baker's temperature rise?
- 3 What do you learn about the problems facing surgeons on the Western Front?